

Machine Learning Operations

Manjunath Prasad H. R.

Assistant Professor

Department of Artificial Intelligence and Machine Learning

Mangalore Institute of Technology and Engineering



Outline

- 1 Course Description
- 2 Course Module Topics
 - Module Topics
 - Module Lab Component
- 3 Course Outcomes
- 4 Textbook
 - Textbook Structure
- 5 Module 1: Why Now and Challenges
 - Defining MLOps and Its Challenges
 - MLOps to Mitigate Risk
 - MLOps for Scale
- 6 Module 1: People of MLOps
- 7 Module 1: Key MLOps Features
 - A Primer on Machine Learning
 - Model Development
 - Productionalization and Deployment

Course Description

- ❶ **Module 1:** Establish foundational understanding of MLOps practices, lifecycle management, and version control workflows using Git/GitHub for collaborative ML project tracking
- ❷ **Module 2:** Execute systematic model development through data exploration, feature engineering, algorithm comparison, and implement reproducible experimentation tracking using MLflow
- ❸ **Module 3:** Automate model training and testing through CI/CD pipelines (GitHub Actions), containerize models using Docker, and implement production deployment strategies
- ❹ **Module 4:** Detect model degradation and drift in production, establish feedback loops for retraining, implement governance frameworks, and ensure regulatory compliance and responsible AI
- ❺ **Module 5:** Apply end-to-end MLOps workflows to real-world scenarios including credit risk classification, recommendation engines with online learning, and continual learning pipelines

Course Description

- ❶ **Module 1:** Establish foundational understanding of MLOps practices, lifecycle management, and version control workflows using Git/GitHub for collaborative ML project tracking
- ❷ **Module 2:** Execute systematic model development through data exploration, feature engineering, algorithm comparison, and implement reproducible experimentation tracking using MLflow
- ❸ **Module 3:** Automate model training and testing through CI/CD pipelines (GitHub Actions), containerize models using Docker, and implement production deployment strategies
- ❹ **Module 4:** Detect model degradation and drift in production, establish feedback loops for retraining, implement governance frameworks, and ensure regulatory compliance and responsible AI
- ❺ **Module 5:** Apply end-to-end MLOps workflows to real-world scenarios including credit risk classification, recommendation engines with online learning, and continual learning pipelines

Course Description

- ❶ **Module 1:** Establish foundational understanding of MLOps practices, lifecycle management, and version control workflows using Git/GitHub for collaborative ML project tracking
- ❷ **Module 2:** Execute systematic model development through data exploration, feature engineering, algorithm comparison, and implement reproducible experimentation tracking using MLflow
- ❸ **Module 3:** Automate model training and testing through CI/CD pipelines (GitHub Actions), containerize models using Docker, and implement production deployment strategies
- ❹ **Module 4:** Detect model degradation and drift in production, establish feedback loops for retraining, implement governance frameworks, and ensure regulatory compliance and responsible AI
- ❺ **Module 5:** Apply end-to-end MLOps workflows to real-world scenarios including credit risk classification, recommendation engines with online learning, and continual learning pipelines

Course Description

- ❶ **Module 1:** Establish foundational understanding of MLOps practices, lifecycle management, and version control workflows using Git/GitHub for collaborative ML project tracking
- ❷ **Module 2:** Execute systematic model development through data exploration, feature engineering, algorithm comparison, and implement reproducible experimentation tracking using MLflow
- ❸ **Module 3:** Automate model training and testing through CI/CD pipelines (GitHub Actions), containerize models using Docker, and implement production deployment strategies
- ❹ **Module 4:** Detect model degradation and drift in production, establish feedback loops for retraining, implement governance frameworks, and ensure regulatory compliance and responsible AI
- ❺ **Module 5:** Apply end-to-end MLOps workflows to real-world scenarios including credit risk classification, recommendation engines with online learning, and continual learning pipelines

Course Description

- ❶ **Module 1:** Establish foundational understanding of MLOps practices, lifecycle management, and version control workflows using Git/GitHub for collaborative ML project tracking
- ❷ **Module 2:** Execute systematic model development through data exploration, feature engineering, algorithm comparison, and implement reproducible experimentation tracking using MLflow
- ❸ **Module 3:** Automate model training and testing through CI/CD pipelines (GitHub Actions), containerize models using Docker, and implement production deployment strategies
- ❹ **Module 4:** Detect model degradation and drift in production, establish feedback loops for retraining, implement governance frameworks, and ensure regulatory compliance and responsible AI
- ❺ **Module 5:** Apply end-to-end MLOps workflows to real-world scenarios including credit risk classification, recommendation engines with online learning, and continual learning pipelines

Course Module 1 Topics - Introduction to MLOps

- ♣ Defining MLOps and Its Challenges
- ♣ MLOps to Mitigate Risk
- ♣ MLOps for Scale
- ♣ People of MLOps
- ♣ Subject Matter Experts
- ♣ Key MLOps Features
- ♣ A Primer on Machine Learning
- ♣ Model Development
- ♣ Productionalization and Deployment
- ♣ Monitoring
- ♣ Iteration
- ♣ Life Cycle Governance

Module 1 Laboratory - Git Version Control

- ♣ Set up GitHub/GitLab for ML project versioning.
- ♣ Familiarize with Git commands.
- ♣ Use Git for tracking model and code changes.

Course Module 2 - Process in Model Development and Versioning

- ♣ Machine Learning Model
- ♣ ML Required Components
- ♣ Different ML Algorithms
- ♣ Different MLOps Challenges
- ♣ Data Exploration
- ♣ Feature Engineering and Selection
- ♣ Experimentation
- ♣ Evaluating and Comparing Models
- ♣ Version Management
- ♣ Reproducibility

Module 2 Laboratory - Experiment Tracking with MLflow

- ♣ Train a simple machine learning model (e.g., Logistic Regression).
- ♣ Log model parameters using MLflow.
- ♣ Log evaluation metrics.
- ♣ Store model artifacts.
- ♣ Compare different experiment runs.

Course Module 3 - Preparing for Production and CI/CD Pipelining

- ♣ Runtime Environments
- ♣ Model Risk Evaluation
- ♣ Quality Assurance for Machine Learning
- ♣ Key Testing Considerations
- ♣ Reproducibility and Auditability
- ♣ Machine Learning Security
- ♣ Model Risk Mitigation
- ♣ Deploying to Production using CI/CD Pipelines
- ♣ Building ML Artifacts
- ♣ Deployment Strategies
- ♣ Containerization
- ♣ Scaling Deployments
- ♣ Requirements and Challenges

Module 3 Laboratory - CI/CD for ML Models

- ♣ Set up a CI/CD pipeline using GitHub Actions.
- ♣ Automate model training.
- ♣ Automate model testing on code commits.
- ♣ Validate reproducible ML workflows.

Course Module 4 - Monitoring, Feedback Loop and Governance

- ♣ Model Retraining
- ♣ Understanding Model Degradation
- ♣ Drift Detection in Practice
- ♣ Feedback Loop
- ♣ Model Governance
- ♣ Matching Governance with Risk Level
- ♣ Current Regulations
- ♣ AI-Specific Regulations
- ♣ Responsible Artificial Intelligence

Module 4 Laboratory - Containerizing the Model

- ♣ Train a simple ML model using Scikit-learn.
- ♣ Save the trained model using Joblib or Pickle.
- ♣ Develop a Flask or FastAPI application.
- ♣ Serve the model as a REST API.
- ♣ Create a Dockerfile.
- ♣ Build the Docker image.
- ♣ Run the Docker container.

Course Module 5 - MLOps in Practice

- ♣ Consumer Credit Risk Management
- ♣ Model Development
- ♣ Model Bias Consideration
- ♣ Production Deployment
- ♣ Marketing Recommendation Engines
- ♣ Data Preparation
- ♣ Experiment Management
- ♣ Model Training
- ♣ Model Deployment
- ♣ Consumption Forecasting

Module 5 Laboratory - Credit Risk Model

- ♣ Use the German Credit Risk or Home Credit Default Risk dataset.
- ♣ Train a classification model.
- ♣ Apply Logistic Regression, Random Forest, or XGBoost.
- ♣ Predict loan default probability.
- ♣ Track the model lifecycle using MLflow.

Module 5 Laboratory - Continual Learning Model

- ♣ Train a recommendation engine using the MovieLens dataset.
- ♣ Build an online learning pipeline.
- ♣ Update the model using new user interactions.
- ♣ Demonstrate continual learning in production.

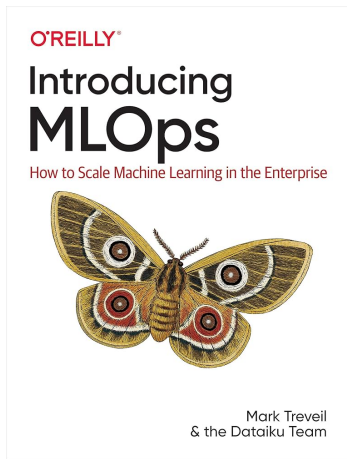
Course Outcomes

At the end of this course, the student will be able to:

- ❶ **UNDERSTAND** the fundamentals of MLOps, machine learning model deployment, and its significance.
- ❷ **COMPARE** various model versioning techniques and experiment tracking methods.
- ❸ **IMPLEMENT** model deployment strategies using Docker, Kubernetes, and cloud solutions.
- ❹ **APPLY** machine learning models in production by detecting performance degradation and data/model drift.
- ❺ **DEVELOP** an end-to-end MLOps workflow by integrating best practices in automation and testing.

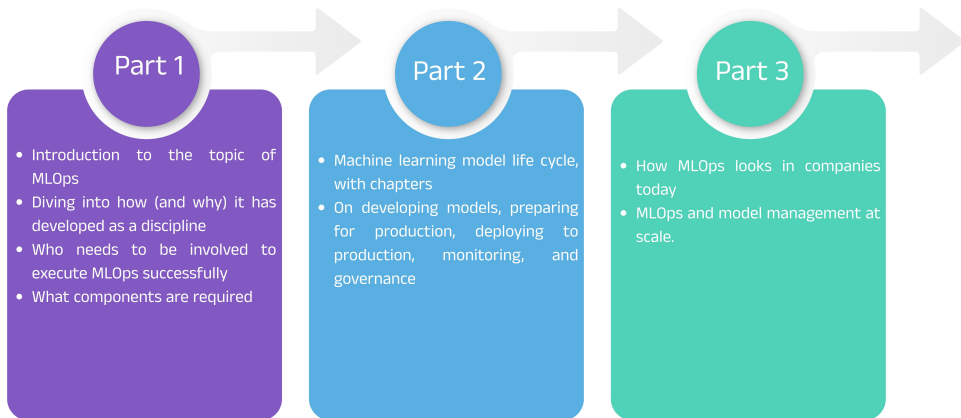
Textbook

Mark Treveil et. al. *Introducing MLOps: How to Scale Machine Learning in the Enterprise*, O'Reilly Media, 2020



Textbook Structure

Mark Treveil et. al. *Introducing MLOps: How to Scale Machine Learning in the Enterprise*, O'Reilly Media, 2020



Technology Stack Used in this Course

Category	Tools
Version Control	Git, GitHub
Programming	Python
ML Libraries	scikit-learn
Experiment Tracking	MLflow
API Development	FastAPI / Flask
Containers	Docker
CI/CD	GitHub Actions
Deployment	Docker Containers
Monitoring	MLflow, Custom Metrics
Data Handling	Pandas, NumPy

Learning Outcome:

- ♣ Gain hands-on experience with an end-to-end MLOps toolchain.
- ♣ Learn how machine learning models are developed, deployed, monitored, and maintained in

Thank You

Introduction to MLOps

Machine Learning Operations (MLOps)

A process that helps organizations and business leaders generate long-term value and reduce the risk associated with data science, machine learning, and AI initiatives.

- ❶ MLOps is quickly becoming a **critical component** of successful data science project deployment in the enterprise.
- ❷ It is a **relatively new concept**, yet it has seemingly skyrocketed into the data science lexicon overnight.
- ❸ This chapter covers:
 - ▶ What MLOps is at a high level and its challenges.
 - ▶ Why it is essential to a successful data science strategy.
 - ▶ Why it is coming to the forefront **now**.

Introduction to MLOps

Machine Learning Operations (MLOps)

A process that helps organizations and business leaders generate long-term value and reduce the risk associated with data science, machine learning, and AI initiatives.

- ❶ MLOps is quickly becoming a **critical component** of successful data science project deployment in the enterprise.
- ❷ It is a **relatively new concept**, yet it has seemingly skyrocketed into the data science lexicon overnight.
- ❸ This chapter covers:
 - ▶ What MLOps is at a high level and its challenges.
 - ▶ Why it is essential to a successful data science strategy.
 - ▶ Why it is coming to the forefront **now**.

Introduction to MLOps

Machine Learning Operations (MLOps)

A process that helps organizations and business leaders generate long-term value and reduce the risk associated with data science, machine learning, and AI initiatives.

- ❶ MLOps is quickly becoming a **critical component** of successful data science project deployment in the enterprise.
- ❷ It is a **relatively new concept**, yet it has seemingly skyrocketed into the data science lexicon overnight.
- ❸ This chapter covers:
 - ▶ What MLOps is at a high level and its challenges.
 - ▶ Why it is essential to a successful data science strategy.
 - ▶ Why it is coming to the forefront **now**.

MLOps Versus ModelOps Versus AIOps

- ❶ **MLOps / ModelOps:** A new discipline that emerged under these names in late 2018 and 2019; the two terms are largely used **interchangeably**.
- ❷ Some argue **ModelOps** is more general than MLOps, since it covers any kind of model (e.g., rule-based models), not only machine learning models.
- ❸ This course focuses on the **machine learning model life cycle**, hence the term “**MLOps**”.
- ❹ **AIOps** is an entirely different topic: solving operational challenges **using AI** (i.e., AI for DevOps).
- ❺ **AIOps example:** Predictive maintenance for network failures, alerting DevOps teams to possible problems before they arise.

MLOps Versus ModelOps Versus AIOps

- ❶ **MLOps / ModelOps:** A new discipline that emerged under these names in late 2018 and 2019; the two terms are largely used **interchangeably**.
- ❷ Some argue **ModelOps is more general** than MLOps, since it covers any kind of model (e.g., rule-based models), not only machine learning models.
- ❸ This course focuses on the **machine learning model life cycle**, hence the term “MLOps”.
- ❹ AIOps is an entirely different topic: solving operational challenges **using AI** (i.e., AI for DevOps).
- ❺ AIOps example: Predictive maintenance for network failures, alerting DevOps teams to possible problems before they arise.

MLOps Versus ModelOps Versus AIOps

- ❶ **MLOps / ModelOps:** A new discipline that emerged under these names in late 2018 and 2019; the two terms are largely used **interchangeably**.
- ❷ Some argue **ModelOps is more general** than MLOps, since it covers any kind of model (e.g., rule-based models), not only machine learning models.
- ❸ This course focuses on the **machine learning model life cycle**, hence the term “**MLOps**”.
- ❹ AIOps is an entirely different topic: solving operational challenges using AI (i.e., AI for DevOps).
- ❺ AIOps example: Predictive maintenance for network failures, alerting DevOps teams to possible problems before they arise.

MLOps Versus ModelOps Versus AIOps

- ❶ **MLOps / ModelOps:** A new discipline that emerged under these names in late 2018 and 2019; the two terms are largely used **interchangeably**.
- ❷ Some argue **ModelOps is more general** than MLOps, since it covers any kind of model (e.g., rule-based models), not only machine learning models.
- ❸ This course focuses on the **machine learning model life cycle**, hence the term “**MLOps**”.
- ❹ **AIOps** is an entirely different topic: solving operational challenges **using AI** (i.e., AI for DevOps).
- ❺ **AIOps example:** Predictive maintenance for network failures, alerting DevOps teams to possible problems before they arise.

MLOps Versus ModelOps Versus AIOps

- ❶ **MLOps / ModelOps:** A new discipline that emerged under these names in late 2018 and 2019; the two terms are largely used **interchangeably**.
- ❷ Some argue **ModelOps is more general** than MLOps, since it covers any kind of model (e.g., rule-based models), not only machine learning models.
- ❸ This course focuses on the **machine learning model life cycle**, hence the term “**MLOps**”.
- ❹ **AIOps** is an entirely different topic: solving operational challenges **using AI** (i.e., AI for DevOps).
- ❺ **AIOps example:** Predictive maintenance for network failures, alerting DevOps teams to possible problems before they arise.

Growth of MLOps

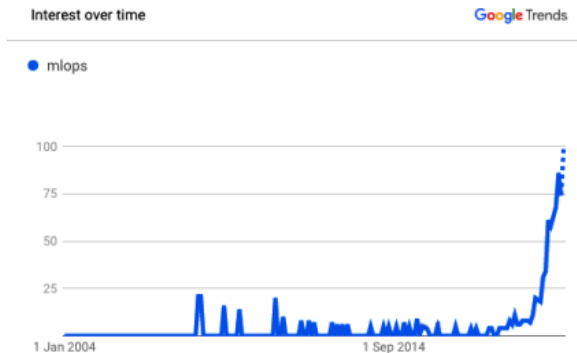


Figure: Representation of the exponential growth of MLOps (not the parallel growth of the term “ModelOps”)

- ❶ The chart shows Google Trends **interest over time** for the term “mlops” from 2004 onwards.
- ❷ For more than a decade the interest remained **close to zero**, with only occasional spikes.
- ❸ From around **2018–2019** the curve rises **exponentially**, reaching its peak (100).
- ❹ This confirms MLOps as a **new and rapidly adopted** discipline in enterprise data science.

Growth of MLOps

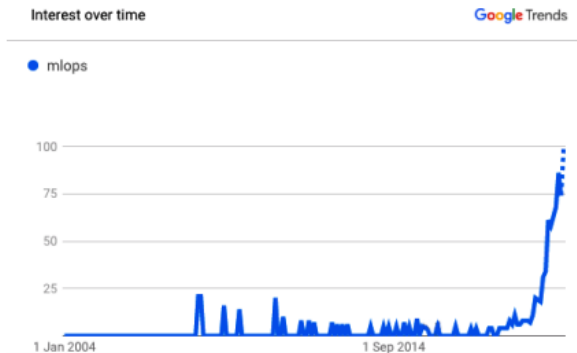


Figure: Representation of the exponential growth of MLOps (not the parallel growth of the term “ModelOps”)

- 1 The chart shows Google Trends **interest over time** for the term “mlops” from 2004 onwards.
- 2 For more than a decade the interest remained **close to zero**, with only occasional spikes.
- 3 From around 2018–2019 the curve rises **exponentially**, reaching its peak (100).
- 4 This confirms MLOps as a **new and rapidly adopted** discipline in enterprise data science.

Growth of MLOps

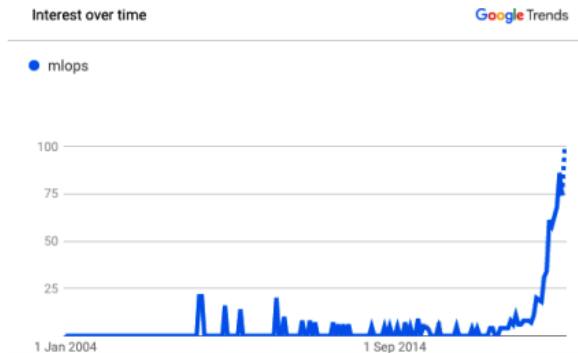


Figure: Representation of the exponential growth of MLOps (not the parallel growth of the term “ModelOps”)

- 1 The chart shows Google Trends **interest over time** for the term “mlops” from 2004 onwards.
- 2 For more than a decade the interest remained **close to zero**, with only occasional spikes.
- 3 From around **2018–2019** the curve rises **exponentially**, reaching its peak (100).
- 4 This confirms MLOps as a **new and rapidly adopted discipline** in enterprise data science.

Growth of MLOps

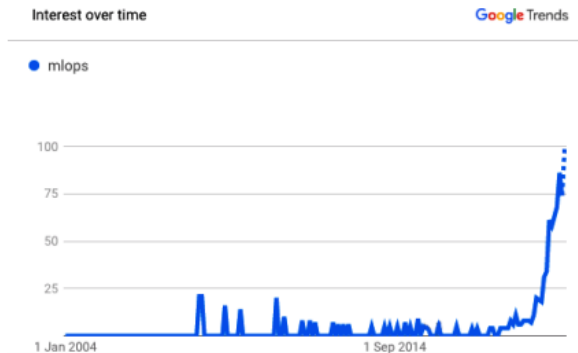


Figure: Representation of the exponential growth of MLOps (not the parallel growth of the term “ModelOps”)

- ❶ The chart shows Google Trends **interest over time** for the term “mlops” from 2004 onwards.
- ❷ For more than a decade the interest remained **close to zero**, with only occasional spikes.
- ❸ From around **2018–2019** the curve rises **exponentially**, reaching its peak (100).
- ❹ This confirms MLOps as a **new and rapidly adopted** discipline in enterprise data science.

Defining MLOps

Definition

At its core, MLOps is the **standardization and streamlining of machine learning life cycle management**.

- ❶ From business problem to ML model, the steps at a very high level **seem straightforward**.
- ❷ For most traditional organizations, developing **multiple ML models** and deploying them in production is **relatively new**.
- ❸ Earlier, the number of models was manageable at a small scale, and there was less interest in understanding models and their dependencies company-wide.
- ❹ With **decision automation** (decisions made without human intervention), models become more critical and **managing model risk** becomes a top-level concern.

Defining MLOps

Definition

At its core, MLOps is the **standardization and streamlining of machine learning life cycle management**.

- ❶ From business problem to ML model, the steps at a very high level **seem straightforward**.
- ❷ For most traditional organizations, developing **multiple ML models** and deploying them in production is **relatively new**.
- ❸ Earlier, the number of models was manageable at a small scale, and there was less interest in understanding models and their dependencies company-wide.
- ❹ With **decision automation** (decisions made without human intervention), models become more critical and **managing model risk** becomes a top-level concern.

Defining MLOps

Definition

At its core, MLOps is the **standardization and streamlining of machine learning life cycle management**.

- ❶ From business problem to ML model, the steps at a very high level **seem straightforward**.
- ❷ For most traditional organizations, developing **multiple ML models** and deploying them in production is **relatively new**.
- ❸ Earlier, the number of models was manageable at a small scale, and there was less interest in understanding models and their dependencies company-wide.
- ❹ With **decision automation** (decisions made without human intervention), models become more critical and **managing model risk** becomes a top-level concern.

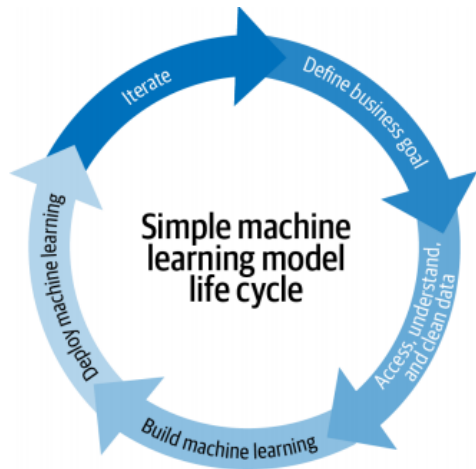
Defining MLOps

Definition

At its core, MLOps is the **standardization and streamlining of machine learning life cycle management**.

- ❶ From business problem to ML model, the steps at a very high level **seem straightforward**.
- ❷ For most traditional organizations, developing **multiple ML models** and deploying them in production is **relatively new**.
- ❸ Earlier, the number of models was manageable at a small scale, and there was less interest in understanding models and their dependencies company-wide.
- ❹ With **decision automation** (decisions made without human intervention), models become more critical and **managing model risk** becomes a top-level concern.

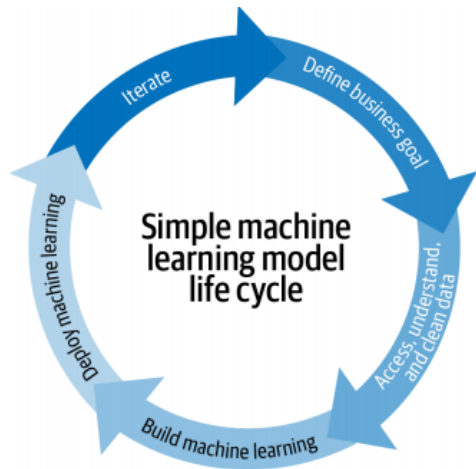
Simple Machine Learning Model Life Cycle



- 1 **Define business goal:** Identify the business problem the model must solve.
 - 2 **Access, understand, and clean data:** Gather and prepare the data required.
 - 3 **Build machine learning:** Train the ML model on the prepared data.
 - 4 **Deploy machine learning:** Push the trained model into use.
 - 5 **Iterate:** Repeat the cycle to improve the model.
- This view underplays the real complexity of needs and tooling in an enterprise (see next figure).

Figure: A simple representation of the ML model

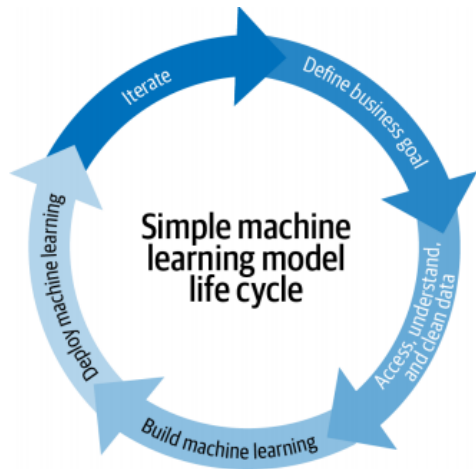
Simple Machine Learning Model Life Cycle



- 1 **Define business goal:** Identify the business problem the model must solve.
 - 2 **Access, understand, and clean data:** Gather and prepare the data required.
 - 3 **Build machine learning:** Train the ML model on the prepared data.
 - 4 **Deploy machine learning:** Push the trained model into use.
 - 5 **Iterate:** Repeat the cycle to improve the model.
- This view underplays the real complexity of needs and tooling in an enterprise (see next figure).

Figure: A simple representation of the ML model

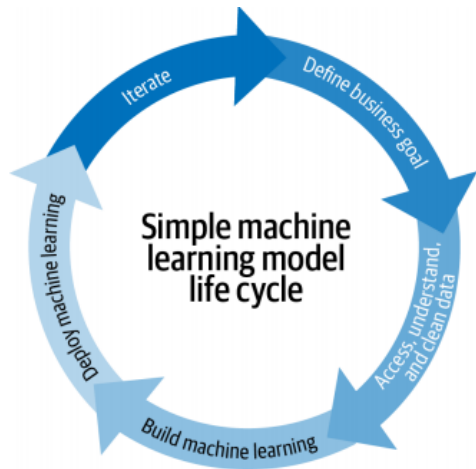
Simple Machine Learning Model Life Cycle



- 1 **Define business goal:** Identify the business problem the model must solve.
 - 2 **Access, understand, and clean data:** Gather and prepare the data required.
 - 3 **Build machine learning:** Train the ML model on the prepared data.
 - 4 **Deploy machine learning:** Push the trained model into use.
 - 5 **Iterate:** Repeat the cycle to improve the model.
- This view underplays the real complexity of needs and tooling in an enterprise (see next figure).

Figure: A simple representation of the ML model

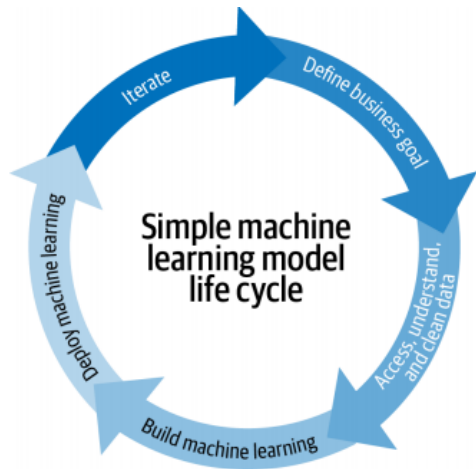
Simple Machine Learning Model Life Cycle



- ❶ **Define business goal:** Identify the business problem the model must solve.
 - ❷ **Access, understand, and clean data:** Gather and prepare the data required.
 - ❸ **Build machine learning:** Train the ML model on the prepared data.
 - ❹ **Deploy machine learning:** Push the trained model into use.
 - ❺ **Iterate:** Repeat the cycle to improve the model.
- This view underplays the real complexity of needs and tooling in an enterprise (see next figure).

Figure: A simple representation of the ML model

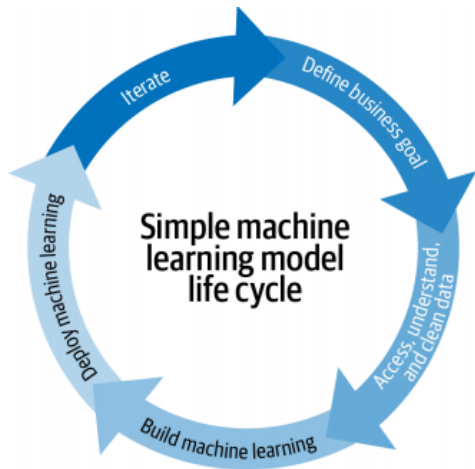
Simple Machine Learning Model Life Cycle



- 1 **Define business goal:** Identify the business problem the model must solve.
 - 2 **Access, understand, and clean data:** Gather and prepare the data required.
 - 3 **Build machine learning:** Train the ML model on the prepared data.
 - 4 **Deploy machine learning:** Push the trained model into use.
 - 5 **Iterate:** Repeat the cycle to improve the model.
- 6 This view underplays the real complexity of needs and tooling in an enterprise (see next figure).

Figure: A simple representation of the ML model

Simple Machine Learning Model Life Cycle



- 1 **Define business goal:** Identify the business problem the model must solve.
- 2 **Access, understand, and clean data:** Gather and prepare the data required.
- 3 **Build machine learning:** Train the ML model on the prepared data.
- 4 **Deploy machine learning:** Push the trained model into use.
- 5 **Iterate:** Repeat the cycle to improve the model.
- 6 This view **underplays** the real complexity of needs and tooling in an enterprise (see next figure).

Figure: A simple representation of the ML model

Challenges: Managing ML Life Cycles at Scale

❶ There are many dependencies:

- ▶ Data is constantly changing, and business needs shift as well.
- ▶ Results must be continually relayed back to the business to ensure the model in production meets the original goal.

❷ Not everyone speaks the same language:

- ▶ Business, data science, and IT teams are involved, but they do not use the same tools or share the same fundamental skills.

❸ Data scientists are not software engineers:

- ▶ They specialize in model building and assessment, not in writing applications.
- ▶ They juggle many roles; with staff turnover, they must manage models **they did not create**.

Challenges: Managing ML Life Cycles at Scale

❶ There are many dependencies:

- ▶ Data is constantly changing, and business needs shift as well.
- ▶ Results must be continually relayed back to the business to ensure the model in production meets the original goal.

❷ Not everyone speaks the same language:

- ▶ Business, data science, and IT teams are involved, but they do not use the same tools or share the same fundamental skills.

❸ Data scientists are not software engineers:

- ▶ They specialize in model building and assessment, not in writing applications.
- ▶ They juggle many roles; with staff turnover, they must manage models **they did not create**.

Challenges: Managing ML Life Cycles at Scale

❶ There are many dependencies:

- ▶ Data is constantly changing, and business needs shift as well.
- ▶ Results must be continually relayed back to the business to ensure the model in production meets the original goal.

❷ Not everyone speaks the same language:

- ▶ Business, data science, and IT teams are involved, but they do not use the same tools or share the same fundamental skills.

❸ Data scientists are not software engineers:

- ▶ They specialize in model building and assessment, not in writing applications.
- ▶ They juggle many roles; with staff turnover, they must manage models **they did not create**.

Realistic Machine Learning Model Life Cycle

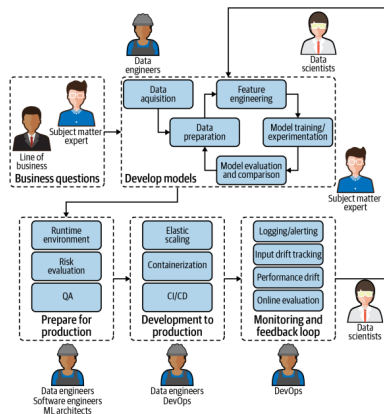


Figure: The realistic picture of an ML model life cycle inside an average organization today

- 1 **Business questions:** Line of business and subject matter experts define the problem.
- 2 **Develop models:** Data acquisition → data preparation → feature engineering → model training/experimentation → model evaluation and comparison (data engineers, data scientists).
- 3 **Prepare for production:** Runtime environment, risk evaluation, QA (data engineers, software engineers, ML architects).
- 4 **Development to production:** Elastic scaling, containerization, CI/CD (data engineers, DevOps).
- 5 **Monitoring and feedback loop:** Logging/alerting, input drift tracking, performance drift, online evaluation (DevOps).
- 6 **Feedback flows back to data scientists and SMEs;** many people with **different skill sets and tools** are involved.

Realistic Machine Learning Model Life Cycle

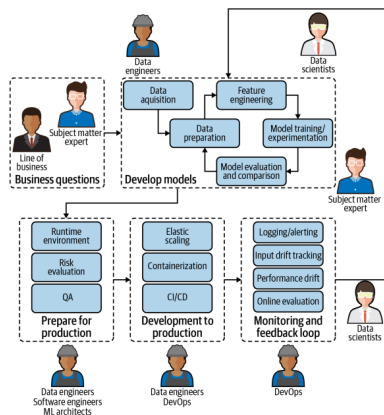


Figure: The realistic picture of an ML model life cycle inside an average organization today

- 1 Business questions:** Line of business and subject matter experts define the problem.
- 2 Develop models:** Data acquisition → data preparation → feature engineering → model training/experimentation → model evaluation and comparison (data engineers, data scientists).
- 3 Prepare for production:** Runtime environment, risk evaluation, QA (data engineers, software engineers, ML architects).
- 4 Development to production:** Elastic scaling, containerization, CI/CD (data engineers, DevOps).
- 5 Monitoring and feedback loop:** Logging/alerting, input drift tracking, performance drift, online evaluation (DevOps).
- 6 Feedback flows back to data scientists and SMEs;** many people with different skill sets and tools are involved.

Realistic Machine Learning Model Life Cycle

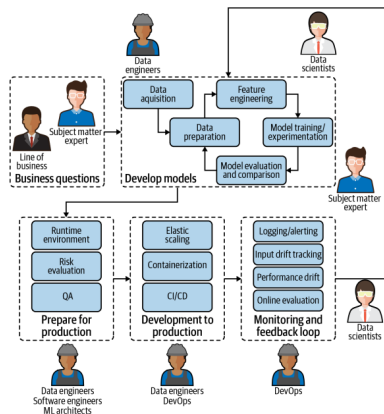


Figure: The realistic picture of an ML model life cycle inside an average organization today

- 1 Business questions:** Line of business and subject matter experts define the problem.
 - 2 Develop models:** Data acquisition → data preparation → feature engineering → model training/experimentation → model evaluation and comparison (data engineers, data scientists).
 - 3 Prepare for production:** Runtime environment, risk evaluation, QA (data engineers, software engineers, ML architects).
 - 4 Development to production:** Elastic scaling, containerization, CI/CD (data engineers, DevOps).
- Monitoring and feedback loop:** Logging/alerting, input drift tracking, performance drift, online evaluation (DevOps).
- Feedback flows back to data scientists and SMEs; many people with different skill sets and tools are involved.

Realistic Machine Learning Model Life Cycle

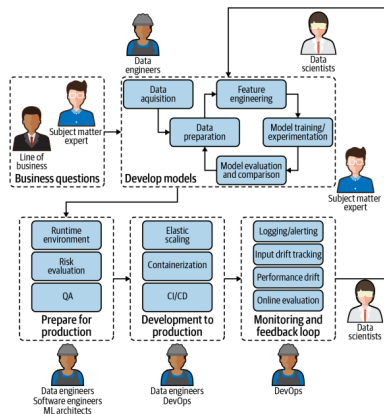


Figure: The realistic picture of an ML model life cycle inside an average organization today

- 1 **Business questions:** Line of business and subject matter experts define the problem.
- 2 **Develop models:** Data acquisition → data preparation → feature engineering → model training/experimentation → model evaluation and comparison (data engineers, data scientists).
- 3 **Prepare for production:** Runtime environment, risk evaluation, QA (data engineers, software engineers, ML architects).
- 4 **Development to production:** Elastic scaling, containerization, CI/CD (data engineers, DevOps).
- 5 **Monitoring and feedback loop:** Logging/alerting, input drift tracking, performance drift, online evaluation (DevOps).

Feedback flows back to data scientists and SMEs; many people with different skill sets and tools are involved.

Realistic Machine Learning Model Life Cycle

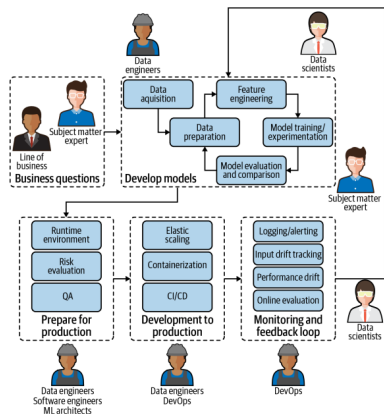


Figure: The realistic picture of an ML model life cycle inside an average organization today

- 1 Business questions:** Line of business and subject matter experts define the problem.
- 2 Develop models:** Data acquisition → data preparation → feature engineering → model training/experimentation → model evaluation and comparison (data engineers, data scientists).
- 3 Prepare for production:** Runtime environment, risk evaluation, QA (data engineers, software engineers, ML architects).
- 4 Development to production:** Elastic scaling, containerization, CI/CD (data engineers, DevOps).
- 5 Monitoring and feedback loop:** Logging/alerting, input drift tracking, performance drift, online evaluation (DevOps).
- 6** Feedback flows back to data scientists and SMEs; many people with **different skill sets and tools** are involved.

MLOps and DevOps

- ❶ MLOps pulls heavily from **DevOps**, which streamlines the practice of software changes and updates.
- ❷ Both center around:
 - ▶ Robust automation and trust between teams.
 - ▶ Collaboration and increased communication between teams.
 - ▶ The end-to-end service life cycle (build, test, release).
 - ▶ Prioritizing continuous delivery and high quality.
- ❸ **Critical difference:** Deploying software code is fundamentally different from deploying ML models.
- ❹ Software code is **relatively static**, whereas data is **always changing**; models must constantly learn and adapt to new inputs.
- ❺ ML models are made up of **both code and data**, which makes MLOps a new and unique discipline.

MLOps and DevOps

- ❶ MLOps pulls heavily from **DevOps**, which streamlines the practice of software changes and updates.
- ❷ Both center around:
 - ▶ Robust automation and trust between teams.
 - ▶ Collaboration and increased communication between teams.
 - ▶ The end-to-end service life cycle (build, test, release).
 - ▶ Prioritizing continuous delivery and high quality.
- ❸ **Critical difference:** Deploying software code is fundamentally different from deploying ML models.
- ❹ Software code is **relatively static**, whereas data is **always changing**; models must constantly learn and adapt to new inputs.
- ❺ ML models are made up of **both code and data**, which makes MLOps a new and unique discipline.

MLOps and DevOps

- ❶ MLOps pulls heavily from **DevOps**, which streamlines the practice of software changes and updates.
- ❷ Both center around:
 - ▶ Robust automation and trust between teams.
 - ▶ Collaboration and increased communication between teams.
 - ▶ The end-to-end service life cycle (build, test, release).
 - ▶ Prioritizing continuous delivery and high quality.
- ❸ **Critical difference:** Deploying software code is fundamentally different from deploying ML models.
- ❹ Software code is **relatively static**, whereas data is **always changing**; models must constantly learn and adapt to new inputs.
- ❺ ML models are made up of **both code and data**, which makes MLOps a new and unique discipline.

MLOps and DevOps

- ❶ MLOps pulls heavily from **DevOps**, which streamlines the practice of software changes and updates.
- ❷ Both center around:
 - ▶ Robust automation and trust between teams.
 - ▶ Collaboration and increased communication between teams.
 - ▶ The end-to-end service life cycle (build, test, release).
 - ▶ Prioritizing continuous delivery and high quality.
- ❸ **Critical difference:** Deploying software code is fundamentally different from deploying ML models.
- ❹ Software code is **relatively static**, whereas data is **always changing**; models must constantly learn and adapt to new inputs.
- ❺ ML models are made up of **both code and data**, which makes MLOps a new and unique discipline.

MLOps and DevOps

- ❶ MLOps pulls heavily from **DevOps**, which streamlines the practice of software changes and updates.
- ❷ Both center around:
 - ▶ Robust automation and trust between teams.
 - ▶ Collaboration and increased communication between teams.
 - ▶ The end-to-end service life cycle (build, test, release).
 - ▶ Prioritizing continuous delivery and high quality.
- ❸ **Critical difference:** Deploying software code is fundamentally different from deploying ML models.
- ❹ Software code is **relatively static**, whereas data is **always changing**; models must constantly learn and adapt to new inputs.
- ❺ ML models are made up of **both code and data**, which makes MLOps a new and unique discipline.

What About DataOps?

- ❶ **DataOps** was introduced in 2014 by IBM.
- ❷ It seeks to provide **business-ready data** quickly, with focus on **data quality** and **metadata management**.
- ❸ **Example:** If data a model relies on changes suddenly, DataOps alerts the business team and notifies the data team to investigate or revert a library upgrade and rebuild the partition.
- ❹ MLOps intersects with DataOps, but goes further with additional key features (Chapter 3).
- ❺ Until recently, teams could manage without centralized processes because they were not deploying ML models at scale.
- ❻ Now, teams seek to formalize a **multi-stage, multi-discipline, multi-phase** process with a framework of MLOps best practices.

What About DataOps?

- ❶ **DataOps** was introduced in 2014 by IBM.
- ❷ It seeks to provide **business-ready data** quickly, with focus on **data quality** and **metadata management**.
- ❸ **Example:** If data a model relies on changes suddenly, DataOps alerts the business team and notifies the data team to investigate or revert a library upgrade and rebuild the partition.
- ❹ MLOps intersects with DataOps, but goes further with additional key features (Chapter 3).
- ❺ Until recently, teams could manage without centralized processes because they were not deploying ML models at scale.
- ❻ Now, teams seek to formalize a **multi-stage, multi-discipline, multi-phase** process with a framework of MLOps best practices.

What About DataOps?

- ❶ **DataOps** was introduced in 2014 by IBM.
- ❷ It seeks to provide **business-ready data** quickly, with focus on **data quality** and **metadata management**.
- ❸ **Example:** If data a model relies on changes suddenly, DataOps alerts the business team and notifies the data team to investigate or revert a library upgrade and rebuild the partition.
- ❹ MLOps intersects with DataOps, but goes further with additional key features (Chapter 3).
- ❺ Until recently, teams could manage without centralized processes because they were not deploying ML models at scale.
- ❻ Now, teams seek to formalize a **multi-stage, multi-discipline, multi-phase** process with a framework of MLOps best practices.

What About DataOps?

- ❶ **DataOps** was introduced in 2014 by IBM.
- ❷ It seeks to provide **business-ready data** quickly, with focus on **data quality** and **metadata management**.
- ❸ **Example:** If data a model relies on changes suddenly, DataOps alerts the business team and notifies the data team to investigate or revert a library upgrade and rebuild the partition.
- ❹ MLOps **intersects** with DataOps, but goes further with additional key features (Chapter 3).
- ❺ Until recently, teams could manage without centralized processes because they were not deploying ML models at scale.
- ❻ Now, teams seek to formalize a **multi-stage, multi-discipline, multi-phase** process with a framework of MLOps best practices.

What About DataOps?

- ❶ **DataOps** was introduced in 2014 by IBM.
- ❷ It seeks to provide **business-ready data** quickly, with focus on **data quality** and **metadata management**.
- ❸ **Example:** If data a model relies on changes suddenly, DataOps alerts the business team and notifies the data team to investigate or revert a library upgrade and rebuild the partition.
- ❹ MLOps **intersects** with DataOps, but goes further with additional key features (Chapter 3).
- ❺ Until recently, teams could manage without centralized processes because they were not deploying ML models at scale.
- ❻ Now, teams seek to formalize a **multi-stage, multi-discipline, multi-phase** process with a framework of MLOps best practices.

What About DataOps?

- ❶ **DataOps** was introduced in 2014 by IBM.
- ❷ It seeks to provide **business-ready data** quickly, with focus on **data quality** and **metadata management**.
- ❸ **Example:** If data a model relies on changes suddenly, DataOps alerts the business team and notifies the data team to investigate or revert a library upgrade and rebuild the partition.
- ❹ MLOps **intersects** with DataOps, but goes further with additional key features (Chapter 3).
- ❺ Until recently, teams could manage without centralized processes because they were not deploying ML models at scale.
- ❻ Now, teams seek to formalize a **multi-stage, multi-discipline, multi-phase** process with a framework of MLOps best practices.

MLOps to Mitigate Risk

- ❶ MLOps is important to any team that has **even one model** in production.
- ❷ Depending on the model, **continuous performance monitoring and adjusting** is essential.
- ❸ By allowing safe and reliable operations, MLOps is key to **mitigating the risks** induced by ML models.
- ❹ MLOps practices come at a **cost**; a proper **cost-benefit evaluation** should be performed for each use case.

MLOps to Mitigate Risk

- ❶ MLOps is important to any team that has **even one model** in production.
- ❷ Depending on the model, **continuous performance monitoring and adjusting** is essential.
- ❸ By allowing safe and reliable operations, MLOps is key to **mitigating the risks** induced by ML models.
- ❹ MLOps practices come at a cost; a proper **cost-benefit evaluation** should be performed for each use case.

MLOps to Mitigate Risk

- ❶ MLOps is important to any team that has **even one model** in production.
- ❷ Depending on the model, **continuous performance monitoring and adjusting** is essential.
- ❸ By allowing safe and reliable operations, MLOps is key to **mitigating the risks** induced by ML models.
- ❹ MLOps practices come at a cost; a proper cost-benefit evaluation should be performed for each use case.

MLOps to Mitigate Risk

- ❶ MLOps is important to any team that has **even one model** in production.
- ❷ Depending on the model, **continuous performance monitoring and adjusting** is essential.
- ❸ By allowing safe and reliable operations, MLOps is key to **mitigating the risks** induced by ML models.
- ❹ MLOps practices come at a **cost**; a proper **cost-benefit evaluation** should be performed for each use case.

Risk Assessment

- ❶ Risks vary widely: a monthly marketing recommendation engine is far lower stakes than a travel site whose **pricing and revenue** depend on an ML model.
- ❷ A risk analysis should cover the risk that:
 - ▶ The model is **unavailable** for a given period of time.
 - ▶ The model returns a **bad prediction** for a given sample.
 - ▶ Model **accuracy or fairness decreases** over time.
 - ▶ The **skills** needed to maintain the model (data science talent) are lost.
- ❸ Risks are usually larger for models **deployed widely** and used outside the organization.
- ❹ Risk assessment should be done at the **beginning** of each project and **reassessed periodically**.

Risk Assessment

- ❶ Risks vary widely: a monthly marketing recommendation engine is far lower stakes than a travel site whose **pricing and revenue** depend on an ML model.
- ❷ A risk analysis should cover the risk that:
 - ▶ The model is **unavailable** for a given period of time.
 - ▶ The model returns a **bad prediction** for a given sample.
 - ▶ Model **accuracy or fairness decreases** over time.
 - ▶ The **skills** needed to maintain the model (data science talent) are lost.
- ❸ Risks are usually larger for models **deployed widely** and used outside the organization.
- ❹ Risk assessment should be done at the **beginning** of each project and **reassessed periodically**.

Risk Assessment

- ❶ Risks vary widely: a monthly marketing recommendation engine is far lower stakes than a travel site whose **pricing and revenue** depend on an ML model.
- ❷ A risk analysis should cover the risk that:
 - ▶ The model is **unavailable** for a given period of time.
 - ▶ The model returns a **bad prediction** for a given sample.
 - ▶ Model **accuracy or fairness decreases** over time.
 - ▶ The **skills** needed to maintain the model (data science talent) are lost.
- ❸ Risks are usually larger for models **deployed widely** and used outside the organization.
- ❹ Risk assessment should be done at the **beginning** of each project and **reassessed periodically**.

Risk Assessment

- ❶ Risks vary widely: a monthly marketing recommendation engine is far lower stakes than a travel site whose **pricing and revenue** depend on an ML model.
- ❷ A risk analysis should cover the risk that:
 - ▶ The model is **unavailable** for a given period of time.
 - ▶ The model returns a **bad prediction** for a given sample.
 - ▶ Model **accuracy or fairness decreases** over time.
 - ▶ The **skills** needed to maintain the model (data science talent) are lost.
- ❸ Risks are usually larger for models **deployed widely** and used outside the organization.
- ❹ Risk assessment should be done at the **beginning** of each project and **reassessed periodically**.

Quantitative Risk Analysis: 5×5 Risk Matrix

5 x 5 risk matrix

Probability	Highly probable	5 Moderate	10 Major	15 Major	20 Severe	25 Severe
	Probable	4 Moderate	8 Moderate	12 Major	16 Major	20 Severe
	Possible	3 Minor	6 Moderate	9 Moderate	12 Major	15 Major
	Unlikely	2 Minor	4 Moderate	6 Moderate	8 Moderate	10 Major
	Rare	1 Minor	2 Minor	3 Minor	5 Moderate	6 Moderate
		Very low	Low	Medium	High	Very high
		Impact				

Figure: A table that helps decision makers with quantitative risk analysis

- 1 Risk is assessed on two metrics:
probability (Rare $\rightarrow$ Highly probable) and **impact** (Very low $\rightarrow$ Very high).
- 2 Each is scored 1–5; their **product** gives the model's **severity** (1–25).
- 3 Severity is graded as **Minor**, **Moderate**, **Major**, or **Severe**.
- 4 Example: Possible (3) $\times$ High (4) = 12 $\rightarrow$ Major.
- 5 Mitigation measures are chosen based on this severity.

Quantitative Risk Analysis: 5×5 Risk Matrix

5 x 5 risk matrix

Probability	Highly probable	5 Moderate	10 Major	15 Major	20 Severe	25 Severe
	Probable	4 Moderate	8 Moderate	12 Major	16 Major	20 Severe
	Possible	3 Minor	6 Moderate	9 Moderate	12 Major	15 Major
	Unlikely	2 Minor	4 Moderate	6 Moderate	8 Moderate	10 Major
	Rare	1 Minor	2 Minor	3 Minor	5 Moderate	6 Moderate
		Very low	Low	Medium	High	Very high
		Impact				

- 1 Risk is assessed on two metrics:
probability (Rare → Highly probable) and **impact** (Very low → Very high).
- 2 Each is scored 1–5; their **product** gives the model's **severity** (1–25).
- 3 Severity is graded as Minor, Moderate, Major, or Severe.
- 4 Example: Possible (3) × High (4) = 12 → Major.
- 5 Mitigation measures are chosen based on this severity.

Figure: A table that helps decision makers with quantitative risk analysis

Quantitative Risk Analysis: 5×5 Risk Matrix

5 x 5 risk matrix

	Highly probable	5 Moderate	10 Major	15 Major	20 Severe	25 Severe
Probability ↑	Probable	4 Moderate	8 Moderate	12 Major	16 Major	20 Severe
	Possible	3 Minor	6 Moderate	9 Moderate	12 Major	15 Major
	Unlikely	2 Minor	4 Moderate	6 Moderate	8 Moderate	10 Major
	Rare	1 Minor	2 Minor	3 Minor	5 Moderate	6 Moderate
		Very low	Low	Medium	High	Very high
		Impact →				

Figure: A table that helps decision makers with quantitative risk analysis

- 1 Risk is assessed on two metrics:
probability (Rare → Highly probable) and **impact** (Very low → Very high).
 - 2 Each is scored 1–5; their **product** gives the model's **severity** (1–25).
 - 3 Severity is graded as **Minor**, **Moderate**, **Major**, or **Severe**.
- ① Example: Possible (3) × High (4) = 12 → Major.
- ② Mitigation measures are chosen based on this severity.

Quantitative Risk Analysis: 5 × 5 Risk Matrix

5 x 5 risk matrix

	Highly probable	5 Moderate	10 Major	15 Major	20 Severe	25 Severe
Probable	4 Moderate	8 Moderate	12 Major	16 Major	20 Severe	
Possible	3 Minor	6 Moderate	9 Moderate	12 Major	15 Major	
Unlikely	2 Minor	4 Moderate	6 Moderate	8 Moderate	10 Major	
Rare	1 Minor	2 Minor	3 Minor	5 Moderate	6 Moderate	
	Very low	Low	Medium	High	Very high	

Figure: A table that helps decision makers with quantitative risk analysis

- 1 Risk is assessed on two metrics:
probability (Rare → Highly probable) and **impact** (Very low → Very high).
 - 2 Each is scored 1–5; their **product** gives the model's **severity** (1–25).
 - 3 Severity is graded as **Minor**, **Moderate**, **Major**, or **Severe**.
 - 4 Example: Possible (3) × High (4) = 12 → Major.
- 2 Mitigation measures are chosen based on this severity.

Quantitative Risk Analysis: 5 × 5 Risk Matrix

5 x 5 risk matrix

	Highly probable	5 Moderate	10 Major	15 Major	20 Severe	25 Severe
Probable	4 Moderate	8 Moderate	12 Major	16 Major	20 Severe	
Possible	3 Minor	6 Moderate	9 Moderate	12 Major	15 Major	
Unlikely	2 Minor	4 Moderate	6 Moderate	8 Moderate	10 Major	
Rare	1 Minor	2 Minor	3 Minor	5 Moderate	6 Moderate	
	Very low	Low	Medium	High	Very high	

- 1 Risk is assessed on two metrics:
probability (Rare → Highly probable) and **impact** (Very low → Very high).
- 2 Each is scored 1–5; their **product** gives the model's **severity** (1–25).
- 3 Severity is graded as **Minor**, **Moderate**, **Major**, or **Severe**.
- 4 Example: Possible (3) × High (4) = 12 → Major.
- 5 **Mitigation measures** are chosen based on this severity.

Figure: A table that helps decision makers with quantitative risk analysis

Risk Mitigation

- ❶ MLOps becomes critical when a centralized team has **more than a handful** of operational models.
- ❷ Without standardization, it is difficult to have a **global view** of the state of all models and take appropriate mitigation measures.
- ❸ Full performance assessment is often only possible in the **production environment**.
- ❹ Models are only as good as their training data; if production data changes, **performance may decrease rapidly**.
- ❺ Model performance is sensitive to the **software environment** (versions of libraries such as scikit-learn, Python, or Linux) it runs in.
- ❻ Deployment is **not the final step**; it is the beginning of monitoring the model and knowing how broadly it is used.

Risk Mitigation

- ❶ MLOps becomes critical when a centralized team has **more than a handful** of operational models.
- ❷ Without standardization, it is difficult to have a **global view** of the state of all models and take appropriate mitigation measures.
- ❸ Full performance assessment is often only possible in the **production environment**.
- ❹ Models are only as good as their training data; if production data changes, **performance may decrease rapidly**.
- ❺ Model performance is sensitive to the **software environment** (versions of libraries such as scikit-learn, Python, or Linux) it runs in.
- ❻ Deployment is **not the final step**; it is the beginning of monitoring the model and knowing how broadly it is used.

Risk Mitigation

- ❶ MLOps becomes critical when a centralized team has **more than a handful** of operational models.
- ❷ Without standardization, it is difficult to have a **global view** of the state of all models and take appropriate mitigation measures.
- ❸ Full performance assessment is often only possible in the **production environment**.
- ❹ Models are only as good as their training data; if production data changes, performance may decrease rapidly.
- ❺ Model performance is sensitive to the **software environment** (versions of libraries such as scikit-learn, Python, or Linux) it runs in.
- ❻ Deployment is **not the final step**; it is the beginning of monitoring the model and knowing how broadly it is used.

Risk Mitigation

- ❶ MLOps becomes critical when a centralized team has **more than a handful** of operational models.
- ❷ Without standardization, it is difficult to have a **global view** of the state of all models and take appropriate mitigation measures.
- ❸ Full performance assessment is often only possible in the **production environment**.
- ❹ Models are only as good as their training data; if production data changes, **performance may decrease rapidly**.
- ❺ Model performance is sensitive to the **software environment** (versions of libraries such as scikit-learn, Python, or Linux) it runs in.
- ❻ Deployment is **not the final step**; it is the beginning of monitoring the model and knowing how broadly it is used.

Risk Mitigation

- ❶ MLOps becomes critical when a centralized team has **more than a handful** of operational models.
- ❷ Without standardization, it is difficult to have a **global view** of the state of all models and take appropriate mitigation measures.
- ❸ Full performance assessment is often only possible in the **production environment**.
- ❹ Models are only as good as their training data; if production data changes, **performance may decrease rapidly**.
- ❺ Model performance is sensitive to the **software environment** (versions of libraries such as scikit-learn, Python, or Linux) it runs in.
- ❻ Deployment is **not the final step**; it is the beginning of monitoring the model and knowing how broadly it is used.

Risk Mitigation

- ❶ MLOps becomes critical when a centralized team has **more than a handful** of operational models.
- ❷ Without standardization, it is difficult to have a **global view** of the state of all models and take appropriate mitigation measures.
- ❸ Full performance assessment is often only possible in the **production environment**.
- ❹ Models are only as good as their training data; if production data changes, **performance may decrease rapidly**.
- ❺ Model performance is sensitive to the **software environment** (versions of libraries such as scikit-learn, Python, or Linux) it runs in.
- ❻ Deployment is **not the final step**; it is the beginning of monitoring the model and knowing how broadly it is used.

MLOps for Responsible AI

① Intentionality:

- ▶ Models are designed and behave in ways aligned with their purpose.
- ▶ Data comes from compliant and unbiased sources, with checks and balances on model bias.
- ▶ Includes **explainability**: results should be explainable by humans.

② Accountability:

- ▶ Centrally controlling, managing, and auditing the enterprise AI effort — **no shadow IT**.
- ▶ Knowing which teams use what data, how, and in which models.
- ▶ Closely tied to **traceability**: where in the pipeline did something go wrong?

MLOps for Responsible AI

❶ Intentionality:

- ▶ Models are designed and behave in ways aligned with their purpose.
- ▶ Data comes from compliant and unbiased sources, with checks and balances on model bias.
- ▶ Includes **explainability**: results should be explainable by humans.

❷ Accountability:

- ▶ Centrally controlling, managing, and auditing the enterprise AI effort — **no shadow IT**.
- ▶ Knowing which teams use what data, how, and in which models.
- ▶ Closely tied to **traceability**: where in the pipeline did something go wrong?

Responsible AI and the Shift of Accountability

- ❶ ML models **lack the transparency** of traditional imperative code.
- ❷ It is harder to understand which features determine a prediction, and to prove **regulatory compliance**.
- ❸ Automation **shifts accountability** from the bottom of the hierarchy to the top.
- ❹ Decisions once made by individuals (product price, loan approval) are now made by a model owned by a **data team manager or executive**.
- ❺ Teams need good MLOps to practice Responsible AI, and Responsible AI **necessitates** MLOps strategies.

Responsible AI and the Shift of Accountability

- ❶ ML models **lack the transparency** of traditional imperative code.
- ❷ It is harder to understand which features determine a prediction, and to prove **regulatory compliance**.
- ❸ Automation **shifts accountability** from the bottom of the hierarchy to the top.
- ❹ Decisions once made by individuals (product price, loan approval) are now made by a model owned by a **data team manager or executive**.
- ❺ Teams need good MLOps to practice Responsible AI, and Responsible AI **necessitates** MLOps strategies.

Responsible AI and the Shift of Accountability

- ❶ ML models **lack the transparency** of traditional imperative code.
- ❷ It is harder to understand which features determine a prediction, and to prove **regulatory compliance**.
- ❸ Automation **shifts accountability** from the bottom of the hierarchy to the top.
- ❹ Decisions once made by individuals (product price, loan approval) are now made by a model owned by a **data team manager or executive**.
- ❺ Teams need good MLOps to practice Responsible AI, and Responsible AI **necessitates** MLOps strategies.

Responsible AI and the Shift of Accountability

- ❶ ML models **lack the transparency** of traditional imperative code.
- ❷ It is harder to understand which features determine a prediction, and to prove **regulatory compliance**.
- ❸ Automation **shifts accountability** from the bottom of the hierarchy to the top.
- ❹ Decisions once made by individuals (product price, loan approval) are now made by a model owned by a **data team manager or executive**.
- ❺ Teams need good MLOps to practice Responsible AI, and Responsible AI necessitates MLOps strategies.

Responsible AI and the Shift of Accountability

- ❶ ML models **lack the transparency** of traditional imperative code.
- ❷ It is harder to understand which features determine a prediction, and to prove **regulatory compliance**.
- ❸ Automation **shifts accountability** from the bottom of the hierarchy to the top.
- ❹ Decisions once made by individuals (product price, loan approval) are now made by a model owned by a **data team manager or executive**.
- ❺ Teams need good MLOps to practice Responsible AI, and Responsible AI **necessitates** MLOps strategies.

MLOps for Scale

- ❶ MLOps is essential to **massively deploying** ML efforts and benefiting from economies of scale.
- ❷ Going from a handful of models to **tens, hundreds, or thousands** requires MLOps discipline.
- ❸ Good MLOps practices help teams, at a minimum, to:
 - ▶ Keep track of **versioning**, especially with experiments in the design phase.
 - ▶ Understand whether **retrained models are better** than previous versions, and promote better models to production.
 - ▶ Ensure at defined periods (daily, monthly, etc.) that model performance is **not degrading** in production.

MLOps for Scale

- ❶ MLOps is essential to **massively deploying** ML efforts and benefiting from economies of scale.
- ❷ Going from a handful of models to **tens, hundreds, or thousands** requires MLOps discipline.
- ❸ Good MLOps practices help teams, at a minimum, to:
 - ▶ Keep track of **versioning**, especially with experiments in the design phase.
 - ▶ Understand whether **retrained models are better** than previous versions, and promote better models to production.
 - ▶ Ensure at defined periods (daily, monthly, etc.) that model performance is **not degrading** in production.

MLOps for Scale

- ❶ MLOps is essential to **massively deploying** ML efforts and benefiting from economies of scale.
- ❷ Going from a handful of models to **tens, hundreds, or thousands** requires MLOps discipline.
- ❸ Good MLOps practices help teams, at a minimum, to:
 - ▶ Keep track of **versioning**, especially with experiments in the design phase.
 - ▶ Understand whether **retrained models are better** than previous versions, and promote better models to production.
 - ▶ Ensure at defined periods (daily, monthly, etc.) that model performance is **not degrading** in production.

Closing Thoughts

- ❶ MLOps practices are **not optional**; they are essential for scaling data science without putting the business at risk.
- ❷ Without MLOps, teams face issues with **model quality and continuity**.
- ❸ Worse, they may deploy models with real **negative business impact** (e.g., biased predictions).
- ❹ Upper management should understand which models are in production and their business effect.
- ❺ MLOps provides **transparency and accountability** down to the whole data pipeline (raw data to final output).

Closing Thoughts

- ❶ MLOps practices are **not optional**; they are essential for scaling data science without putting the business at risk.
- ❷ Without MLOps, teams face issues with **model quality and continuity**.
- ❸ Worse, they may deploy models with real **negative business impact** (e.g., biased predictions).
- ❹ Upper management should understand which models are in production and their business effect.
- ❺ MLOps provides **transparency and accountability** down to the whole data pipeline (raw data to final output).

Closing Thoughts

- ❶ MLOps practices are **not optional**; they are essential for scaling data science without putting the business at risk.
- ❷ Without MLOps, teams face issues with **model quality and continuity**.
- ❸ Worse, they may deploy models with real **negative business impact** (e.g., biased predictions).
- ❹ Upper management should understand which models are in production and their business effect.
- ❺ MLOps provides **transparency and accountability** down to the whole data pipeline (raw data to final output).

Closing Thoughts

- ❶ MLOps practices are **not optional**; they are essential for scaling data science without putting the business at risk.
- ❷ Without MLOps, teams face issues with **model quality and continuity**.
- ❸ Worse, they may deploy models with real **negative business impact** (e.g., biased predictions).
- ❹ Upper management should understand which models are in production and their business effect.
- ❺ MLOps provides **transparency and accountability** down to the whole data pipeline (raw data to final output).

Closing Thoughts

- ❶ MLOps practices are **not optional**; they are essential for scaling data science without putting the business at risk.
- ❷ Without MLOps, teams face issues with **model quality and continuity**.
- ❸ Worse, they may deploy models with real **negative business impact** (e.g., biased predictions).
- ❹ Upper management should understand which models are in production and their business effect.
- ❺ MLOps provides **transparency and accountability** down to the whole data pipeline (raw data to final output).

People of MLOps

- ❶ ML models are built primarily by data scientists, but MLOps involves **many roles** across the organization (see the realistic life cycle figure).
- ❷ Each role has a different **skill set**, different **tools**, and a different **stake** in the model.
- ❸ Key roles in MLOps:
 - ▶ Subject Matter Experts (SMEs)
 - ▶ Data Scientists and Data Engineers
 - ▶ Software Engineers and DevOps
 - ▶ Model Risk Manager / Auditor
 - ▶ Machine Learning Architect

People of MLOps

- ➊ ML models are built primarily by data scientists, but MLOps involves **many roles** across the organization (see the realistic life cycle figure).
- ➋ Each role has a different **skill set**, different **tools**, and a different **stake** in the model.
- ➌ Key roles in MLOps:
 - ▶ Subject Matter Experts (SMEs)
 - ▶ Data Scientists and Data Engineers
 - ▶ Software Engineers and DevOps
 - ▶ Model Risk Manager / Auditor
 - ▶ Machine Learning Architect

People of MLOps

- ❶ ML models are built primarily by data scientists, but MLOps involves **many roles** across the organization (see the realistic life cycle figure).
- ❷ Each role has a different **skill set**, different **tools**, and a different **stake** in the model.
- ❸ Key roles in MLOps:
 - ▶ Subject Matter Experts (SMEs)
 - ▶ Data Scientists and Data Engineers
 - ▶ Software Engineers and DevOps
 - ▶ Model Risk Manager / Auditor
 - ▶ Machine Learning Architect

Roles in MLOps and Their Responsibilities

Role	Role in the ML Model Life Cycle
Subject Matter Experts	Provide business questions, goals, and KPIs; evaluate whether model performance meets business needs.
Data Scientists	Build models that address the business question; deliver operationalizable models; assess model quality with SMEs.
Data Engineers	Optimize the retrieval and use of data to power ML models.
Software Engineers	Integrate ML models into the company's applications and systems.
DevOps	Build and test operational systems for security, performance, and availability; manage CI/CD pipelines.
Model Risk Manager/Auditor	Minimize overall risk from models in production; ensure compliance before deployment.
ML Architect	Ensure a scalable and flexible environment for ML pipelines from design to monitoring.

Subject Matter Experts (SMEs)

- ❶ SMEs (line of business) **start the life cycle** by framing the business question.
- ❷ They define **goals and KPIs** against which the model will ultimately be judged.
- ❸ They bring **domain knowledge** needed during data exploration and feature engineering.
- ❹ They evaluate whether model results are **useful for the business** and not just statistically good.
- ❺ They take part in the **feedback loop**: monitoring results are relayed back to SMEs to confirm the model still meets the original goal.

Subject Matter Experts (SMEs)

- ➊ SMEs (line of business) **start the life cycle** by framing the business question.
- ➋ They define **goals and KPIs** against which the model will ultimately be judged.
- ➌ They bring **domain knowledge** needed during data exploration and feature engineering.
- ➍ They evaluate whether model results are **useful for the business** and not just statistically good.
- ➎ They take part in the **feedback loop**: monitoring results are relayed back to SMEs to confirm the model still meets the original goal.

Subject Matter Experts (SMEs)

- ❶ SMEs (line of business) **start the life cycle** by framing the business question.
- ❷ They define **goals and KPIs** against which the model will ultimately be judged.
- ❸ They bring **domain knowledge** needed during data exploration and feature engineering.
- ❹ They evaluate whether model results are **useful for the business** and not just statistically good.
- ❺ They take part in the **feedback loop**: monitoring results are relayed back to SMEs to confirm the model still meets the original goal.

Subject Matter Experts (SMEs)

- ❶ SMEs (line of business) **start the life cycle** by framing the business question.
- ❷ They define **goals and KPIs** against which the model will ultimately be judged.
- ❸ They bring **domain knowledge** needed during data exploration and feature engineering.
- ❹ They evaluate whether model results are **useful for the business** and not just statistically good.
- ❺ They take part in the **feedback loop**: monitoring results are relayed back to SMEs to confirm the model still meets the original goal.

Subject Matter Experts (SMEs)

- ❶ SMEs (line of business) **start the life cycle** by framing the business question.
- ❷ They define **goals and KPIs** against which the model will ultimately be judged.
- ❸ They bring **domain knowledge** needed during data exploration and feature engineering.
- ❹ They evaluate whether model results are **useful for the business** and not just statistically good.
- ❺ They take part in the **feedback loop**: monitoring results are relayed back to SMEs to confirm the model still meets the original goal.

Data Scientists and Data Engineers

❶ Data Scientists:

- ▶ Explore data, engineer features, train, evaluate, and compare models.
- ▶ Must deliver models that are **operationalizable**, reproducible, and explainable.
- ▶ Monitor model quality and drive retraining in the feedback loop.

❷ Data Engineers:

- ▶ Handle data acquisition and preparation pipelines.
- ▶ Ensure data is accessible, reliable, and available at the scale and speed required.
- ▶ Support preparing for production and deployment.

Data Scientists and Data Engineers

❶ Data Scientists:

- ▶ Explore data, engineer features, train, evaluate, and compare models.
- ▶ Must deliver models that are **operationalizable**, reproducible, and explainable.
- ▶ Monitor model quality and drive retraining in the feedback loop.

❷ Data Engineers:

- ▶ Handle data acquisition and preparation pipelines.
- ▶ Ensure data is accessible, reliable, and available at the scale and speed required.
- ▶ Support preparing for production and deployment.

Software Engineers and DevOps

❶ Software Engineers:

- ▶ Integrate ML models into applications and business systems.
- ▶ Ensure models work seamlessly with non-ML components.

❷ DevOps:

- ▶ Build and operate production systems: runtime, containerization, elastic scaling.
- ▶ Own CI/CD pipelines for automated testing and deployment.
- ▶ Monitor resource usage, latency, logging, and alerting in production.

Software Engineers and DevOps

❶ Software Engineers:

- ▶ Integrate ML models into applications and business systems.
- ▶ Ensure models work seamlessly with non-ML components.

❷ DevOps:

- ▶ Build and operate production systems: runtime, containerization, elastic scaling.
- ▶ Own **CI/CD pipelines** for automated testing and deployment.
- ▶ Monitor resource usage, latency, logging, and alerting in production.

Model Risk Manager/Auditor and ML Architect

❶ Model Risk Manager / Auditor:

- ▶ Minimizes the overall risk to the company from models in production.
- ▶ Ensures compliance with internal and external (regulatory) requirements.
- ▶ Reviews documentation, validation, and sign-offs before deployment.

❷ Machine Learning Architect:

- ▶ Designs a scalable and flexible environment for ML pipelines.
- ▶ Selects and introduces technologies that improve performance in production.
- ▶ Bridges data science, data engineering, and IT infrastructure.

Model Risk Manager/Auditor and ML Architect

❶ Model Risk Manager / Auditor:

- ▶ Minimizes the overall risk to the company from models in production.
- ▶ Ensures compliance with internal and external (regulatory) requirements.
- ▶ Reviews documentation, validation, and sign-offs before deployment.

❷ Machine Learning Architect:

- ▶ Designs a scalable and flexible environment for ML pipelines.
- ▶ Selects and introduces technologies that improve performance in production.
- ▶ Bridges data science, data engineering, and IT infrastructure.

Key MLOps Features

- ❶ MLOps affects many roles and many parts of the ML life cycle.
- ❷ The five key components of MLOps are:
 - ❶ Development
 - ❷ Deployment
 - ❸ Monitoring
 - ❹ Iteration
 - ❺ Governance
- ❸ These are introduced here at a high level and form the foundation for Chapters 4–8.

Key MLOps Features

- ❶ MLOps affects many roles and many parts of the ML life cycle.
- ❷ The **five key components** of MLOps are:
 - ❶ Development
 - ❷ Deployment
 - ❸ Monitoring
 - ❹ Iteration
 - ❺ Governance
- ❸ These are introduced here at a high level and form the foundation for Chapters 4–8.

Key MLOps Features

- ❶ MLOps affects many roles and many parts of the ML life cycle.
- ❷ The **five key components** of MLOps are:
 - ❶ Development
 - ❷ Deployment
 - ❸ Monitoring
 - ❹ Iteration
 - ❺ Governance
- ❸ These are introduced here at a high level and form the foundation for Chapters 4–8.

A Primer on Machine Learning

Machine Learning

The science of computer algorithms that automatically **learn and improve from experience** rather than being explicitly programmed.

- ❶ Algorithm selection has a **direct impact** on MLOps processes.
- ❷ Algorithms analyze sample data (**training data**) to build a model that makes predictions.
- ❸ **Examples:** Identifying an electricity meter type from a photo; recommending insurance products based on behavior of similar customers.
- ❹ On **unseen data**, the model predicts assuming it is related to the previous data.
- ❺ Models range from simple **decision trees** and **logistic regression** to complex **deep learning** models.

A Primer on Machine Learning

Machine Learning

The science of computer algorithms that automatically **learn and improve from experience** rather than being explicitly programmed.

- ❶ Algorithm selection has a **direct impact** on MLOps processes.
- ❷ Algorithms analyze sample data (**training data**) to build a model that makes predictions.
- ❸ **Examples:** Identifying an electricity meter type from a photo; recommending insurance products based on behavior of similar customers.
- ❹ On **unseen data**, the model predicts assuming it is related to the previous data.
- ❺ Models range from simple **decision trees** and **logistic regression** to complex **deep learning** models.

A Primer on Machine Learning

Machine Learning

The science of computer algorithms that automatically **learn and improve from experience** rather than being explicitly programmed.

- ❶ Algorithm selection has a **direct impact** on MLOps processes.
- ❷ Algorithms analyze sample data (**training data**) to build a model that makes predictions.
- ❸ **Examples:** Identifying an electricity meter type from a photo; recommending insurance products based on behavior of similar customers.
- ❹ On **unseen data**, the model predicts assuming it is related to the previous data.
- ❺ Models range from simple **decision trees** and **logistic regression** to complex **deep learning** models.

A Primer on Machine Learning

Machine Learning

The science of computer algorithms that automatically **learn and improve from experience** rather than being explicitly programmed.

- ❶ Algorithm selection has a **direct impact** on MLOps processes.
- ❷ Algorithms analyze sample data (**training data**) to build a model that makes predictions.
- ❸ **Examples:** Identifying an electricity meter type from a photo; recommending insurance products based on behavior of similar customers.
- ❹ On **unseen data**, the model predicts assuming it is related to the previous data.
- ❺ Models range from simple **decision trees** and **logistic regression** to complex **deep learning** models.

A Primer on Machine Learning

Machine Learning

The science of computer algorithms that automatically **learn and improve from experience** rather than being explicitly programmed.

- ➊ Algorithm selection has a **direct impact** on MLOps processes.
- ➋ Algorithms analyze sample data (**training data**) to build a model that makes predictions.
- ➌ **Examples:** Identifying an electricity meter type from a photo; recommending insurance products based on behavior of similar customers.
- ➍ On **unseen data**, the model predicts assuming it is related to the previous data.
- ➎ Models range from simple **decision trees** and **logistic regression** to complex **deep learning** models.

Model Development: Establishing Business Objectives

- ❶ Development starts with a **business objective**, e.g., reducing fraudulent transactions to $< 0.1\%$ or identifying faces in social media photos.
- ❷ Objectives come with performance targets, infrastructure requirements, and cost constraints, captured as **KPIs**.
- ❸ KPIs enable the business performance of models in production to be **monitored**.
- ❹ ML projects are part of a larger project that impacts technologies, processes, and people, so objectives include **change management**.
- ❺ The required degree of **transparency** strongly influences the choice of algorithm and the need to explain predictions.

Model Development: Establishing Business Objectives

- ➊ Development starts with a **business objective**, e.g., reducing fraudulent transactions to $< 0.1\%$ or identifying faces in social media photos.
- ➋ Objectives come with performance targets, infrastructure requirements, and cost constraints, captured as **KPIs**.
- ➌ KPIs enable the business performance of models in production to be **monitored**.
- ➍ ML projects are part of a larger project that impacts technologies, processes, and people, so objectives include **change management**.
- ➎ The required degree of **transparency** strongly influences the choice of algorithm and the need to explain predictions.

Model Development: Establishing Business Objectives

- ➊ Development starts with a **business objective**, e.g., reducing fraudulent transactions to $< 0.1\%$ or identifying faces in social media photos.
- ➋ Objectives come with performance targets, infrastructure requirements, and cost constraints, captured as **KPIs**.
- ➌ KPIs enable the business performance of models in production to be **monitored**.
- ➍ ML projects are part of a larger project that impacts technologies, processes, and people, so objectives include **change management**.
- ➎ The required degree of **transparency** strongly influences the choice of algorithm and the need to explain predictions.

Model Development: Establishing Business Objectives

- ➊ Development starts with a **business objective**, e.g., reducing fraudulent transactions to $< 0.1\%$ or identifying faces in social media photos.
- ➋ Objectives come with performance targets, infrastructure requirements, and cost constraints, captured as **KPIs**.
- ➌ KPIs enable the business performance of models in production to be **monitored**.
- ➍ ML projects are part of a larger project that impacts technologies, processes, and people, so objectives include **change management**.
- ➎ The required degree of **transparency** strongly influences the choice of algorithm and the need to explain predictions.

Model Development: Establishing Business Objectives

- ➊ Development starts with a **business objective**, e.g., reducing fraudulent transactions to $< 0.1\%$ or identifying faces in social media photos.
- ➋ Objectives come with performance targets, infrastructure requirements, and cost constraints, captured as **KPIs**.
- ➌ KPIs enable the business performance of models in production to be **monitored**.
- ➍ ML projects are part of a larger project that impacts technologies, processes, and people, so objectives include **change management**.
- ➎ The required degree of **transparency** strongly influences the choice of algorithm and the need to explain predictions.

Data Sources and Exploratory Data Analysis

- ❶ SMEs and data scientists begin by searching for suitable input data — often the **most arduous** part of the journey.

- ❷ **Key questions for finding data:**

- ▶ What relevant datasets are available? Are they accurate and reliable?
- ▶ How can stakeholders access the data? What **features** can be made by combining sources?
- ▶ Will the data be available in real time?
- ▶ Is labeling with “ground truth” needed, or does unsupervised learning make sense? At what cost?

- ❸ **More questions:**

- ▶ What platform should be used? How will data be updated after deployment?
- ▶ Will the model's use reduce the **representativeness** of the data?
- ▶ How will the KPIs be measured?

Data Sources and Exploratory Data Analysis

- ❶ SMEs and data scientists begin by searching for suitable input data — often the **most arduous** part of the journey.
- ❷ **Key questions for finding data:**
 - ▶ What relevant datasets are available? Are they accurate and reliable?
 - ▶ How can stakeholders access the data? What **features** can be made by combining sources?
 - ▶ Will the data be available in real time?
 - ▶ Is labeling with “ground truth” needed, or does unsupervised learning make sense? At what cost?
- ❸ **More questions:**
 - ▶ What platform should be used? How will data be updated after deployment?
 - ▶ Will the model’s use reduce the **representativeness** of the data?
 - ▶ How will the KPIs be measured?

Data Sources and Exploratory Data Analysis

- ❶ SMEs and data scientists begin by searching for suitable input data — often the **most arduous** part of the journey.
- ❷ **Key questions for finding data:**
 - ▶ What relevant datasets are available? Are they accurate and reliable?
 - ▶ How can stakeholders access the data? What **features** can be made by combining sources?
 - ▶ Will the data be available in real time?
 - ▶ Is labeling with “ground truth” needed, or does unsupervised learning make sense? At what cost?
- ❸ **More questions:**
 - ▶ What platform should be used? How will data be updated after deployment?
 - ▶ Will the model’s use reduce the **representativeness** of the data?
 - ▶ How will the KPIs be measured?

Data Governance Questions and EDA

❶ Data governance constraints raise more questions:

- ▶ Can the datasets be used for this purpose? What are the terms of use?
- ▶ Is there **PII** that must be redacted or anonymized?
- ▶ Are there features (e.g., gender) that legally cannot be used?
- ▶ Are minority populations represented well enough for equivalent performance?

❷ Exploratory Data Analysis (EDA) helps to:

- ▶ Build hypotheses about the data.
- ▶ Identify data cleaning requirements.
- ▶ Inform the selection of potentially significant features.

❸ EDA can be done **visually** for intuition or **statistically** for rigor.

Data Governance Questions and EDA

❶ **Data governance constraints** raise more questions:

- ▶ Can the datasets be used for this purpose? What are the terms of use?
- ▶ Is there **PII** that must be redacted or anonymized?
- ▶ Are there features (e.g., gender) that legally cannot be used?
- ▶ Are minority populations represented well enough for equivalent performance?

❷ **Exploratory Data Analysis (EDA)** helps to:

- ▶ Build hypotheses about the data.
- ▶ Identify data cleaning requirements.
- ▶ Inform the selection of potentially significant features.

❸ EDA can be done **visually** for intuition or **statistically** for rigor.

Data Governance Questions and EDA

❶ **Data governance constraints** raise more questions:

- ▶ Can the datasets be used for this purpose? What are the terms of use?
- ▶ Is there **PII** that must be redacted or anonymized?
- ▶ Are there features (e.g., gender) that legally cannot be used?
- ▶ Are minority populations represented well enough for equivalent performance?

❷ **Exploratory Data Analysis (EDA)** helps to:

- ▶ Build hypotheses about the data.
- ▶ Identify data cleaning requirements.
- ▶ Inform the selection of potentially significant features.

❸ EDA can be done **visually** for intuition or **statistically** for rigor.

Feature Engineering and Selection; Training and Evaluation

- ❶ **Feature engineering** transforms raw data into “features” that better represent the underlying problem.
- ❷ Features are **arrays of numbers of fixed size** — the only object ML algorithms understand.
- ❸ It includes **data cleansing**, often the largest part of an ML project in time spent.
- ❹ **Training** is iterative: test algorithms, generate features, adapt selections, tune hyperparameters; it is the most **compute-intensive** step.
- ❺ An **experiment tracking tool** records data, features, parameters, and metrics so experiments can be compared side-by-side.
- ❻ The best solution needs **quantitative** (accuracy, error) and **qualitative** (explainability, ease of deployment) criteria.

Feature Engineering and Selection; Training and Evaluation

- ❶ **Feature engineering** transforms raw data into “features” that better represent the underlying problem.
- ❷ Features are **arrays of numbers of fixed size** — the only object ML algorithms understand.
- ❸ It includes **data cleansing**, often the largest part of an ML project in time spent.
- ❹ **Training** is iterative: test algorithms, generate features, adapt selections, tune hyperparameters; it is the most **compute-intensive** step.
- ❺ An **experiment tracking tool** records data, features, parameters, and metrics so experiments can be compared side-by-side.
- ❻ The best solution needs **quantitative** (accuracy, error) and **qualitative** (explainability, ease of deployment) criteria.

Feature Engineering and Selection; Training and Evaluation

- ❶ **Feature engineering** transforms raw data into “features” that better represent the underlying problem.
- ❷ Features are **arrays of numbers of fixed size** — the only object ML algorithms understand.
- ❸ It includes **data cleansing**, often the largest part of an ML project in time spent.
- ❹ **Training** is iterative: test algorithms, generate features, adapt selections, tune hyperparameters; it is the most **compute-intensive** step.
- ❺ An **experiment tracking tool** records data, features, parameters, and metrics so experiments can be compared side-by-side.
- ❻ The best solution needs **quantitative** (accuracy, error) and **qualitative** (explainability, ease of deployment) criteria.

Feature Engineering and Selection; Training and Evaluation

- ❶ **Feature engineering** transforms raw data into “features” that better represent the underlying problem.
- ❷ Features are **arrays of numbers of fixed size** — the only object ML algorithms understand.
- ❸ It includes **data cleansing**, often the largest part of an ML project in time spent.
- ❹ **Training** is iterative: test algorithms, generate features, adapt selections, tune hyperparameters; it is the most **compute-intensive** step.
- ❺ An **experiment tracking tool** records data, features, parameters, and metrics so experiments can be compared side-by-side.
- ❻ The best solution needs **quantitative** (accuracy, error) and **qualitative** (explainability, ease of deployment) criteria.

Feature Engineering and Selection; Training and Evaluation

- ❶ **Feature engineering** transforms raw data into “features” that better represent the underlying problem.
- ❷ Features are **arrays of numbers of fixed size** — the only object ML algorithms understand.
- ❸ It includes **data cleansing**, often the largest part of an ML project in time spent.
- ❹ **Training** is iterative: test algorithms, generate features, adapt selections, tune hyperparameters; it is the most **compute-intensive** step.
- ❺ An **experiment tracking tool** records data, features, parameters, and metrics so experiments can be compared side-by-side.
- ❻ The best solution needs **quantitative** (accuracy, error) and **qualitative** (explainability, ease of deployment) criteria.

Feature Engineering and Selection; Training and Evaluation

- ❶ **Feature engineering** transforms raw data into “features” that better represent the underlying problem.
- ❷ Features are **arrays of numbers of fixed size** — the only object ML algorithms understand.
- ❸ It includes **data cleansing**, often the largest part of an ML project in time spent.
- ❹ **Training** is iterative: test algorithms, generate features, adapt selections, tune hyperparameters; it is the most **compute-intensive** step.
- ❺ An **experiment tracking tool** records data, features, parameters, and metrics so experiments can be compared side-by-side.
- ❻ The best solution needs **quantitative** (accuracy, error) and **qualitative** (explainability, ease of deployment) criteria.

Reproducibility

- ❶ Many experiments are short-lived, but **significant versions** of a model must be saved.
- ❷ Reproducibility aims to save enough about the environment so the model can be reproduced with the same results from scratch.
- ❸ Without it, data scientists cannot confidently iterate on models.
- ❹ Nor can they hand the model to DevOps to check that the lab result is faithfully reproduced in production.
- ❺ True reproducibility requires version control of all assets and parameters, including training/evaluation data and the software environment.

Reproducibility

- ❶ Many experiments are short-lived, but **significant versions** of a model must be saved.
- ❷ **Reproducibility** aims to save enough about the environment so the model can be **reproduced with the same results from scratch**.
- ❸ Without it, data scientists cannot confidently iterate on models.
- ❹ Nor can they hand the model to **DevOps** to check that the lab result is faithfully reproduced in production.
- ❺ True reproducibility requires **version control of all assets and parameters**, including training/evaluation data and the software environment.

Reproducibility

- ❶ Many experiments are short-lived, but **significant versions** of a model must be saved.
- ❷ **Reproducibility** aims to save enough about the environment so the model can be **reproduced with the same results from scratch**.
- ❸ Without it, data scientists cannot confidently iterate on models.
- ❹ Nor can they hand the model to **DevOps** to check that the lab result is faithfully reproduced in production.
- ❺ True reproducibility requires **version control of all assets and parameters**, including training/evaluation data and the software environment.

Reproducibility

- ❶ Many experiments are short-lived, but **significant versions** of a model must be saved.
- ❷ **Reproducibility** aims to save enough about the environment so the model can be **reproduced with the same results from scratch**.
- ❸ Without it, data scientists cannot confidently iterate on models.
- ❹ Nor can they hand the model to **DevOps** to check that the lab result is faithfully reproduced in production.
- ❺ True reproducibility requires version control of all assets and parameters, including training/evaluation data and the software environment.

Reproducibility

- ❶ Many experiments are short-lived, but **significant versions** of a model must be saved.
- ❷ **Reproducibility** aims to save enough about the environment so the model can be **reproduced with the same results from scratch**.
- ❸ Without it, data scientists cannot confidently iterate on models.
- ❹ Nor can they hand the model to **DevOps** to check that the lab result is faithfully reproduced in production.
- ❺ True reproducibility requires **version control of all assets and parameters**, including training/evaluation data and the software environment.

Responsible AI in Model Development

- ❶ DevOps must understand how to **verify** the model: what it does, how to test it, expected results.
- ❷ Regulated industries must document how the model was built and tuned; critical models may be **independently rebuilt**.
- ❸ **Documentation** is the standard solution; automated model documentation helps, but some must be written by hand.
- ❹ Standard performance measures do not explain **how** predictions are made.
- ❺ The “how” is needed to sanity-check the model, improve features, and ensure **fairness** (sex, age, race) — the field of **explainability**.

Responsible AI in Model Development

- ❶ DevOps must understand how to **verify** the model: what it does, how to test it, expected results.
- ❷ Regulated industries must document how the model was built and tuned; critical models may be **independently rebuilt**.
- ❸ **Documentation** is the standard solution; automated model documentation helps, but some must be written by hand.
- ❹ Standard performance measures do not explain **how** predictions are made.
- ❺ The “how” is needed to sanity-check the model, improve features, and ensure **fairness** (sex, age, race) — the field of **explainability**.

Responsible AI in Model Development

- ❶ DevOps must understand how to **verify** the model: what it does, how to test it, expected results.
- ❷ Regulated industries must document how the model was built and tuned; critical models may be **independently rebuilt**.
- ❸ **Documentation** is the standard solution; automated model documentation helps, but some must be written by hand.
- ❹ Standard performance measures do not explain **how** predictions are made.
- ❺ The “how” is needed to sanity-check the model, improve features, and ensure **fairness** (sex, age, race) — the field of **explainability**.

Responsible AI in Model Development

- ① DevOps must understand how to **verify** the model: what it does, how to test it, expected results.
- ② Regulated industries must document how the model was built and tuned; critical models may be **independently rebuilt**.
- ③ **Documentation** is the standard solution; automated model documentation helps, but some must be written by hand.
- ④ Standard performance measures do not explain **how** predictions are made.
- ⑤ The “how” is needed to sanity-check the model, improve features, and ensure **fairness** (sex, age, race) — the field of **explainability**.

Responsible AI in Model Development

- ① DevOps must understand how to **verify** the model: what it does, how to test it, expected results.
- ② Regulated industries must document how the model was built and tuned; critical models may be **independently rebuilt**.
- ③ **Documentation** is the standard solution; automated model documentation helps, but some must be written by hand.
- ④ Standard performance measures do not explain **how** predictions are made.
- ⑤ The “how” is needed to sanity-check the model, improve features, and ensure **fairness** (sex, age, race) — the field of **explainability**.

Explainability Techniques

- ❶ **Partial dependence plots:** Look at the marginal impact of features on the predicted outcome.
- ❷ **Subpopulation analyses:** Look at how the model treats specific subpopulations; the basis of many fairness analyses.
- ❸ **Individual model predictions (e.g., Shapley values):** Explain how each feature value contributes to a specific prediction.
- ❹ **What-if analysis:** Helps the user understand the sensitivity of the prediction to its inputs.
- ❺ **MLOps work done up front in development makes models easier to manage later, especially in production.**

Explainability Techniques

- ❶ **Partial dependence plots:** Look at the marginal impact of features on the predicted outcome.
- ❷ **Subpopulation analyses:** Look at how the model treats specific subpopulations; the basis of many fairness analyses.
- ❸ **Individual model predictions (e.g., Shapley values):** Explain how each feature value contributes to a specific prediction.
- ❹ **What-if analysis:** Helps the user understand the sensitivity of the prediction to its inputs.
- ❺ **MLOps work done up front in development makes models easier to manage later, especially in production.**

Explainability Techniques

- ❶ **Partial dependence plots:** Look at the marginal impact of features on the predicted outcome.
- ❷ **Subpopulation analyses:** Look at how the model treats specific subpopulations; the basis of many fairness analyses.
- ❸ **Individual model predictions (e.g., Shapley values):** Explain how each feature value contributes to a specific prediction.
- ❹ **What-if analysis:** Helps the user understand the sensitivity of the prediction to its inputs.
- ❺ **MLOps work done up front in development makes models easier to manage later, especially in production.**

Explainability Techniques

- ❶ **Partial dependence plots:** Look at the marginal impact of features on the predicted outcome.
- ❷ **Subpopulation analyses:** Look at how the model treats specific subpopulations; the basis of many fairness analyses.
- ❸ **Individual model predictions (e.g., Shapley values):** Explain how each feature value contributes to a specific prediction.
- ❹ **What-if analysis:** Helps the user understand the sensitivity of the prediction to its inputs.
- ❺ MLOps work done up front in development makes models easier to manage later, especially in production.

Explainability Techniques

- ❶ **Partial dependence plots:** Look at the marginal impact of features on the predicted outcome.
- ❷ **Subpopulation analyses:** Look at how the model treats specific subpopulations; the basis of many fairness analyses.
- ❸ **Individual model predictions (e.g., Shapley values):** Explain how each feature value contributes to a specific prediction.
- ❹ **What-if analysis:** Helps the user understand the sensitivity of the prediction to its inputs.
- ❺ MLOps work done up front in development makes models **easier to manage** later, especially in production.

Productionalization and Deployment

- ❶ Presents an **entirely different** set of technical challenges than model development.
- ❷ It is the domain of the **software engineer** and the **DevOps** team.
- ❸ Information exchange between data scientists and these teams must not be underestimated.
- ❹ Without effective collaboration, **delays or failures to deploy** are inevitable.

Productionalization and Deployment

- ❶ Presents an **entirely different** set of technical challenges than model development.
- ❷ It is the domain of the **software engineer** and the **DevOps team**.
- ❸ Information exchange between data scientists and these teams must not be underestimated.
- ❹ Without effective collaboration, **delays or failures to deploy** are inevitable.

Productionalization and Deployment

- ❶ Presents an **entirely different** set of technical challenges than model development.
- ❷ It is the domain of the **software engineer** and the **DevOps team**.
- ❸ Information exchange between data scientists and these teams must not be underestimated.
- ❹ Without effective collaboration, delays or failures to deploy are inevitable.

Productionalization and Deployment

- ❶ Presents an **entirely different** set of technical challenges than model development.
- ❷ It is the domain of the **software engineer** and the **DevOps team**.
- ❸ Information exchange between data scientists and these teams must not be underestimated.
- ❹ Without effective collaboration, **delays or failures to deploy** are inevitable.

Model Deployment Types and Contents

- ❶ **Model-as-a-service (live-scoring model):** Deployed in a framework providing a **REST API endpoint** that responds to requests in real time.
- ❷ **Embedded model:** Packaged into an application which is then published, e.g., an application providing **batch-scoring**.
- ❸ A deployed model comprises **code** (Python, R, or Java) and **data artifacts**, with version dependencies on runtimes and packages.
- ❹ Different versions in production **may cause predictions to differ**.
- ❺ **Portable formats** (PMML, PFA, ONNX, POJO) reduce dependencies, but support limited algorithms and may behave subtly differently.

Model Deployment Types and Contents

- ❶ **Model-as-a-service (live-scoring model):** Deployed in a framework providing a **REST API endpoint** that responds to requests in real time.
- ❷ **Embedded model:** Packaged into an application which is then published, e.g., an application providing **batch-scoring**.
- ❸ A deployed model comprises **code** (Python, R, or Java) and **data artifacts**, with version dependencies on runtimes and packages.
- ❹ Different versions in production **may cause predictions to differ**.
- ❺ **Portable formats** (PMML, PFA, ONNX, POJO) reduce dependencies, but support limited algorithms and may behave subtly differently.

Model Deployment Types and Contents

- ❶ **Model-as-a-service (live-scoring model):** Deployed in a framework providing a **REST API endpoint** that responds to requests in real time.
- ❷ **Embedded model:** Packaged into an application which is then published, e.g., an application providing **batch-scoring**.
- ❸ A deployed model comprises **code** (Python, R, or Java) and **data artifacts**, with version dependencies on runtimes and packages.
- ❹ Different versions in production may cause predictions to differ.
- ❺ Portable formats (PMML, PFA, ONNX, POJO) reduce dependencies, but support limited algorithms and may behave subtly differently.

Model Deployment Types and Contents

- ❶ **Model-as-a-service (live-scoring model):** Deployed in a framework providing a **REST API endpoint** that responds to requests in real time.
- ❷ **Embedded model:** Packaged into an application which is then published, e.g., an application providing **batch-scoring**.
- ❸ A deployed model comprises **code** (Python, R, or Java) and **data artifacts**, with version dependencies on runtimes and packages.
- ❹ Different versions in production **may cause predictions to differ**.
- ❺ **Portable formats** (PMML, PFA, ONNX, POJO) reduce dependencies, but support limited algorithms and may behave subtly differently.

Model Deployment Types and Contents

- ❶ **Model-as-a-service (live-scoring model):** Deployed in a framework providing a **REST API endpoint** that responds to requests in real time.
- ❷ **Embedded model:** Packaged into an application which is then published, e.g., an application providing **batch-scoring**.
- ❸ A deployed model comprises **code** (Python, R, or Java) and **data artifacts**, with version dependencies on runtimes and packages.
- ❹ Different versions in production **may cause predictions to differ**.
- ❺ **Portable formats** (PMML, PFA, ONNX, POJO) reduce dependencies, but support limited algorithms and may behave subtly differently.

Containerization

- ❶ An increasingly popular solution to **dependency headaches** when deploying ML models.
- ❷ Container technologies such as **Docker** are lightweight alternatives to virtual machines.
- ❸ Applications run in **independent, self-contained environments** matching each model's exact requirements.
- ❹ Enable new models to be seamlessly deployed using the **blue-green deployment** technique.
- ❺ Compute can be **scaled elastically** using multiple containers.
- ❻ Orchestration of many containers is handled by **Kubernetes**, in the cloud or on-premise.

Containerization

- ❶ An increasingly popular solution to **dependency headaches** when deploying ML models.
- ❷ Container technologies such as **Docker** are lightweight alternatives to virtual machines.
- ❸ Applications run in **independent, self-contained environments** matching each model's exact requirements.
- ❹ Enable new models to be seamlessly deployed using the **blue-green deployment** technique.
- ❺ Compute can be **scaled elastically** using multiple containers.
- ❻ Orchestration of many containers is handled by **Kubernetes**, in the cloud or on-premise.

Containerization

- ❶ An increasingly popular solution to **dependency headaches** when deploying ML models.
- ❷ Container technologies such as **Docker** are lightweight alternatives to virtual machines.
- ❸ Applications run in **independent, self-contained environments** matching each model's exact requirements.
- ❹ Enable new models to be seamlessly deployed using the **blue-green deployment** technique.
- ❺ Compute can be **scaled elastically** using multiple containers.
- ❻ Orchestration of many containers is handled by **Kubernetes**, in the cloud or on-premise.

Containerization

- ❶ An increasingly popular solution to **dependency headaches** when deploying ML models.
- ❷ Container technologies such as **Docker** are lightweight alternatives to virtual machines.
- ❸ Applications run in **independent, self-contained environments** matching each model's exact requirements.
- ❹ Enable new models to be seamlessly deployed using the **blue-green deployment** technique.
- ❺ Compute can be **scaled elastically** using multiple containers.
- ❻ Orchestration of many containers is handled by **Kubernetes**, in the cloud or on-premise.

Containerization

- ❶ An increasingly popular solution to **dependency headaches** when deploying ML models.
- ❷ Container technologies such as **Docker** are lightweight alternatives to virtual machines.
- ❸ Applications run in **independent, self-contained environments** matching each model's exact requirements.
- ❹ Enable new models to be seamlessly deployed using the **blue-green deployment** technique.
- ❺ Compute can be **scaled elastically** using multiple containers.
- ❻ Orchestration of many containers is handled by **Kubernetes**, in the cloud or on-premise.

Containerization

- ❶ An increasingly popular solution to **dependency headaches** when deploying ML models.
- ❷ Container technologies such as **Docker** are lightweight alternatives to virtual machines.
- ❸ Applications run in **independent, self-contained environments** matching each model's exact requirements.
- ❹ Enable new models to be seamlessly deployed using the **blue-green deployment** technique.
- ❺ Compute can be **scaled elastically** using multiple containers.
- ❻ Orchestration of many containers is handled by **Kubernetes**, in the cloud or on-premise.

Model Deployment Requirements

- ❶ **Rapid, automated deployment** is always preferred to labor-intensive processes.
- ❷ For **short-lifetime, self-service** applications, a fully automated single-step push-to-production may be adequate (resources capped by Linux cgroups; simple UIs with Flask).
- ❸ For **customer-facing, mission-critical** use cases, a robust **CI/CD pipeline** is required.
- ❹ In heavily regulated industries (finance, pharmaceuticals), checks are extensive and may involve **manual intervention**.
- ❺ MLOps, like DevOps, aims to **automate CI/CD** as far as possible: faster deployment, more regression testing, fewer errors.

Model Deployment Requirements

- ❶ **Rapid, automated deployment** is always preferred to labor-intensive processes.
- ❷ For **short-lifetime, self-service** applications, a fully automated single-step push-to-production may be adequate (resources capped by Linux cgroups; simple UIs with Flask).
- ❸ For **customer-facing, mission-critical** use cases, a robust **CI/CD pipeline** is required.
- ❹ In heavily regulated industries (finance, pharmaceuticals), checks are extensive and may involve **manual intervention**.
- ❺ MLOps, like DevOps, aims to **automate CI/CD** as far as possible: faster deployment, more regression testing, fewer errors.

Model Deployment Requirements

- ❶ **Rapid, automated deployment** is always preferred to labor-intensive processes.
- ❷ For **short-lifetime, self-service** applications, a fully automated single-step push-to-production may be adequate (resources capped by Linux cgroups; simple UIs with Flask).
- ❸ For **customer-facing, mission-critical** use cases, a robust **CI/CD pipeline** is required.
- ❹ In heavily regulated industries (finance, pharmaceuticals), checks are extensive and may involve **manual intervention**.
- ❺ MLOps, like DevOps, aims to **automate CI/CD** as far as possible: faster deployment, more regression testing, fewer errors.

Model Deployment Requirements

- ❶ **Rapid, automated deployment** is always preferred to labor-intensive processes.
- ❷ For **short-lifetime, self-service** applications, a fully automated single-step push-to-production may be adequate (resources capped by Linux cgroups; simple UIs with Flask).
- ❸ For **customer-facing, mission-critical** use cases, a robust **CI/CD pipeline** is required.
- ❹ In heavily regulated industries (finance, pharmaceuticals), checks are extensive and may involve **manual intervention**.
- ❺ MLOps, like DevOps, aims to **automate CI/CD** as far as possible: faster deployment, more regression testing, fewer errors.

Model Deployment Requirements

- ❶ **Rapid, automated deployment** is always preferred to labor-intensive processes.
- ❷ For **short-lifetime, self-service** applications, a fully automated single-step push-to-production may be adequate (resources capped by Linux cgroups; simple UIs with Flask).
- ❸ For **customer-facing, mission-critical** use cases, a robust **CI/CD pipeline** is required.
- ❹ In heavily regulated industries (finance, pharmaceuticals), checks are extensive and may involve **manual intervention**.
- ❺ MLOps, like DevOps, aims to **automate CI/CD** as far as possible: faster deployment, more regression testing, fewer errors.

Robust CI/CD Pipeline for Mission-Critical Models

- ❶ Ensuring all coding, documentation, and sign-off standards have been met.
- ❷ Re-creating the model in something approaching the production environment.
- ❸ Revalidating the model accuracy.
- ❹ Performing explainability checks.
- ❺ Ensuring all governance requirements have been met.
- ❻ Checking the quality of any data artifacts.
- ❼ Testing resource usage under load.
- ❽ Embedding into a more complex application, including integration tests.

Robust CI/CD Pipeline for Mission-Critical Models

- ❶ Ensuring all coding, documentation, and sign-off standards have been met.
- ❷ Re-creating the model in something approaching the production environment.
- ❸ Revalidating the model accuracy.
- ❹ Performing explainability checks.
- ❺ Ensuring all governance requirements have been met.
- ❻ Checking the quality of any data artifacts.
- ❼ Testing resource usage under load.
- ❽ Embedding into a more complex application, including integration tests.

Robust CI/CD Pipeline for Mission-Critical Models

- ❶ Ensuring all coding, documentation, and sign-off standards have been met.
- ❷ Re-creating the model in something approaching the production environment.
- ❸ Revalidating the model accuracy.
- ❹ Performing explainability checks.
- ❺ Ensuring all governance requirements have been met.
- ❻ Checking the quality of any data artifacts.
- ❼ Testing resource usage under load.
- ❽ Embedding into a more complex application, including integration tests.

Robust CI/CD Pipeline for Mission-Critical Models

- ❶ Ensuring all coding, documentation, and sign-off standards have been met.
- ❷ Re-creating the model in something approaching the production environment.
- ❸ Revalidating the model accuracy.
- ❹ Performing explainability checks.
- ❺ Ensuring all governance requirements have been met.
- ❻ Checking the quality of any data artifacts.
- ❼ Testing resource usage under load.
- ❽ Embedding into a more complex application, including integration tests.

Robust CI/CD Pipeline for Mission-Critical Models

- ❶ Ensuring all coding, documentation, and sign-off standards have been met.
- ❷ Re-creating the model in something approaching the production environment.
- ❸ Revalidating the model accuracy.
- ❹ Performing explainability checks.
- ❺ Ensuring all governance requirements have been met.
- ❻ Checking the quality of any data artifacts.
- ❼ Testing resource usage under load.
- ❽ Embedding into a more complex application, including integration tests.

Robust CI/CD Pipeline for Mission-Critical Models

- ➊ Ensuring all coding, documentation, and sign-off standards have been met.
- ➋ Re-creating the model in something approaching the production environment.
- ➌ Revalidating the model accuracy.
- ➍ Performing explainability checks.
- ➎ Ensuring all governance requirements have been met.
- ➏ Checking the quality of any data artifacts.
- ➐ Testing resource usage under load.
- ➑ Embedding into a more complex application, including integration tests.

Robust CI/CD Pipeline for Mission-Critical Models

- ➊ Ensuring all coding, documentation, and sign-off standards have been met.
- ➋ Re-creating the model in something approaching the production environment.
- ➌ Revalidating the model accuracy.
- ➍ Performing explainability checks.
- ➎ Ensuring all governance requirements have been met.
- ➏ Checking the quality of any data artifacts.
- ➐ Testing resource usage under load.
- ➑ Embedding into a more complex application, including integration tests.

Robust CI/CD Pipeline for Mission-Critical Models

- ❶ Ensuring all coding, documentation, and sign-off standards have been met.
- ❷ Re-creating the model in something approaching the production environment.
- ❸ Revalidating the model accuracy.
- ❹ Performing explainability checks.
- ❺ Ensuring all governance requirements have been met.
- ❻ Checking the quality of any data artifacts.
- ❼ Testing resource usage under load.
- ❽ Embedding into a more complex application, including integration tests.

Monitoring

- ❶ After deployment, the model must **continue to perform well** over time.
- ❷ Good performance means different things to DevOps, data scientists, and the business.
- ❸ DevOps concerns:
 - ▶ Is the model getting the job done **quickly enough**?
 - ▶ Is it using a sensible amount of **memory and processing time**?
- ❹ This is traditional IT performance monitoring; ML resource demands are similar to traditional software.
- ❺ Scalability matters, e.g., when **retraining in production**; deep learning needs more resources than decision trees.

Monitoring

- ➊ After deployment, the model must **continue to perform well** over time.
- ➋ Good performance means different things to **DevOps, data scientists, and the business.**
- ➌ DevOps concerns:
 - ▶ Is the model getting the job done **quickly enough**?
 - ▶ Is it using a sensible amount of **memory and processing time**?
- ➍ This is traditional IT performance monitoring; ML resource demands are similar to traditional software.
- ➎ Scalability matters, e.g., when **retraining in production**; deep learning needs more resources than decision trees.

Monitoring

- ❶ After deployment, the model must **continue to perform well** over time.
- ❷ Good performance means different things to **DevOps, data scientists, and the business**.
- ❸ **DevOps concerns:**
 - ▶ Is the model getting the job done **quickly enough**?
 - ▶ Is it using a sensible amount of **memory and processing time**?
- ❹ This is traditional IT performance monitoring; ML resource demands are similar to traditional software.
- ❺ Scalability matters, e.g., when **retraining in production**; deep learning needs more resources than decision trees.

Monitoring

- ❶ After deployment, the model must **continue to perform well** over time.
- ❷ Good performance means different things to **DevOps, data scientists, and the business**.
- ❸ **DevOps concerns:**
 - ▶ Is the model getting the job done **quickly enough**?
 - ▶ Is it using a sensible amount of **memory and processing time**?
- ❹ This is traditional IT performance monitoring; ML resource demands are similar to traditional software.
- ❺ Scalability matters, e.g., when **retraining in production**; deep learning needs more resources than decision trees.

Monitoring

- ❶ After deployment, the model must **continue to perform well** over time.
- ❷ Good performance means different things to **DevOps, data scientists, and the business**.
- ❸ **DevOps concerns:**
 - ▶ Is the model getting the job done **quickly enough**?
 - ▶ Is it using a sensible amount of **memory and processing time**?
- ❹ This is traditional IT performance monitoring; ML resource demands are similar to traditional software.
- ❺ Scalability matters, e.g., when **retraining in production**; deep learning needs more resources than decision trees.

Data Scientist Concerns

- ❶ ML models **degrade over time**, since they are models of the data they were trained on — a problem not faced by traditional software.
- ❷ The real world does not stand still:
 - ▶ A fraud model trained six months ago will not reflect a **new type of fraud**.
 - ▶ An ad model becomes less relevant if a website attracts a **younger user base**.
- ❸ When performance becomes unacceptable, **retraining** is necessary.
- ❹ Retraining frequency depends on how fast the world changes, accuracy needed, and ease of building/deploying a better model.
- ❺ Two approaches to detect degradation: **ground truth** and **input drift**.

Data Scientist Concerns

- ① ML models **degrade over time**, since they are models of the data they were trained on — a problem not faced by traditional software.
- ② The real world does not stand still:
 - ▶ A fraud model trained six months ago will not reflect a **new type of fraud**.
 - ▶ An ad model becomes less relevant if a website attracts a **younger user base**.
- ③ When performance becomes unacceptable, **retraining** is necessary.
- ④ Retraining frequency depends on how fast the world changes, accuracy needed, and ease of building/deploying a better model.
- ⑤ Two approaches to detect degradation: **ground truth** and **input drift**.

Data Scientist Concerns

- ❶ ML models **degrade over time**, since they are models of the data they were trained on — a problem not faced by traditional software.
- ❷ The real world does not stand still:
 - ▶ A fraud model trained six months ago will not reflect a **new type of fraud**.
 - ▶ An ad model becomes less relevant if a website attracts a **younger user base**.
- ❸ When performance becomes unacceptable, **retraining** is necessary.
- ❹ Retraining frequency depends on how fast the world changes, accuracy needed, and ease of building/deploying a better model.
- ❺ Two approaches to detect degradation: **ground truth** and **input drift**.

Data Scientist Concerns

- ❶ ML models **degrade over time**, since they are models of the data they were trained on — a problem not faced by traditional software.
- ❷ The real world does not stand still:
 - ▶ A fraud model trained six months ago will not reflect a **new type of fraud**.
 - ▶ An ad model becomes less relevant if a website attracts a **younger user base**.
- ❸ When performance becomes unacceptable, **retraining** is necessary.
- ❹ Retraining frequency depends on how fast the world changes, accuracy needed, and ease of building/deploying a better model.
- ❺ Two approaches to detect degradation: **ground truth** and **input drift**.

Data Scientist Concerns

- ❶ ML models **degrade over time**, since they are models of the data they were trained on — a problem not faced by traditional software.
- ❷ The real world does not stand still:
 - ▶ A fraud model trained six months ago will not reflect a **new type of fraud**.
 - ▶ An ad model becomes less relevant if a website attracts a **younger user base**.
- ❸ When performance becomes unacceptable, **retraining** is necessary.
- ❹ Retraining frequency depends on how fast the world changes, accuracy needed, and ease of building/deploying a better model.
- ❺ Two approaches to detect degradation: **ground truth** and **input drift**.

Ground Truth and Input Drift

- ❶ **Ground truth:** The correct answer to the question the model was asked (e.g., “Is this transaction actually fraudulent?”).
- ❷ Sometimes obtained **rapidly** (ad clicks within seconds); often **slowly** (fraud reported up to a month later, or never).
- ❸ Slow ground truth prevents accurate daily monitoring; if rapid feedback is needed, use input drift.
- ❹ **Input drift:** If recent requests differ distinctly from training data, model performance is likely compromised.
- ❺ All required data **already exists** — no need to wait for ground truth.
- ❻ Drift identification brings **agility** to enterprise AI efforts (detailed in Chapter 7).

Ground Truth and Input Drift

- ❶ **Ground truth:** The correct answer to the question the model was asked (e.g., “Is this transaction actually fraudulent?”).
- ❷ Sometimes obtained **rapidly** (ad clicks within seconds); often **slowly** (fraud reported up to a month later, or never).
- ❸ Slow ground truth prevents accurate daily monitoring; if rapid feedback is needed, use input drift.
- ❹ **Input drift:** If recent requests differ distinctly from training data, model performance is likely compromised.
- ❺ All required data **already exists** — no need to wait for ground truth.
- ❻ Drift identification brings **agility** to enterprise AI efforts (detailed in Chapter 7).

Ground Truth and Input Drift

- ❶ **Ground truth:** The correct answer to the question the model was asked (e.g., “Is this transaction actually fraudulent?”).
- ❷ Sometimes obtained **rapidly** (ad clicks within seconds); often **slowly** (fraud reported up to a month later, or never).
- ❸ Slow ground truth prevents accurate daily monitoring; if rapid feedback is needed, use input drift.
- ❹ **Input drift:** If recent requests differ distinctly from training data, model performance is likely compromised.
- ❺ All required data **already exists** — no need to wait for ground truth.
- ❻ Drift identification brings **agility** to enterprise AI efforts (detailed in Chapter 7).

Ground Truth and Input Drift

- ❶ **Ground truth:** The correct answer to the question the model was asked (e.g., “Is this transaction actually fraudulent?”).
- ❷ Sometimes obtained **rapidly** (ad clicks within seconds); often **slowly** (fraud reported up to a month later, or never).
- ❸ Slow ground truth prevents accurate daily monitoring; if rapid feedback is needed, use input drift.
- ❹ **Input drift:** If recent requests differ distinctly from training data, model performance is likely compromised.
- ❺ All required data **already exists** — no need to wait for ground truth.
- ❻ Drift identification brings **agility** to enterprise AI efforts (detailed in Chapter 7).

Ground Truth and Input Drift

- ❶ **Ground truth:** The correct answer to the question the model was asked (e.g., “Is this transaction actually fraudulent?”).
- ❷ Sometimes obtained **rapidly** (ad clicks within seconds); often **slowly** (fraud reported up to a month later, or never).
- ❸ Slow ground truth prevents accurate daily monitoring; if rapid feedback is needed, use input drift.
- ❹ **Input drift:** If recent requests differ distinctly from training data, model performance is likely compromised.
- ❺ All required data **already exists** — no need to wait for ground truth.
- ❻ Drift identification brings **agility** to enterprise AI efforts (detailed in Chapter 7).

Ground Truth and Input Drift

- ❶ **Ground truth:** The correct answer to the question the model was asked (e.g., “Is this transaction actually fraudulent?”).
- ❷ Sometimes obtained **rapidly** (ad clicks within seconds); often **slowly** (fraud reported up to a month later, or never).
- ❸ Slow ground truth prevents accurate daily monitoring; if rapid feedback is needed, use input drift.
- ❹ **Input drift:** If recent requests differ distinctly from training data, model performance is likely compromised.
- ❺ All required data **already exists** — no need to wait for ground truth.
- ❻ Drift identification brings **agility** to enterprise AI efforts (detailed in Chapter 7).

Business Concerns

- ❶ The business takes a **holistic** view:
 - ▶ Is the model delivering **value** to the enterprise?
 - ▶ Do benefits outweigh the **cost** of developing and deploying it?
- ❷ The original **KPIs** should be monitored, automatically where possible — rarely trivial.
- ❸ Even monitoring fraud rate does not answer: what is the **net gain in dollars**?
- ❹ In the absence of a “dollar-o-meter,” monitoring **business KPIs** is the best option.
- ❺ Choose a **baseline** that isolates ML value, e.g., compare against a **rule-based** model built from subject matter expertise.

Business Concerns

- ❶ The business takes a **holistic** view:
 - ▶ Is the model delivering **value** to the enterprise?
 - ▶ Do benefits outweigh the **cost** of developing and deploying it?
- ❷ The original **KPIs** should be monitored, automatically where possible — rarely trivial.
 - ❸ Even monitoring fraud rate does not answer: what is the **net gain in dollars**?
 - ❹ In the absence of a “dollar-o-meter,” monitoring **business KPIs** is the best option.
 - ❺ Choose a **baseline** that isolates ML value, e.g., compare against a **rule-based** model built from subject matter expertise.

Business Concerns

- ❶ The business takes a **holistic** view:
 - ▶ Is the model delivering **value** to the enterprise?
 - ▶ Do benefits outweigh the **cost** of developing and deploying it?
- ❷ The original **KPIs** should be monitored, automatically where possible — rarely trivial.
- ❸ Even monitoring fraud rate does not answer: what is the **net gain in dollars**?
- ❹ In the absence of a “dollar-o-meter,” monitoring **business KPIs** is the best option.
- ❺ Choose a **baseline** that isolates ML value, e.g., compare against a **rule-based** model built from subject matter expertise.

Business Concerns

- ❶ The business takes a **holistic** view:
 - ▶ Is the model delivering **value** to the enterprise?
 - ▶ Do benefits outweigh the **cost** of developing and deploying it?
- ❷ The original **KPIs** should be monitored, automatically where possible — rarely trivial.
- ❸ Even monitoring fraud rate does not answer: what is the **net gain in dollars**?
- ❹ In the absence of a “dollar-o-meter,” monitoring **business KPIs** is the best option.
- ❺ Choose a **baseline** that isolates ML value, e.g., compare against a **rule-based** model built from subject matter expertise.

Business Concerns

- ❶ The business takes a **holistic** view:
 - ▶ Is the model delivering **value** to the enterprise?
 - ▶ Do benefits outweigh the **cost** of developing and deploying it?
- ❷ The original **KPIs** should be monitored, automatically where possible — rarely trivial.
- ❸ Even monitoring fraud rate does not answer: what is the **net gain in dollars**?
- ❹ In the absence of a “dollar-o-meter,” monitoring **business KPIs** is the best option.
- ❺ Choose a **baseline** that isolates ML value, e.g., compare against a **rule-based** model built from subject matter expertise.

Iteration and Life Cycle

- ❶ Developing and deploying **improved versions** is essential and one of the more challenging parts of MLOps.
- ❷ Reasons for a new version: **model drift**, refined business objectives/KPIs, or a **better model design**.
- ❸ In fast-moving environments, daily retraining and redeployment are often **automated**.
- ❹ Pitfalls even in simple retraining:
 - ▶ Does the new training data look as expected? (automated validation)
 - ▶ Is the data complete and consistent?
 - ▶ Are feature distributions similar to the previous training set?
- ❺ Compare new and live model metrics on the **same dataset**; if variation is wide, seek **manual intervention**.

Iteration and Life Cycle

- ❶ Developing and deploying **improved versions** is essential and one of the more challenging parts of MLOps.
- ❷ Reasons for a new version: **model drift**, refined business objectives/KPIs, or a **better model design**.
- ❸ In fast-moving environments, daily retraining and redeployment are often **automated**.
- ❹ Pitfalls even in simple retraining:
 - ▶ Does the new training data look as expected? (automated validation)
 - ▶ Is the data complete and consistent?
 - ▶ Are feature distributions similar to the previous training set?
- ❺ Compare new and live model metrics on the **same dataset**; if variation is wide, seek **manual intervention**.

Iteration and Life Cycle

- ➊ Developing and deploying **improved versions** is essential and one of the more challenging parts of MLOps.
- ➋ Reasons for a new version: **model drift**, refined business objectives/KPIs, or a **better model design**.
- ➌ In fast-moving environments, daily retraining and redeployment are often **automated**.
- ➍ Pitfalls even in simple retraining:
 - ▶ Does the new training data look as expected? (automated validation)
 - ▶ Is the data complete and consistent?
 - ▶ Are feature distributions similar to the previous training set?
- ➎ Compare new and live model metrics on the **same dataset**; if variation is wide, seek **manual intervention**.

Iteration and Life Cycle

- ❶ Developing and deploying **improved versions** is essential and one of the more challenging parts of MLOps.
- ❷ Reasons for a new version: **model drift**, refined business objectives/KPIs, or a **better model design**.
- ❸ In fast-moving environments, daily retraining and redeployment are often **automated**.
- ❹ Pitfalls even in simple retraining:
 - ▶ Does the new training data look as expected? (automated validation)
 - ▶ Is the data complete and consistent?
 - ▶ Are feature distributions similar to the previous training set?
- ❺ Compare new and live model metrics on the same dataset; if variation is wide, seek **manual intervention**.

Iteration and Life Cycle

- ❶ Developing and deploying **improved versions** is essential and one of the more challenging parts of MLOps.
- ❷ Reasons for a new version: **model drift**, refined business objectives/KPIs, or a **better model design**.
- ❸ In fast-moving environments, daily retraining and redeployment are often **automated**.
- ❹ Pitfalls even in simple retraining:
 - ▶ Does the new training data look as expected? (automated validation)
 - ▶ Is the data complete and consistent?
 - ▶ Are feature distributions similar to the previous training set?
- ❺ Compare new and live model metrics on the **same dataset**; if variation is wide, seek **manual intervention**.

Retraining Scenarios

- ❶ Even “simple” retraining needs multiple datasets: scoring data reconciliation (with ground truth), cleaning/validation, the previous model, and checks.
- ❷ Other scenarios are more complicated, making **automated redeployment unlikely**.
- ❸ **Drift-motivated retraining:** If new labeled data exists, retraining with it has the best cost-benefit ratio.
- ❹ If ground truth is slow, there may be **little new labeled data**.
- ❺ Data scientists must understand the drift, **adjust existing training data**, and design mitigations (e.g., remove a feature, resample rows).
- ❻ Time needed grows with the amount of **modeling debt**.

Retraining Scenarios

- ❶ Even “simple” retraining needs multiple datasets: scoring data reconciliation (with ground truth), cleaning/validation, the previous model, and checks.
- ❷ Other scenarios are more complicated, making **automated redeployment unlikely**.
- ❸ **Drift-motivated retraining:** If new labeled data exists, retraining with it has the best cost-benefit ratio.
- ❹ If ground truth is slow, there may be little new labeled data.
- ❺ Data scientists must understand the drift, **adjust existing training data**, and design mitigations (e.g., remove a feature, resample rows).
- ❻ Time needed grows with the amount of **modeling debt**.

Retraining Scenarios

- ❶ Even “simple” retraining needs multiple datasets: scoring data reconciliation (with ground truth), cleaning/validation, the previous model, and checks.
- ❷ Other scenarios are more complicated, making **automated redeployment unlikely**.
- ❸ **Drift-motivated retraining:** If new labeled data exists, retraining with it has the best cost-benefit ratio.
- ❹ If ground truth is slow, there may be little new labeled data.
- ❺ Data scientists must understand the drift, **adjust existing training data**, and design mitigations (e.g., remove a feature, resample rows).
- ❻ Time needed grows with the amount of **modeling debt**.

Retraining Scenarios

- ❶ Even “simple” retraining needs multiple datasets: scoring data reconciliation (with ground truth), cleaning/validation, the previous model, and checks.
- ❷ Other scenarios are more complicated, making **automated redeployment unlikely**.
- ❸ **Drift-motivated retraining:** If new labeled data exists, retraining with it has the best cost-benefit ratio.
- ❹ If ground truth is slow, there may be **little new labeled data**.
- ❺ Data scientists must understand the drift, **adjust existing training data**, and design mitigations (e.g., remove a feature, resample rows).
- ❻ Time needed grows with the amount of **modeling debt**.

Retraining Scenarios

- ❶ Even “simple” retraining needs multiple datasets: scoring data reconciliation (with ground truth), cleaning/validation, the previous model, and checks.
- ❷ Other scenarios are more complicated, making **automated redeployment unlikely**.
- ❸ **Drift-motivated retraining:** If new labeled data exists, retraining with it has the best cost-benefit ratio.
- ❹ If ground truth is slow, there may be **little new labeled data**.
- ❺ Data scientists must understand the drift, **adjust existing training data**, and design mitigations (e.g., remove a feature, resample rows).
- ❻ Time needed grows with the amount of modeling debt.

Retraining Scenarios

- ❶ Even “simple” retraining needs multiple datasets: scoring data reconciliation (with ground truth), cleaning/validation, the previous model, and checks.
- ❷ Other scenarios are more complicated, making **automated redeployment unlikely**.
- ❸ **Drift-motivated retraining**: If new labeled data exists, retraining with it has the best cost-benefit ratio.
- ❹ If ground truth is slow, there may be **little new labeled data**.
- ❺ Data scientists must understand the drift, **adjust existing training data**, and design mitigations (e.g., remove a feature, resample rows).
- ❻ Time needed grows with the amount of **modeling debt**.

The Feedback Loop: Shadow Testing and A/B Testing

- ❶ The live scoring and retraining environments are usually **distinct**, so evaluation in retraining may be compromised.
- ❷ **Shadow testing**: New version runs alongside the incumbent; the incumbent answers, the new version's results are only **logged**, then compared statistically.
- ❸ Shadow scoring gives SMEs visibility on future versions for a **smoother transition**.
- ❹ When outcomes depend on the user's reaction (ad clicks), **A/B testing** is more common.
- ❺ **A/B testing**: Requests are **split** between the two models; each request is processed by only one model.

The Feedback Loop: Shadow Testing and A/B Testing

- ❶ The live scoring and retraining environments are usually **distinct**, so evaluation in retraining may be compromised.
- ❷ **Shadow testing:** New version runs alongside the incumbent; the incumbent answers, the new version's results are only **logged**, then compared statistically.
- ❸ Shadow scoring gives SMEs visibility on future versions for a **smoother transition**.
- ❹ When outcomes depend on the user's reaction (ad clicks), **A/B testing** is more common.
- ❺ **A/B testing:** Requests are **split** between the two models; each request is processed by only one model.

The Feedback Loop: Shadow Testing and A/B Testing

- ❶ The live scoring and retraining environments are usually **distinct**, so evaluation in retraining may be compromised.
- ❷ **Shadow testing:** New version runs alongside the incumbent; the incumbent answers, the new version's results are only **logged**, then compared statistically.
- ❸ Shadow scoring gives SMEs visibility on future versions for a **smoother transition**.
- ❹ When outcomes depend on the user's reaction (ad clicks), **A/B testing** is more common.
- ❺ **A/B testing:** Requests are **split** between the two models; each request is processed by only one model.

The Feedback Loop: Shadow Testing and A/B Testing

- ❶ The live scoring and retraining environments are usually **distinct**, so evaluation in retraining may be compromised.
- ❷ **Shadow testing:** New version runs alongside the incumbent; the incumbent answers, the new version's results are only **logged**, then compared statistically.
- ❸ Shadow scoring gives SMEs visibility on future versions for a **smoother transition**.
- ❹ When outcomes depend on the user's reaction (ad clicks), **A/B testing** is more common.
- ❺ **A/B testing:** Requests are **split** between the two models; each request is processed by only one model.

The Feedback Loop: Shadow Testing and A/B Testing

- ❶ The live scoring and retraining environments are usually **distinct**, so evaluation in retraining may be compromised.
- ❷ **Shadow testing:** New version runs alongside the incumbent; the incumbent answers, the new version's results are only **logged**, then compared statistically.
- ❸ Shadow scoring gives SMEs visibility on future versions for a **smoother transition**.
- ❹ When outcomes depend on the user's reaction (ad clicks), **A/B testing** is more common.
- ❺ **A/B testing:** Requests are **split** between the two models; each request is processed by only one model.

A/B Testing Variants

- ❶ Statistically meaningful A/B conclusions require **careful planning**.
- ❷ **Fixed-horizon test**: Wait until a predetermined number of samples is tested; “peeking” early is **unreliable**.
- ❸ In a live commercial environment, every bad prediction costs money, so not stopping early can be **expensive**.
- ❹ **Bayesian / multi-armed bandit tests** aim to draw conclusions **more quickly**.
- ❺ Multi-armed bandit testing is **adaptive**: traffic to underperforming models is reduced, lowering business cost.

A/B Testing Variants

- ❶ Statistically meaningful A/B conclusions require **careful planning**.
- ❷ **Fixed-horizon test:** Wait until a predetermined number of samples is tested; “peeking” early is **unreliable**.
- ❸ In a live commercial environment, every bad prediction costs money, so not stopping early can be **expensive**.
- ❹ **Bayesian / multi-armed bandit tests** aim to draw conclusions **more quickly**.
- ❺ Multi-armed bandit testing is **adaptive**: traffic to underperforming models is reduced, lowering business cost.

A/B Testing Variants

- ❶ Statistically meaningful A/B conclusions require **careful planning**.
- ❷ **Fixed-horizon test**: Wait until a predetermined number of samples is tested; “peeking” early is **unreliable**.
- ❸ In a live commercial environment, every bad prediction costs money, so not stopping early can be **expensive**.
- ❹ Bayesian / multi-armed bandit tests aim to draw conclusions more quickly.
- ❺ Multi-armed bandit testing is **adaptive**: traffic to underperforming models is reduced, lowering business cost.

A/B Testing Variants

- ❶ Statistically meaningful A/B conclusions require **careful planning**.
- ❷ **Fixed-horizon test**: Wait until a predetermined number of samples is tested; “peeking” early is **unreliable**.
- ❸ In a live commercial environment, every bad prediction costs money, so not stopping early can be **expensive**.
- ❹ **Bayesian / multi-armed bandit tests** aim to draw conclusions **more quickly**.
- ❺ Multi-armed bandit testing is **adaptive**: traffic to underperforming models is reduced, lowering business cost.

A/B Testing Variants

- ❶ Statistically meaningful A/B conclusions require **careful planning**.
- ❷ **Fixed-horizon test**: Wait until a predetermined number of samples is tested; “peeking” early is **unreliable**.
- ❸ In a live commercial environment, every bad prediction costs money, so not stopping early can be **expensive**.
- ❹ **Bayesian / multi-armed bandit tests** aim to draw conclusions **more quickly**.
- ❺ Multi-armed bandit testing is **adaptive**: traffic to underperforming models is reduced, lowering business cost.

Iterating on the Edge

- ❶ Iterating on models deployed to **millions of devices** (phones, sensors, cars) poses different challenges.
- ❷ **Centralized training:** Relay feedback to a central point. Tesla's autopilot (500,000+ cars) retrains about 50 neural networks using 70,000 GPU hours.
- ❸ **Federated learning:** Google's GBoard retrains locally on each phone and sends a summary of improvements, which are averaged into the shared model.
- ❹ **Benefits of federated learning:**
 - ▶ Personal data need not be collected centrally.
 - ▶ Improved model is usable immediately on each phone.
 - ▶ Overall power consumption goes down.

Iterating on the Edge

- ❶ Iterating on models deployed to **millions of devices** (phones, sensors, cars) poses different challenges.
- ❷ **Centralized training:** Relay feedback to a central point. Tesla's autopilot (500,000+ cars) retrains about 50 neural networks using 70,000 GPU hours.
- ❸ **Federated learning:** Google's GBoard retrains locally on each phone and sends a summary of improvements, which are averaged into the shared model.
- ❹ **Benefits of federated learning:**
 - ▶ Personal data need not be collected centrally.
 - ▶ Improved model is usable immediately on each phone.
 - ▶ Overall power consumption goes down.

Iterating on the Edge

- ❶ Iterating on models deployed to **millions of devices** (phones, sensors, cars) poses different challenges.
- ❷ **Centralized training:** Relay feedback to a central point. Tesla's autopilot (500,000+ cars) retrains about 50 neural networks using 70,000 GPU hours.
- ❸ **Federated learning:** Google's GBoard retrains locally on each phone and sends a summary of improvements, which are averaged into the shared model.
- ❹ Benefits of federated learning:
 - ▶ Personal data need not be collected centrally.
 - ▶ Improved model is usable immediately on each phone.
 - ▶ Overall power consumption goes down.

Iterating on the Edge

- ❶ Iterating on models deployed to **millions of devices** (phones, sensors, cars) poses different challenges.
- ❷ **Centralized training:** Relay feedback to a central point. Tesla's autopilot (500,000+ cars) retrains about 50 neural networks using 70,000 GPU hours.
- ❸ **Federated learning:** Google's GBoard retrains locally on each phone and sends a summary of improvements, which are averaged into the shared model.
- ❹ Benefits of federated learning:
 - ▶ Personal data need not be collected centrally.
 - ▶ Improved model is usable immediately on each phone.
 - ▶ Overall power consumption goes down.

Governance

Governance

The set of controls placed on a business to ensure that it delivers on its responsibilities to all stakeholders — shareholders, employees, the public, and national governments.

- ❶ Responsibilities include **financial, legal, and ethical** obligations, all underpinned by **fairness**.
- ❷ Legal obligations existed before ML: financial regulations protect against fraud; pharmaceutical rules safeguard public health.
- ❸ Broader legislation protects vulnerable sectors and ensures a level playing field on sex, race, age, or religion.
- ❹ Data regulations such as **GDPR (2016, EU)** and **CCPA (2018, California)** have had immense impact on ML.

Governance

Governance

The set of controls placed on a business to ensure that it delivers on its responsibilities to all stakeholders — shareholders, employees, the public, and national governments.

- ➊ Responsibilities include **financial, legal, and ethical** obligations, all underpinned by **fairness**.
- ➋ **Legal obligations** existed before ML: financial regulations protect against fraud; pharmaceutical rules safeguard public health.
- ➌ Broader legislation protects vulnerable sectors and ensures a level playing field on sex, race, age, or religion.
- ➍ Data regulations such as **GDPR (2016, EU)** and **CCPA (2018, California)** have had immense impact on ML.

Governance

Governance

The set of controls placed on a business to ensure that it delivers on its responsibilities to all stakeholders — shareholders, employees, the public, and national governments.

- ❶ Responsibilities include **financial, legal, and ethical** obligations, all underpinned by **fairness**.
- ❷ **Legal obligations** existed before ML: financial regulations protect against fraud; pharmaceutical rules safeguard public health.
- ❸ Broader legislation protects vulnerable sectors and ensures a level playing field on sex, race, age, or religion.
- ❹ Data regulations such as **GDPR (2016, EU)** and **CCPA (2018, California)** have had immense impact on ML.

Governance

Governance

The set of controls placed on a business to ensure that it delivers on its responsibilities to all stakeholders — shareholders, employees, the public, and national governments.

- ➊ Responsibilities include **financial, legal, and ethical** obligations, all underpinned by **fairness**.
- ➋ **Legal obligations** existed before ML: financial regulations protect against fraud; pharmaceutical rules safeguard public health.
- ➌ Broader legislation protects vulnerable sectors and ensures a level playing field on sex, race, age, or religion.
- ➍ Data regulations such as **GDPR (2016, EU)** and **CCPA (2018, California)** have had immense impact on ML.

GDPR Principles

- ❶ GDPR sets principles for **processing personal data** (collection, storage, alteration, use); CCPA closely mirrors them.
- ❷ The principles are:
 - ▶ Lawfulness, fairness, and transparency
 - ▶ Purpose limitation
 - ▶ Data minimization
 - ▶ Accuracy
 - ▶ Storage limitation
 - ▶ Integrity and confidentiality (security)
 - ▶ Accountability
- ❸ The EU is leading planned legislation defining acceptable uses of AI; businesses must heed yet more regulation.

GDPR Principles

- ❶ GDPR sets principles for **processing personal data** (collection, storage, alteration, use); CCPA closely mirrors them.
- ❷ The principles are:
 - ▶ Lawfulness, fairness, and transparency
 - ▶ Purpose limitation
 - ▶ Data minimization
 - ▶ Accuracy
 - ▶ Storage limitation
 - ▶ Integrity and confidentiality (security)
 - ▶ Accountability
- ❸ The EU is leading planned legislation defining acceptable uses of AI; businesses must heed yet more regulation.

GDPR Principles

- ❶ GDPR sets principles for **processing personal data** (collection, storage, alteration, use); CCPA closely mirrors them.
- ❷ The principles are:
 - ▶ Lawfulness, fairness, and transparency
 - ▶ Purpose limitation
 - ▶ Data minimization
 - ▶ Accuracy
 - ▶ Storage limitation
 - ▶ Integrity and confidentiality (security)
 - ▶ Accountability
- ❸ The **EU** is leading planned legislation defining acceptable uses of AI; businesses must heed yet more regulation.

Moral Responsibility and Governance Categories

- ❶ Businesses increasingly care about moral responsibility beyond legislation, as seen in **ESG** performance indicators.
- ❷ Trust matters to consumers and shareholders; businesses engage with **Responsible AI**.
- ❸ Governance in MLOps is challenging: complex processes, opaque technology, fundamental data dependence.
- ❹ **Data governance**: A framework for ensuring appropriate use and management of data.
- ❺ **Process governance**: Well-defined processes ensuring all governance considerations are addressed at the right point in the life cycle, with a full and accurate record.

Moral Responsibility and Governance Categories

- ❶ Businesses increasingly care about moral responsibility beyond legislation, as seen in **ESG** performance indicators.
- ❷ **Trust** matters to consumers and shareholders; businesses engage with **Responsible AI**.
- ❸ Governance in MLOps is challenging: complex processes, opaque technology, fundamental data dependence.
- ❹ **Data governance**: A framework for ensuring appropriate use and management of data.
- ❺ **Process governance**: Well-defined processes ensuring all governance considerations are addressed at the right point in the life cycle, with a full and accurate record.

Moral Responsibility and Governance Categories

- ❶ Businesses increasingly care about moral responsibility beyond legislation, as seen in **ESG** performance indicators.
- ❷ **Trust** matters to consumers and shareholders; businesses engage with **Responsible AI**.
- ❸ Governance in MLOps is challenging: complex processes, opaque technology, fundamental data dependence.
- ❹ **Data governance**: A framework for ensuring appropriate use and management of data.
- ❺ **Process governance**: Well-defined processes ensuring all governance considerations are addressed at the right point in the life cycle, with a full and accurate record.

Moral Responsibility and Governance Categories

- ❶ Businesses increasingly care about moral responsibility beyond legislation, as seen in **ESG** performance indicators.
- ❷ **Trust** matters to consumers and shareholders; businesses engage with **Responsible AI**.
- ❸ Governance in MLOps is challenging: complex processes, opaque technology, fundamental data dependence.
- ❹ **Data governance**: A framework for ensuring appropriate use and management of data.
- ❺ **Process governance**: Well-defined processes ensuring all governance considerations are addressed at the right point in the life cycle, with a full and accurate record.

Moral Responsibility and Governance Categories

- ❶ Businesses increasingly care about moral responsibility beyond legislation, as seen in **ESG** performance indicators.
- ❷ **Trust** matters to consumers and shareholders; businesses engage with **Responsible AI**.
- ❸ Governance in MLOps is challenging: complex processes, opaque technology, fundamental data dependence.
- ❹ **Data governance**: A framework for ensuring appropriate use and management of data.
- ❺ **Process governance**: Well-defined processes ensuring all governance considerations are addressed at the right point in the life cycle, with a full and accurate record.

Data Governance

❶ Addresses questions such as:

- ▶ What is the data's **provenance**? Under what terms was it collected?
- ▶ Is the data accurate and up to date?
- ▶ Is there **PII** or sensitive data that should not be used?

❷ Understanding **data lineage**, especially at feature level, is complex but essential for GDPR compliance.

❸ Anonymizing or pseudo-anonymizing data is **not always sufficient**; individuals may still be singled out.

❹ **Accidental bias**: A recruitment model discriminated against women by learning from all-female schools' underrepresentation in upper management (Amazon, 2018).

Data Governance

- ❶ Addresses questions such as:
 - ▶ What is the data's **provenance**? Under what terms was it collected?
 - ▶ Is the data accurate and up to date?
 - ▶ Is there **PII** or sensitive data that should not be used?
- ❷ Understanding **data lineage**, especially at feature level, is complex but essential for GDPR compliance.
- ❸ Anonymizing or pseudo-anonymizing data is **not always sufficient**; individuals may still be singled out.
- ❹ **Accidental bias**: A recruitment model discriminated against women by learning from all-female schools' underrepresentation in upper management (Amazon, 2018).

Data Governance

- ❶ Addresses questions such as:
 - ▶ What is the data's **provenance**? Under what terms was it collected?
 - ▶ Is the data accurate and up to date?
 - ▶ Is there **PII** or sensitive data that should not be used?
- ❷ Understanding **data lineage**, especially at feature level, is complex but essential for GDPR compliance.
- ❸ Anonymizing or pseudo-anonymizing data is **not always sufficient**; individuals may still be singled out.
- ❹ **Accidental bias**: A recruitment model discriminated against women by learning from all-female schools' underrepresentation in upper management (Amazon, 2018).

Data Governance

- ❶ Addresses questions such as:
 - ▶ What is the data's **provenance**? Under what terms was it collected?
 - ▶ Is the data accurate and up to date?
 - ▶ Is there **PII** or sensitive data that should not be used?
- ❷ Understanding **data lineage**, especially at feature level, is complex but essential for GDPR compliance.
- ❸ Anonymizing or pseudo-anonymizing data is **not always sufficient**; individuals may still be singled out.
- ❹ **Accidental bias**: A recruitment model discriminated against women by learning from all-female schools' underrepresentation in upper management (Amazon, 2018).

Data Lineage Tools

❶ Data governance tools are still **in their infancy**.

❷ Most focus on two data lineage questions:

- ▶ Where did the information in this dataset **come from**, and how can I use it?
- ▶ How is this dataset **used**, and what are the downstream implications if I change it?

❸ Neither is easy to answer fully in real-world pipelines.

❹ **Example:** If a Python function processes several datasets in memory to produce one output, from what information was each cell derived?

Data Lineage Tools

- ❶ Data governance tools are still **in their infancy**.
- ❷ Most focus on two data lineage questions:
 - ▶ Where did the information in this dataset **come from**, and how can I use it?
 - ▶ How is this dataset **used**, and what are the downstream implications if I change it?
- ❸ Neither is easy to answer fully in real-world pipelines.
- ❹ **Example:** If a Python function processes several datasets in memory to produce one output, from what information was each cell derived?

Data Lineage Tools

- ❶ Data governance tools are still **in their infancy**.
- ❷ Most focus on two data lineage questions:
 - ▶ Where did the information in this dataset **come from**, and how can I use it?
 - ▶ How is this dataset **used**, and what are the downstream implications if I change it?
- ❸ Neither is easy to answer fully in real-world pipelines.
- ❹ Example: If a Python function processes several datasets in memory to produce one output, from what information was each cell derived?

Data Lineage Tools

- ❶ Data governance tools are still **in their infancy**.
- ❷ Most focus on two data lineage questions:
 - ▶ Where did the information in this dataset **come from**, and how can I use it?
 - ▶ How is this dataset **used**, and what are the downstream implications if I change it?
- ❸ Neither is easy to answer fully in real-world pipelines.
- ❹ **Example:** If a Python function processes several datasets in memory to produce one output, from what information was each cell derived?

Process Governance

- ❶ Formalizes MLOps steps and associates actions: **reviews, sign-offs**, and capture of supporting materials.

- ❷ **Aims:**

- ▶ Every governance consideration is made at the correct time (e.g., no deployment until validation passes).
- ▶ Enable **oversight** by auditors, risk managers, and compliance officers.

- ❸ **Why it is hard:**

- ▶ ML processes are rarely easy to define; understanding is spread across teams.
- ▶ Every team must adopt it wholeheartedly.
- ▶ Heavy-weight processes will be **subverted** by teams.

- ❹ Most common today in heavily regulated sectors such as **finance**.

Process Governance

- ❶ Formalizes MLOps steps and associates actions: **reviews**, **sign-offs**, and capture of supporting materials.
- ❷ **Aims:**
 - ▶ Every governance consideration is made at the correct time (e.g., no deployment until validation passes).
 - ▶ Enable **oversight** by auditors, risk managers, and compliance officers.
- ❸ **Why it is hard:**
 - ▶ ML processes are rarely easy to define; understanding is spread across teams.
 - ▶ Every team must adopt it wholeheartedly.
 - ▶ Heavy-weight processes will be **subverted** by teams.
- ❹ Most common today in heavily regulated sectors such as **finance**.

Process Governance

- ❶ Formalizes MLOps steps and associates actions: **reviews**, **sign-offs**, and capture of supporting materials.
- ❷ **Aims:**
 - ▶ Every governance consideration is made at the correct time (e.g., no deployment until validation passes).
 - ▶ Enable **oversight** by auditors, risk managers, and compliance officers.
- ❸ **Why it is hard:**
 - ▶ ML processes are rarely easy to define; understanding is spread across teams.
 - ▶ Every team must adopt it wholeheartedly.
 - ▶ Heavy-weight processes will be **subverted** by teams.

❹ Most common today in heavily regulated sectors such as finance.

Process Governance

- ❶ Formalizes MLOps steps and associates actions: **reviews**, **sign-offs**, and capture of supporting materials.
- ❷ **Aims:**
 - ▶ Every governance consideration is made at the correct time (e.g., no deployment until validation passes).
 - ▶ Enable **oversight** by auditors, risk managers, and compliance officers.
- ❸ **Why it is hard:**
 - ▶ ML processes are rarely easy to define; understanding is spread across teams.
 - ▶ Every team must adopt it wholeheartedly.
 - ▶ Heavy-weight processes will be **subverted** by teams.
- ❹ Most common today in heavily regulated sectors such as **finance**.

Closing Thoughts

- ❶ MLOps features and processes **cannot be ignored** by data teams or the data-driven organization.
- ❷ MLOps is **not a checklist item** (“yes, we do MLOps!”).
- ❸ It is a complex interplay of **technologies, processes, and people**.
- ❹ It requires **discipline and time** to do correctly.
- ❺ The following chapters go deeper into each life cycle component, starting with **Developing Models** (Module 2).

Closing Thoughts

- ❶ MLOps features and processes **cannot be ignored** by data teams or the data-driven organization.
- ❷ MLOps is **not a checklist item** (“yes, we do MLOps!”).
- ❸ It is a complex interplay of **technologies, processes, and people**.
- ❹ It requires **discipline and time** to do correctly.
- ❺ The following chapters go deeper into each life cycle component, starting with **Developing Models** (Module 2).

Closing Thoughts

- ❶ MLOps features and processes **cannot be ignored** by data teams or the data-driven organization.
- ❷ MLOps is **not a checklist item** (“yes, we do MLOps!”).
- ❸ It is a complex interplay of **technologies, processes, and people**.
- ❹ It requires **discipline and time** to do correctly.
- ❺ The following chapters go deeper into each life cycle component, starting with **Developing Models** (Module 2).

Closing Thoughts

- ❶ MLOps features and processes **cannot be ignored** by data teams or the data-driven organization.
- ❷ MLOps is **not a checklist item** (“yes, we do MLOps!”).
- ❸ It is a complex interplay of **technologies, processes, and people**.
- ❹ It requires **discipline and time** to do correctly.
- ❺ The following chapters go deeper into each life cycle component, starting with **Developing Models** (Module 2).

Closing Thoughts

- ❶ MLOps features and processes **cannot be ignored** by data teams or the data-driven organization.
- ❷ MLOps is **not a checklist item** (“yes, we do MLOps!”).
- ❸ It is a complex interplay of **technologies, processes, and people**.
- ❹ It requires **discipline and time** to do correctly.
- ❺ The following chapters go deeper into each life cycle component, starting with **Developing Models** (Module 2).

Developing Models

- ❶ Anyone serious about MLOps needs at least a **cursory understanding** of the model development process.
- ❷ Model development is one element of the **larger ML project life cycle** .
- ❸ Depending on the situation, it ranges from **quite simple** to **extremely complex** .
- ❹ It **dictates the constraints** of subsequent usage, monitoring, and maintenance of models.
- ❺ This module covers model development **in the context of MLOps** : building models so that MLOps considerations are easier to implement later.

Developing Models

- ❶ Anyone serious about MLOps needs at least a **cursory understanding** of the model development process.
- ❷ Model development is one element of the **larger ML project life cycle** .
- ❸ Depending on the situation, it ranges from **quite simple** to **extremely complex** .
- ❹ It **dictates the constraints** of subsequent usage, monitoring, and maintenance of models.
- ❺ This module covers model development **in the context of MLOps** : building models so that MLOps considerations are easier to implement later.

Developing Models

- ❶ Anyone serious about MLOps needs at least a **cursory understanding** of the model development process.
- ❷ Model development is one element of the **larger ML project life cycle** .
- ❸ Depending on the situation, it ranges from **quite simple** to **extremely complex** .
- ❹ It **dictates the constraints** of subsequent usage, monitoring, and maintenance of models.
- ❺ This module covers model development **in the context of MLOps** : building models so that MLOps considerations are easier to implement later.

Developing Models

- ❶ Anyone serious about MLOps needs at least a **cursory understanding** of the model development process.
- ❷ Model development is one element of the **larger ML project life cycle** .
- ❸ Depending on the situation, it ranges from **quite simple** to **extremely complex** .
- ❹ It **dictates the constraints** of subsequent usage, monitoring, and maintenance of models.
- ❺ This module covers model development in the context of MLOps: building models so that MLOps considerations are easier to implement later.

Developing Models

- ❶ Anyone serious about MLOps needs at least a **cursory understanding** of the model development process.
- ❷ Model development is one element of the **larger ML project life cycle**.
- ❸ Depending on the situation, it ranges from **quite simple** to **extremely complex**.
- ❹ It **dictates the constraints** of subsequent usage, monitoring, and maintenance of models.
- ❺ This module covers model development **in the context of MLOps**: building models so that MLOps considerations are easier to implement later.

Model Development in the ML Project Life Cycle

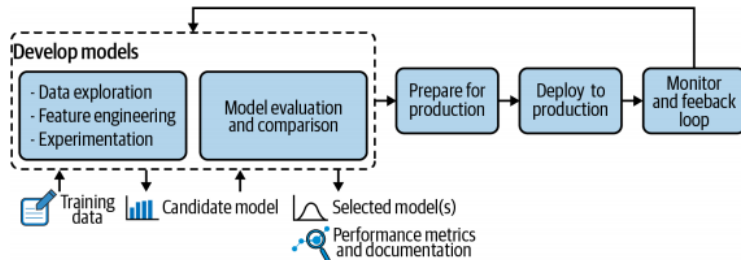


Figure: Model development highlighted in the larger context of the ML project life cycle

- ❶ **Develop models:** Data exploration, feature engineering, experimentation, followed by model evaluation and comparison.
- ❷ **Inputs/outputs:** Training data goes in; candidate models are produced; selected model(s) come out with performance metrics and documentation.
- ❸ Selected models flow to prepare for production → deploy to production → monitor and feedback loop, which loops back to development.

Model Development in the ML Project Life Cycle

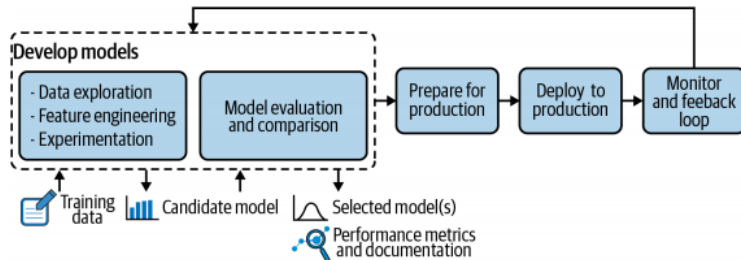


Figure: Model development highlighted in the larger context of the ML project life cycle

- ❶ **Develop models:** Data exploration, feature engineering, experimentation, followed by model evaluation and comparison.
- ❷ **Inputs/outputs:** Training data goes in; candidate models are produced; selected model(s) come out with performance metrics and documentation.
- ❸ Selected models flow to prepare for production → deploy to production → monitor and feedback loop, which loops back to development.

Model Development in the ML Project Life Cycle

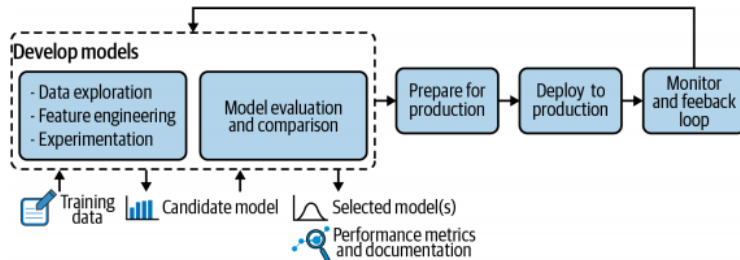


Figure: Model development highlighted in the larger context of the ML project life cycle

- ❶ **Develop models:** Data exploration, feature engineering, experimentation, followed by model evaluation and comparison.
- ❷ **Inputs/outputs:** Training data goes in; candidate models are produced; selected model(s) come out with performance metrics and documentation.
- ❸ Selected models flow to **prepare for production** → **deploy to production** → **monitor and feedback loop**, which loops back to development.

Implications of Development Choices

- ❶ The implications of the **data collection process** are straightforward: a model can easily become **stale**.
- ❷ For other parts of the model, the effects may be **less obvious**.
- ❸ **Feature creation example:** Feeding a **date** versus a flag for **public holiday** can change performance, and also imposes different constraints on updating the model.
- ❹ **Metrics example:** The metrics used for evaluating and comparing models may enable **automatic switching** to the best version later.

Implications of Development Choices

- ❶ The implications of the **data collection process** are straightforward: a model can easily become **stale**.
- ❷ For other parts of the model, the effects may be **less obvious**.
- ❸ **Feature creation example:** Feeding a **date** versus a **flag for public holiday** can change performance, and also imposes different constraints on updating the model.
- ❹ **Metrics example:** The metrics used for evaluating and comparing models may enable **automatic switching** to the best version later.

Implications of Development Choices

- ❶ The implications of the **data collection process** are straightforward: a model can easily become **stale**.
- ❷ For other parts of the model, the effects may be **less obvious**.
- ❸ **Feature creation example:** Feeding a **date** versus a flag for **public holiday** can change performance, and also imposes different constraints on updating the model.
- ❹ **Metrics example:** The metrics used for evaluating and comparing models may enable automatic switching to the best version later.

Implications of Development Choices

- ❶ The implications of the **data collection process** are straightforward: a model can easily become **stale**.
- ❷ For other parts of the model, the effects may be **less obvious**.
- ❸ **Feature creation example:** Feeding a **date** versus a flag for **public holiday** can change performance, and also imposes different constraints on updating the model.
- ❹ **Metrics example:** The metrics used for evaluating and comparing models may enable **automatic switching** to the best version later.

What Is a Machine Learning Model? — In Theory

- ❶ An ML model is a **projection of reality**: a partial and approximate representation of some aspects of a real thing or process.
- ❷ Once trained, it boils down to a **mathematical formula** that yields a result (a probability or a raw value) when fed inputs.
- ❸ Models are based on **statistical theory**; ML algorithms are the **tools that build models** from training data.
- ❹ Training data represents the world **as it was at collection time**.
- ❺ Hence models can predict well **when the future looks like the past**.

What Is a Machine Learning Model? — In Theory

- ➊ An ML model is a **projection of reality**: a partial and approximate representation of some aspects of a real thing or process.
- ➋ Once trained, it boils down to a **mathematical formula** that yields a result (a probability or a raw value) when fed inputs.
- ➌ Models are based on statistical theory; ML algorithms are the tools that build models from training data.
- ➍ Training data represents the world as it was at collection time.
- ➎ Hence models can predict well when the future looks like the past.

What Is a Machine Learning Model? — In Theory

- ➊ An ML model is a **projection of reality**: a partial and approximate representation of some aspects of a real thing or process.
- ➋ Once trained, it boils down to a **mathematical formula** that yields a result (a probability or a raw value) when fed inputs.
- ➌ Models are based on **statistical theory**; ML algorithms are the **tools that build models** from training data.
- ➍ Training data represents the world as it was at collection time.
- ➎ Hence models can predict well when the future looks like the past.

What Is a Machine Learning Model? — In Theory

- ❶ An ML model is a **projection of reality**: a partial and approximate representation of some aspects of a real thing or process.
- ❷ Once trained, it boils down to a **mathematical formula** that yields a result (a probability or a raw value) when fed inputs.
- ❸ Models are based on **statistical theory**; ML algorithms are the **tools that build models** from training data.
- ❹ Training data represents the world **as it was at collection time**.
- ❺ Hence models can predict well when the future looks like the past.

What Is a Machine Learning Model? — In Theory

- ➊ An ML model is a **projection of reality**: a partial and approximate representation of some aspects of a real thing or process.
- ➋ Once trained, it boils down to a **mathematical formula** that yields a result (a probability or a raw value) when fed inputs.
- ➌ Models are based on **statistical theory**; ML algorithms are the **tools that build models** from training data.
- ➍ Training data represents the world **as it was at collection time**.
- ➎ Hence models can predict well **when the future looks like the past**.

Generalization Capacity and Examples

- ❶ **Generalization capacity:** The ability to predict accurately for cases not exactly like the input data.
- ❷ Even outputs like **horses with zebra stripes** (CycleGAN) come from modeling a probability distribution.
- ❸ **House price example:**
 - ▶ Inputs: surface area, bedrooms, bathrooms, year of construction, location.
 - ▶ Contextual inputs: housing market state, seller urgency.
 - ▶ With enough historical data and stable market conditions, a reasonable estimate is possible.
- ❹ **Health example:** Classification models predict the **probability** of developing a disease, sometimes with a **confidence interval**.

Generalization Capacity and Examples

- ➊ **Generalization capacity:** The ability to predict accurately for cases not exactly like the input data.
- ➋ Even outputs like **horses with zebra stripes** (CycleGAN) come from modeling a probability distribution.
- ➌ **House price example:**
 - ▶ Inputs: surface area, bedrooms, bathrooms, year of construction, location.
 - ▶ Contextual inputs: housing market state, seller urgency.
 - ▶ With enough historical data and stable market conditions, a reasonable estimate is possible.
- ➍ **Health example:** Classification models predict the **probability** of developing a disease, sometimes with a **confidence interval**.

Generalization Capacity and Examples

- ❶ **Generalization capacity:** The ability to predict accurately for cases not exactly like the input data.
- ❷ Even outputs like **horses with zebra stripes** (CycleGAN) come from modeling a probability distribution.
- ❸ **House price example:**
 - ▶ Inputs: surface area, bedrooms, bathrooms, year of construction, location.
 - ▶ Contextual inputs: housing market state, seller urgency.
 - ▶ With enough historical data and stable market conditions, a reasonable estimate is possible.
- ❹ **Health example:** Classification models predict the probability of developing a disease, sometimes with a confidence interval.

Generalization Capacity and Examples

- ❶ **Generalization capacity:** The ability to predict accurately for cases not exactly like the input data.
- ❷ Even outputs like **horses with zebra stripes** (CycleGAN) come from modeling a probability distribution.
- ❸ **House price example:**
 - ▶ Inputs: surface area, bedrooms, bathrooms, year of construction, location.
 - ▶ Contextual inputs: housing market state, seller urgency.
 - ▶ With enough historical data and stable market conditions, a reasonable estimate is possible.
- ❹ **Health example:** Classification models predict the **probability** of developing a disease, sometimes with a **confidence interval**.

What Is a Machine Learning Model? — In Practice

- ❶ A model is the **set of parameters** necessary to rebuild and apply the formula.
- ❷ It is usually **stateless and deterministic**: the same inputs give the same outputs (exception: online learning).
- ❸ It includes **all transformations** from input data to the final formula, plus derived data (classification or decision).
- ❹ In practice, whether ML-based or not, a model is a **computable mathematical function** applied to input data, one row at a time.
- ❺ **House price example**: Zip codes may be replaced by derived inputs (average income, population density, amenities); this transformation is **part of the model**.

What Is a Machine Learning Model? — In Practice

- ❶ A model is the **set of parameters** necessary to rebuild and apply the formula.
- ❷ It is usually **stateless and deterministic**: the same inputs give the same outputs (exception: online learning).
- ❸ It includes **all transformations** from input data to the final formula, plus derived data (classification or decision).
- ❹ In practice, whether ML-based or not, a model is a **computable mathematical function** applied to input data, one row at a time.
- ❺ **House price example**: Zip codes may be replaced by derived inputs (average income, population density, amenities); this transformation is **part of the model**.

What Is a Machine Learning Model? — In Practice

- ❶ A model is the **set of parameters** necessary to rebuild and apply the formula.
- ❷ It is usually **stateless and deterministic**: the same inputs give the same outputs (exception: online learning).
- ❸ It includes **all transformations** from input data to the final formula, plus derived data (classification or decision).
- ❹ In practice, whether ML-based or not, a model is a **computable mathematical function** applied to input data, one row at a time.
- ❺ **House price example**: Zip codes may be replaced by derived inputs (average income, population density, amenities); this transformation is **part of the model**.

What Is a Machine Learning Model? — In Practice

- ❶ A model is the **set of parameters** necessary to rebuild and apply the formula.
- ❷ It is usually **stateless and deterministic**: the same inputs give the same outputs (exception: online learning).
- ❸ It includes **all transformations** from input data to the final formula, plus derived data (classification or decision).
- ❹ In practice, whether ML-based or not, a model is **a computable mathematical function** applied to input data, one row at a time.
- ❺ **House price example**: Zip codes may be replaced by derived inputs (average income, population density, amenities); this transformation is part of the model.

What Is a Machine Learning Model? — In Practice

- ❶ A model is the **set of parameters** necessary to rebuild and apply the formula.
- ❷ It is usually **stateless and deterministic**: the same inputs give the same outputs (exception: online learning).
- ❸ It includes **all transformations** from input data to the final formula, plus derived data (classification or decision).
- ❹ In practice, whether ML-based or not, a model is **a computable mathematical function** applied to input data, one row at a time.
- ❺ **House price example**: Zip codes may be replaced by derived inputs (average income, population density, amenities); this transformation is **part of the model**.

Richer Model Outputs and Packaging

- ❶ Outputs can be **richer than a single number**.
- ❷ **Fraud detection:** Outputs a probability (maybe with a confidence interval); a **fine-tuned threshold** decides fraud classification based on costs.
- ❸ Some models include **recommendations or decisions**: which product to show, which treatment gives the most probable recovery.
- ❹ All transformations and associated data are part of the model, but are **not always bundled** as a single artifact.
- ❺ Different parts have **varying constraints** (different refresh rates, external sources, etc.).

Richer Model Outputs and Packaging

- ❶ Outputs can be **richer than a single number**.
- ❷ **Fraud detection:** Outputs a probability (maybe with a confidence interval); a **fine-tuned threshold** decides fraud classification based on costs.
- ❸ Some models include **recommendations or decisions**: which product to show, which treatment gives the most probable recovery.
- ❹ All transformations and associated data are part of the model, but are **not always bundled** as a single artifact.
- ❺ Different parts have **varying constraints** (different refresh rates, external sources, etc.).

Richer Model Outputs and Packaging

- ❶ Outputs can be **richer than a single number**.
- ❷ **Fraud detection:** Outputs a probability (maybe with a confidence interval); a **fine-tuned threshold** decides fraud classification based on costs.
- ❸ Some models include **recommendations or decisions**: which product to show, which treatment gives the most probable recovery.
- ❹ All transformations and associated data are part of the model, but are **not always bundled** as a single artifact.
- ❺ Different parts have **varying constraints** (different refresh rates, external sources, etc.).

Richer Model Outputs and Packaging

- ❶ Outputs can be **richer than a single number**.
- ❷ **Fraud detection:** Outputs a probability (maybe with a confidence interval); a **fine-tuned threshold** decides fraud classification based on costs.
- ❸ Some models include **recommendations or decisions**: which product to show, which treatment gives the most probable recovery.
- ❹ All transformations and associated data are part of the model, but are **not always bundled** as a single artifact.
- ❺ Different parts have **varying constraints** (different refresh rates, external sources, etc.).

Richer Model Outputs and Packaging

- ➊ Outputs can be **richer than a single number**.
- ➋ **Fraud detection:** Outputs a probability (maybe with a confidence interval); a **fine-tuned threshold** decides fraud classification based on costs.
- ➌ Some models include **recommendations or decisions**: which product to show, which treatment gives the most probable recovery.
- ➍ All transformations and associated data are part of the model, but are **not always bundled** as a single artifact.
- ➎ Different parts have **varying constraints** (different refresh rates, external sources, etc.).

Required Components of an ML Model (Table 4-1)

ML Component	Description
Training data	Usually labeled with examples of what is modeled (supervised learning). Must be good data — e.g., WWII damaged-plane data suffered from survivor bias .
Performance metric	What the model seeks to optimize; chosen carefully to avoid the cobra effect . If 95% of data is class A, raw accuracy rewards always predicting A.
ML algorithm	Models work in various ways with pros and cons; selection depends on priorities: performance, stability, interpretability, computation cost.
Hyperparameters	Configurations for ML algorithms: the ways the algorithm goes to find its parameters. E.g., depth of a decision tree.
Evaluation dataset	Data different from the training set, used to evaluate performance on unseen data (generalization).

❶ The number and complexity of these components makes good MLOps **challenging**.

❷ Algorithm choice is yet another piece of the puzzle.

Required Components of an ML Model (Table 4-1)

ML Component	Description
Training data	Usually labeled with examples of what is modeled (supervised learning). Must be good data — e.g., WWII damaged-plane data suffered from survivor bias .
Performance metric	What the model seeks to optimize; chosen carefully to avoid the cobra effect . If 95% of data is class A, raw accuracy rewards always predicting A.
ML algorithm	Models work in various ways with pros and cons; selection depends on priorities: performance, stability, interpretability, computation cost.
Hyperparameters	Configurations for ML algorithms: the ways the algorithm goes to find its parameters. E.g., depth of a decision tree.
Evaluation dataset	Data different from the training set, used to evaluate performance on unseen data (generalization).

- ❶ The number and complexity of these components makes good MLOps **challenging**.
- ❷ **Algorithm choice** is yet another piece of the puzzle.

Required Components — Explained

- ❶ **Training data:** Quality and representativeness decide model quality; biased samples (survivor bias) lead to wrong conclusions.
- ❷ **Performance metric:** Defines “good”; a poorly chosen metric produces unintended behavior (cobra effect).
- ❸ **ML algorithm:** Contains the basic formula; must suit the task and priorities.
- ❹ **Parameters vs hyperparameters:** Parameters are learned operations/operands of the formula; hyperparameters guide how they are found.
- ❺ **Evaluation dataset:** Separate from training data to measure generalization.

Required Components — Explained

- ❶ **Training data:** Quality and representativeness decide model quality; biased samples (survivor bias) lead to wrong conclusions.
- ❷ **Performance metric:** Defines “good”; a poorly chosen metric produces unintended behavior (cobra effect).
- ❸ **ML algorithm:** Contains the basic formula; must suit the task and priorities.
- ❹ **Parameters vs hyperparameters:** Parameters are learned operations/operands of the formula; hyperparameters guide how they are found.
- ❺ **Evaluation dataset:** Separate from training data to measure generalization.

Required Components — Explained

- ❶ **Training data:** Quality and representativeness decide model quality; biased samples (survivor bias) lead to wrong conclusions.
- ❷ **Performance metric:** Defines “good”; a poorly chosen metric produces unintended behavior (cobra effect).
- ❸ **ML algorithm:** Contains the basic formula; must suit the task and priorities.
- ❹ **Parameters vs hyperparameters:** Parameters are learned operations/operands of the formula; hyperparameters guide how they are found.
- ❺ **Evaluation dataset:** Separate from training data to measure generalization.

Required Components — Explained

- ❶ **Training data:** Quality and representativeness decide model quality; biased samples (survivor bias) lead to wrong conclusions.
- ❷ **Performance metric:** Defines “good”; a poorly chosen metric produces unintended behavior (cobra effect).
- ❸ **ML algorithm:** Contains the basic formula; must suit the task and priorities.
- ❹ **Parameters vs hyperparameters:** Parameters are **learned** operations/operands of the formula; hyperparameters **guide how** they are found.
- ❺ **Evaluation dataset:** Separate from training data to measure generalization.

Required Components — Explained

- ❶ **Training data:** Quality and representativeness decide model quality; biased samples (survivor bias) lead to wrong conclusions.
- ❷ **Performance metric:** Defines “good”; a poorly chosen metric produces unintended behavior (cobra effect).
- ❸ **ML algorithm:** Contains the basic formula; must suit the task and priorities.
- ❹ **Parameters vs hyperparameters:** Parameters are **learned** operations/operands of the formula; hyperparameters **guide how** they are found.
- ❺ **Evaluation dataset:** Separate from training data to measure generalization.

MLOps Considerations by Algorithm Type (Table 4-2)

Algorithm Type	Name	MLOps Considerations
Linear	Linear regression	Tendency for overfitting.
	Logistic regression	Tendency for overfitting.
Tree-based	Decision tree	Can be unstable — small data changes can greatly change the tree structure.
	Random forest	Predictions difficult to understand (Responsible AI challenge); relatively slow to output predictions.
	Gradient boosting	Predictions difficult to understand; small changes in features or training set can radically change the model.
Deep learning	Neural networks	Almost impossible to interpret; extremely slow to train; need a lot of power and data. Would a simpler model work just as well?

❶ All algorithms model patterns in past data; they differ in characteristics and **MLOps challenges**.

❷ Quality and relevance of past experience are the key factors in effectiveness.

MLOps Considerations by Algorithm Type (Table 4-2)

Algorithm Type	Name	MLOps Considerations
Linear	Linear regression	Tendency for overfitting.
	Logistic regression	Tendency for overfitting.
Tree-based	Decision tree	Can be unstable — small data changes can greatly change the tree structure.
	Random forest	Predictions difficult to understand (Responsible AI challenge); relatively slow to output predictions.
	Gradient boosting	Predictions difficult to understand; small changes in features or training set can radically change the model.
Deep learning	Neural networks	Almost impossible to interpret; extremely slow to train; need a lot of power and data. Would a simpler model work just as well?

- ❶ All algorithms model patterns in past data; they differ in characteristics and **MLOps challenges**.
- ❷ Quality and relevance of past experience are the **key factors** in effectiveness.

Algorithm Choice: Governance and Computing Power

- ❶ **Governance** may drive algorithm choice: regulated environments needing explanations (financial services) avoid opaque models like neural networks and favor **decision trees**.
- ❷ Often the trade-off is on **cost**, not performance: simpler techniques need more costly **manual feature engineering**.
- ❸ **Computing power**: ML development is **inversely proportional** to the cost of computing power.
- ❹ New algorithms arose as compute became available; random forest and gradient boosting were expensive 20 years ago.
- ❺ They brought **ease of use**, lowering development cost and enabling new use cases.
- ❻ Most algorithms still operate with **all training data in memory**, not distributed big data.

Algorithm Choice: Governance and Computing Power

- ❶ **Governance** may drive algorithm choice: regulated environments needing explanations (financial services) avoid opaque models like neural networks and favor **decision trees**.
- ❷ Often the trade-off is on **cost**, not performance: simpler techniques need more costly **manual feature engineering**.
- ❸ **Computing power**: ML development is **inversely proportional** to the cost of computing power.
- ❹ New algorithms arose as compute became available; random forest and gradient boosting were expensive 20 years ago.
- ❺ They brought **ease of use**, lowering development cost and enabling new use cases.
- ❻ Most algorithms still operate with **all training data in memory**, not distributed big data.

Algorithm Choice: Governance and Computing Power

- ❶ **Governance** may drive algorithm choice: regulated environments needing explanations (financial services) avoid opaque models like neural networks and favor **decision trees**.
- ❷ Often the trade-off is on **cost**, not performance: simpler techniques need more costly **manual feature engineering**.
- ❸ **Computing power**: ML development is **inversely proportional** to the cost of computing power.
- ❹ New algorithms arose as compute became available; random forest and gradient boosting were expensive 20 years ago.
- ❺ They brought **ease of use**, lowering development cost and enabling new use cases.
- ❻ Most algorithms still operate with **all training data in memory**, not distributed big data.

Algorithm Choice: Governance and Computing Power

- ❶ **Governance** may drive algorithm choice: regulated environments needing explanations (financial services) avoid opaque models like neural networks and favor **decision trees**.
- ❷ Often the trade-off is on **cost**, not performance: simpler techniques need more costly **manual feature engineering**.
- ❸ **Computing power**: ML development is **inversely proportional** to the cost of computing power.
- ❹ New algorithms arose as compute became available; random forest and gradient boosting were expensive 20 years ago.
- ❺ They brought **ease of use**, lowering development cost and enabling new use cases.
- ❻ Most algorithms still operate with **all training data in memory**, not distributed big data.

Algorithm Choice: Governance and Computing Power

- ➊ **Governance** may drive algorithm choice: regulated environments needing explanations (financial services) avoid opaque models like neural networks and favor **decision trees**.
- ➋ Often the trade-off is on **cost**, not performance: simpler techniques need more costly **manual feature engineering**.
- ➌ **Computing power**: ML development is **inversely proportional** to the cost of computing power.
- ➍ New algorithms arose as compute became available; random forest and gradient boosting were expensive 20 years ago.
- ➎ They brought **ease of use**, lowering development cost and enabling new use cases.
- ➏ Most algorithms still operate with **all training data in memory**, not distributed big data.

Algorithm Choice: Governance and Computing Power

- ➊ **Governance** may drive algorithm choice: regulated environments needing explanations (financial services) avoid opaque models like neural networks and favor **decision trees**.
- ➋ Often the trade-off is on **cost**, not performance: simpler techniques need more costly **manual feature engineering**.
- ➌ **Computing power**: ML development is **inversely proportional** to the cost of computing power.
- ➍ New algorithms arose as compute became available; random forest and gradient boosting were expensive 20 years ago.
- ➎ They brought **ease of use**, lowering development cost and enabling new use cases.
- ➏ Most algorithms still operate with **all training data in memory**, not distributed big data.

Data Exploration

- ❶ Data scientists must first grasp **what the data looks like**; a model is only as good as its training data.
- ❷ Issues: **incompleteness, inaccuracy, inconsistency**.
- ❸ Exploration processes include:
 - ▶ Documenting how data was collected and assumptions made.
 - ▶ Summary statistics: column domains, missing values, mistakes, outliers.
 - ▶ Examining data distributions.
 - ▶ Cleaning, filling, reshaping, filtering, clipping, sampling.
 - ▶ Checking correlations, statistical tests on subpopulations, fitting distributions.
 - ▶ Comparing with other data or literature.
- ❹ **Domain knowledge** is essential, e.g., industrial sensor data needs equipment expertise to spot abnormal outliers.

Data Exploration

- ❶ Data scientists must first grasp **what the data looks like**; a model is only as good as its training data.
- ❷ Issues: **incompleteness, inaccuracy, inconsistency**.
- ❸ Exploration processes include:
 - ▶ Documenting how data was collected and assumptions made.
 - ▶ Summary statistics: column domains, missing values, mistakes, outliers.
 - ▶ Examining data distributions.
 - ▶ Cleaning, filling, reshaping, filtering, clipping, sampling.
 - ▶ Checking correlations, statistical tests on subpopulations, fitting distributions.
 - ▶ Comparing with other data or literature.
- ❹ **Domain knowledge** is essential, e.g., industrial sensor data needs equipment expertise to spot abnormal outliers.

Data Exploration

- ❶ Data scientists must first grasp **what the data looks like**; a model is only as good as its training data.
- ❷ Issues: **incompleteness, inaccuracy, inconsistency**.
- ❸ Exploration processes include:
 - ▶ Documenting how data was collected and assumptions made.
 - ▶ Summary statistics: column domains, missing values, mistakes, outliers.
 - ▶ Examining data distributions.
 - ▶ Cleaning, filling, reshaping, filtering, clipping, sampling.
 - ▶ Checking correlations, statistical tests on subpopulations, fitting distributions.
 - ▶ Comparing with other data or literature.
- ❹ **Domain knowledge is essential**, e.g., industrial sensor data needs equipment expertise to spot abnormal outliers.

Data Exploration

- ❶ Data scientists must first grasp **what the data looks like**; a model is only as good as its training data.
- ❷ Issues: **incompleteness, inaccuracy, inconsistency**.
- ❸ Exploration processes include:
 - ▶ Documenting how data was collected and assumptions made.
 - ▶ Summary statistics: column domains, missing values, mistakes, outliers.
 - ▶ Examining data distributions.
 - ▶ Cleaning, filling, reshaping, filtering, clipping, sampling.
 - ▶ Checking correlations, statistical tests on subpopulations, fitting distributions.
 - ▶ Comparing with other data or literature.
- ❹ **Domain knowledge** is essential, e.g., industrial sensor data needs equipment expertise to spot abnormal outliers.

Feature Engineering and Selection

- ❶ **Features** are how data is presented to a model, informing it on things it may not infer by itself.

Category	Description
Derivatives	Infer new information from existing information — e.g., what day of the week is this date?
Enrichment	Add new external information — e.g., is this day a public holiday?
Encoding	Present the same information differently — e.g., day of week or weekday versus weekend.
Combination	Link features together — e.g., backlog size weighted by the complexity of its items.

- ❷ **Process duration example:** From a date, derive day of week or distance to the next public holiday; consider location-specific business calendars.
- ❸ **House price example:** Use average income and population density instead of zip code to generalize better.

Feature Engineering and Selection

- ❶ **Features** are how data is presented to a model, informing it on things it may not infer by itself.

Category	Description
Derivatives	Infer new information from existing information — e.g., what day of the week is this date?
Enrichment	Add new external information — e.g., is this day a public holiday?
Encoding	Present the same information differently — e.g., day of week or weekday versus weekend.
Combination	Link features together — e.g., backlog size weighted by the complexity of its items.

- ❷ **Process duration example:** From a date, derive day of week or distance to the next public holiday; consider location-specific business calendars.
- ❸ **House price example:** Use average income and population density instead of zip code to generalize better.

Feature Engineering and Selection

- ❶ **Features** are how data is presented to a model, informing it on things it may not infer by itself.

Category	Description
Derivatives	Infer new information from existing information — e.g., what day of the week is this date?
Enrichment	Add new external information — e.g., is this day a public holiday?
Encoding	Present the same information differently — e.g., day of week or weekday versus weekend.
Combination	Link features together — e.g., backlog size weighted by the complexity of its items.

- ❷ **Process duration example:** From a date, derive day of week or distance to the next public holiday; consider location-specific business calendars.
- ❸ **House price example:** Use average income and population density instead of zip code to generalize better.

Feature Engineering Techniques

- ❶ A market exists for **complementary data** beyond open data; enrichment services save time and effort.
- ❷ **Impact coding:** Replace a modality by the **average target value** for that modality (at some information loss).
- ❸ Most algorithms require a **table of numbers**: each row a sample, all from the same dataset.
- ❹ **One-hot encoding:** A feature with values Raspberry, Blueberry, Strawberry becomes three yes/no features.
- ❺ **Embeddings:** Deep learning transforms text and images into tables of numbers, enabling transfer learning.

Feature Engineering Techniques

- ❶ A market exists for **complementary data** beyond open data; enrichment services save time and effort.
- ❷ **Impact coding:** Replace a modality by the **average target value** for that modality (at some information loss).
- ❸ Most algorithms require a **table of numbers**: each row a sample, all from the same dataset.
- ❹ **One-hot encoding:** A feature with values Raspberry, Blueberry, Strawberry becomes three yes/no features.
- ❺ **Embeddings:** Deep learning transforms text and images into tables of numbers, enabling transfer learning.

Feature Engineering Techniques

- ❶ A market exists for **complementary data** beyond open data; enrichment services save time and effort.
- ❷ **Impact coding:** Replace a modality by the **average target value** for that modality (at some information loss).
- ❸ Most algorithms require a **table of numbers**: each row a sample, all from the same dataset.
- ❹ **One-hot encoding:** A feature with values Raspberry, Blueberry, Strawberry becomes three yes/no features.
- ❺ **Embeddings:** Deep learning transforms text and images into tables of numbers, enabling transfer learning.

Feature Engineering Techniques

- ❶ A market exists for **complementary data** beyond open data; enrichment services save time and effort.
- ❷ **Impact coding:** Replace a modality by the **average target value** for that modality (at some information loss).
- ❸ Most algorithms require a **table of numbers**: each row a sample, all from the same dataset.
- ❹ **One-hot encoding:** A feature with values Raspberry, Blueberry, Strawberry becomes three yes/no features.
- ❺ **Embeddings:** Deep learning transforms text and images into tables of numbers, enabling transfer learning.

Feature Engineering Techniques

- ❶ A market exists for **complementary data** beyond open data; enrichment services save time and effort.
- ❷ **Impact coding:** Replace a modality by the **average target value** for that modality (at some information loss).
- ❸ Most algorithms require a **table of numbers**: each row a sample, all from the same dataset.
- ❹ **One-hot encoding:** A feature with values Raspberry, Blueberry, Strawberry becomes three yes/no features.
- ❺ **Embeddings:** Deep learning transforms text and images into tables of numbers, enabling **transfer learning**.

Transfer Learning

Transfer Learning

The technique of using information gained from solving one problem in solving a different problem.

- ❶ Significantly **accelerates learning** of second or subsequent tasks.
- ❷ Very popular in **deep learning**, where training resources can be enormous.
- ❸ **Example:** A model trained on images without forks may still give a useful embedding for a fork detector, since it learned to detect similar **human-made objects**.

Transfer Learning

Transfer Learning

The technique of using information gained from solving one problem in solving a different problem.

- ❶ Significantly **accelerates learning** of second or subsequent tasks.
- ❷ Very popular in **deep learning**, where training resources can be enormous.
- ❸ **Example:** A model trained on images without forks may still give a useful embedding for a fork detector, since it learned to detect similar **human-made objects**.

Transfer Learning

Transfer Learning

The technique of using information gained from solving one problem in solving a different problem.

- ① Significantly **accelerates learning** of second or subsequent tasks.
- ② Very popular in **deep learning**, where training resources can be enormous.
- ③ **Example:** A model trained on images without forks may still give a useful embedding for a fork detector, since it learned to detect similar **human-made objects**.

How Feature Selection Impacts MLOps Strategy

- ❶ More features may give more accuracy, more fairness, or compensate for missing information.
- ❷ **Downsides affecting MLOps:**
 - ▶ The model becomes more **expensive to compute**.
 - ▶ More features need more inputs and **more maintenance**.
 - ▶ Some loss of **stability**.
 - ▶ The number of features can raise **privacy concerns**.
- ❸ **Automated feature selection** uses heuristics: correlation with the target, or training a simple model on a subset to find the strongest predictors.
- ❹ Tools like **Auto-sklearn** or **AutoML** cross-reference features against the target to keep the most useful ones.

How Feature Selection Impacts MLOps Strategy

- ❶ More features may give more accuracy, more fairness, or compensate for missing information.
- ❷ **Downsides** affecting MLOps:
 - ▶ The model becomes more **expensive to compute**.
 - ▶ More features need more inputs and **more maintenance**.
 - ▶ Some loss of **stability**.
 - ▶ The number of features can raise **privacy concerns**.
- ❸ **Automated feature selection** uses heuristics: correlation with the target, or training a simple model on a subset to find the strongest predictors.
- ❹ Tools like **Auto-sklearn** or **AutoML** cross-reference features against the target to keep the most useful ones.

How Feature Selection Impacts MLOps Strategy

- ❶ More features may give more accuracy, more fairness, or compensate for missing information.
- ❷ **Downsides** affecting MLOps:
 - ▶ The model becomes more **expensive to compute**.
 - ▶ More features need more inputs and **more maintenance**.
 - ▶ Some loss of **stability**.
 - ▶ The number of features can raise **privacy concerns**.
- ❸ **Automated feature selection** uses heuristics: correlation with the target, or training a simple model on a subset to find the strongest predictors.
- ❹ Tools like Auto-sklearn or AutoML cross-reference features against the target to keep the most useful ones.

How Feature Selection Impacts MLOps Strategy

- ❶ More features may give more accuracy, more fairness, or compensate for missing information.
- ❷ **Downsides** affecting MLOps:
 - ▶ The model becomes more **expensive to compute**.
 - ▶ More features need more inputs and **more maintenance**.
 - ▶ Some loss of **stability**.
 - ▶ The number of features can raise **privacy concerns**.
- ❸ **Automated feature selection** uses heuristics: correlation with the target, or training a simple model on a subset to find the strongest predictors.
- ❹ Tools like **Auto-sklearn** or AutoML cross-reference features against the target to keep the most useful ones.

Human Choices and Feature Stores

- ❶ Some choices need **human intervention**, e.g., collecting additional information.
- ❷ **Business-friendly features** improve performance, ease adoption, simplify explanations, and reduce **modeling debt**.
- ❸ Trade-offs exist between time spent understanding the model, expected value, and risk.
- ❹ **Feature stores**: Central repositories of features for business entities, created once and **reused**.
- ❺ They combine an **offline** part (slower, more powerful) and an **online** part (fast, real-time), kept consistent.
- ❻ ML is often the “**high-interest credit card of technical debt**”; feature stores are one way to reverse this.

Human Choices and Feature Stores

- ❶ Some choices need **human intervention**, e.g., collecting additional information.
- ❷ **Business-friendly features** improve performance, ease adoption, simplify explanations, and reduce **modeling debt**.
- ❸ Trade-offs exist between time spent understanding the model, expected value, and risk.
- ❹ **Feature stores**: Central repositories of features for business entities, created once and reused.
- ❺ They combine an **offline** part (slower, more powerful) and an **online** part (fast, real-time), kept consistent.
- ❻ ML is often the “**high-interest credit card of technical debt**”; feature stores are one way to reverse this.

Human Choices and Feature Stores

- ❶ Some choices need **human intervention**, e.g., collecting additional information.
- ❷ **Business-friendly features** improve performance, ease adoption, simplify explanations, and reduce **modeling debt**.
- ❸ Trade-offs exist between time spent understanding the model, expected value, and risk.
- ❹ **Feature stores**: Central repositories of features for business entities, created once and reused.
- ❺ They combine an **offline** part (slower, more powerful) and an **online** part (fast, real-time), kept consistent.
- ❻ ML is often the “**high-interest credit card of technical debt**”; feature stores are one way to reverse this.

Human Choices and Feature Stores

- ❶ Some choices need **human intervention**, e.g., collecting additional information.
- ❷ **Business-friendly features** improve performance, ease adoption, simplify explanations, and reduce **modeling debt**.
- ❸ Trade-offs exist between time spent understanding the model, expected value, and risk.
- ❹ **Feature stores**: Central repositories of features for business entities, created once and **reused**.
 - ❶ They combine an **offline** part (slower, more powerful) and an **online** part (fast, real-time), kept consistent.
 - ❷ ML is often the “high-interest credit card of technical debt”; feature stores are one way to reverse this.

Human Choices and Feature Stores

- ❶ Some choices need **human intervention**, e.g., collecting additional information.
- ❷ **Business-friendly features** improve performance, ease adoption, simplify explanations, and reduce **modeling debt**.
- ❸ Trade-offs exist between time spent understanding the model, expected value, and risk.
- ❹ **Feature stores**: Central repositories of features for business entities, created once and **reused**.
- ❺ They combine an **offline** part (slower, more powerful) and an **online** part (fast, real-time), kept consistent.
- ❻ ML is often the “high-interest credit card of technical debt”; feature stores are one way to reverse this.

Human Choices and Feature Stores

- ❶ Some choices need **human intervention**, e.g., collecting additional information.
- ❷ **Business-friendly features** improve performance, ease adoption, simplify explanations, and reduce **modeling debt**.
- ❸ Trade-offs exist between time spent understanding the model, expected value, and risk.
- ❹ **Feature stores**: Central repositories of features for business entities, created once and **reused**.
- ❺ They combine an **offline** part (slower, more powerful) and an **online** part (fast, real-time), kept consistent.
- ❻ ML is often the “**high-interest credit card of technical debt**”; feature stores are one way to reverse this.

Experimentation

- ❶ Experimentation occurs **throughout** model development; important decisions are justified by experiments or research.
- ❷ It can take many shapes: full predictive models, statistical tests, charting data.
- ❸ Goals of experimentation:
 - ▶ Assess how useful a model can be built given the Table 4-1 components.
 - ▶ Find the best modeling parameters (algorithms, hyperparameters, preprocessing).
 - ▶ Tune the **bias/variance trade-off** for a given training cost.
 - ▶ Balance model improvement against computation cost — how good is **good enough**?

Experimentation

- ❶ Experimentation occurs **throughout** model development; important decisions are justified by experiments or research.
- ❷ It can take many shapes: full predictive models, statistical tests, charting data.
- ❸ **Goals of experimentation:**
 - ▶ Assess how useful a model can be built given the Table 4-1 components.
 - ▶ Find the best modeling parameters (algorithms, hyperparameters, preprocessing).
 - ▶ Tune the **bias/variance trade-off** for a given training cost.
 - ▶ Balance model improvement against computation cost — how good is good enough?

Experimentation

- ❶ Experimentation occurs **throughout** model development; important decisions are justified by experiments or research.
- ❷ It can take many shapes: full predictive models, statistical tests, charting data.
- ❸ **Goals of experimentation:**
 - ▶ Assess how useful a model can be built given the Table 4-1 components.
 - ▶ Find the best modeling parameters (algorithms, hyperparameters, preprocessing).
 - ▶ Tune the **bias/variance trade-off** for a given training cost.
 - ▶ Balance model improvement against computation cost — how good is **good enough**?

Bias and Variance

- ❶ **High bias (underfitting):** Fails to discover rules that could be learned from training data, due to reductive assumptions making it **overly simplistic**.
- ❷ **High variance (overfitting):** Sees patterns in noise and predicts every variation, becoming complex and **not generalizing** beyond training data.
- ❸ Experimentation aims to find the right balance between the two for the problem at hand.

Bias and Variance

- ❶ **High bias (underfitting):** Fails to discover rules that could be learned from training data, due to reductive assumptions making it **overly simplistic**.
- ❷ **High variance (overfitting):** Sees patterns in noise and predicts every variation, becoming complex and **not generalizing** beyond training data.
- ❸ Experimentation aims to find the right balance between the two for the problem at hand.

Bias and Variance

- ❶ **High bias (underfitting):** Fails to discover rules that could be learned from training data, due to reductive assumptions making it **overly simplistic**.
- ❷ **High variance (overfitting):** Sees patterns in noise and predicts every variation, becoming complex and **not generalizing** beyond training data.
- ❸ Experimentation aims to find the right balance between the two for the problem at hand.

Experimentation Tools and Budget

- ❶ Tools test possibilities **semiautomatically**: define the search space from prior knowledge and constraints (computation, budget).
- ❷ **Stratified modeling**: Train one model per store (e.g., demand forecasting) instead of one model for all stores.
- ❸ Trying all combinations becomes **intractable**; define a **time/computation budget** and an acceptability threshold.
- ❹ The process may need repeating whenever data or problem constraints **change significantly**.
- ❺ All experiments, assumptions, and conclusions may need to be **rerun and reexamined**.
- ❻ Platforms automate workflows, preserve processing for repeatability, and support **version control and experimental branching**.

Experimentation Tools and Budget

- ❶ Tools test possibilities **semiautomatically**: define the search space from prior knowledge and constraints (computation, budget).
- ❷ **Stratified modeling**: Train one model per store (e.g., demand forecasting) instead of one model for all stores.
- ❸ Trying all combinations becomes **intractable**; define a **time/computation budget** and an **acceptability threshold**.
- ❹ The process may need repeating whenever data or problem constraints **change significantly**.
- ❺ All experiments, assumptions, and conclusions may need to be **rerun and reexamined**.
- ❻ Platforms automate workflows, preserve processing for repeatability, and support **version control and experimental branching**.

Experimentation Tools and Budget

- ❶ Tools test possibilities **semiautomatically**: define the search space from prior knowledge and constraints (computation, budget).
- ❷ **Stratified modeling**: Train one model per store (e.g., demand forecasting) instead of one model for all stores.
- ❸ Trying all combinations becomes **intractable**; define a **time/computation budget** and an acceptability threshold.
- ❹ The process may need repeating whenever data or problem constraints **change significantly**.
- ❺ All experiments, assumptions, and conclusions may need to be **rerun and reexamined**.
- ❻ Platforms automate workflows, preserve processing for repeatability, and support **version control and experimental branching**.

Experimentation Tools and Budget

- ❶ Tools test possibilities **semiautomatically**: define the search space from prior knowledge and constraints (computation, budget).
- ❷ **Stratified modeling**: Train one model per store (e.g., demand forecasting) instead of one model for all stores.
- ❸ Trying all combinations becomes **intractable**; define a **time/computation budget** and an acceptability threshold.
- ❹ The process may need repeating whenever data or problem constraints **change significantly**.
- ❺ All experiments, assumptions, and conclusions may need to be rerun and reexamined.
- ❻ Platforms automate workflows, preserve processing for repeatability, and support version control and experimental branching.

Experimentation Tools and Budget

- ❶ Tools test possibilities **semiautomatically**: define the search space from prior knowledge and constraints (computation, budget).
- ❷ **Stratified modeling**: Train one model per store (e.g., demand forecasting) instead of one model for all stores.
- ❸ Trying all combinations becomes **intractable**; define a **time/computation budget** and an acceptability threshold.
- ❹ The process may need repeating whenever data or problem constraints **change significantly**.
- ❺ All experiments, assumptions, and conclusions may need to be **rerun and reexamined**.
- ❻ Platforms automate workflows, preserve processing for repeatability, and support version control and experimental branching.

Experimentation Tools and Budget

- ❶ Tools test possibilities **semiautomatically**: define the search space from prior knowledge and constraints (computation, budget).
- ❷ **Stratified modeling**: Train one model per store (e.g., demand forecasting) instead of one model for all stores.
- ❸ Trying all combinations becomes **intractable**; define a **time/computation budget** and an acceptability threshold.
- ❹ The process may need repeating whenever data or problem constraints **change significantly**.
- ❺ All experiments, assumptions, and conclusions may need to be **rerun and reexamined**.
- ❻ Platforms automate workflows, preserve processing for repeatability, and support **version control and experimental branching**.

Evaluating and Comparing Models

- ❶ George E. P. Box: **all models are wrong, but some are useful.**
- ❷ A model should be “**good enough to be useful**” while avoiding the **uncanny valley** (good overall, catastrophic for a subset such as an underrepresented population).
- ❸ Evaluate in context and **compare with what existed before** (previous model or rules-based process).
- ❹ A disappointing absolute performance can still help, e.g., a slightly better demand forecast may give **huge cost savings**.
- ❺ A **perfect score is suspicious**: it may indicate **data leakage** or **overfitting**.

Evaluating and Comparing Models

- ❶ George E. P. Box: **all models are wrong, but some are useful.**
- ❷ A model should be “**good enough to be useful**” while avoiding the **uncanny valley** (good overall, catastrophic for a subset such as an underrepresented population).
- ❸ Evaluate in context and compare with what existed before (previous model or rules-based process).
- ❹ A disappointing absolute performance can still help, e.g., a slightly better demand forecast may give huge cost savings.
- ❺ A perfect score is suspicious: it may indicate data leakage or overfitting.

Evaluating and Comparing Models

- ❶ George E. P. Box: **all models are wrong, but some are useful.**
- ❷ A model should be “**good enough to be useful**” while avoiding the **uncanny valley** (good overall, catastrophic for a subset such as an underrepresented population).
- ❸ Evaluate in context and **compare with what existed before** (previous model or rules-based process).
- ❹ A disappointing absolute performance can still help, e.g., a slightly better demand forecast may give huge cost savings.
- ❺ A perfect score is suspicious: it may indicate data leakage or overfitting.

Evaluating and Comparing Models

- ❶ George E. P. Box: **all models are wrong, but some are useful.**
- ❷ A model should be “**good enough to be useful**” while avoiding the **uncanny valley** (good overall, catastrophic for a subset such as an underrepresented population).
- ❸ Evaluate in context and **compare with what existed before** (previous model or rules-based process).
- ❹ A disappointing absolute performance can still help, e.g., a slightly better demand forecast may give **huge cost savings**.
- ❺ A perfect score is suspicious: it may indicate data leakage or overfitting.

Evaluating and Comparing Models

- ❶ George E. P. Box: **all models are wrong, but some are useful.**
- ❷ A model should be “**good enough to be useful**” while avoiding the **uncanny valley** (good overall, catastrophic for a subset such as an underrepresented population).
- ❸ Evaluate in context and **compare with what existed before** (previous model or rules-based process).
- ❹ A disappointing absolute performance can still help, e.g., a slightly better demand forecast may give **huge cost savings**.
- ❺ A **perfect score is suspicious**: it may indicate **data leakage** or **overfitting**.

Choosing Evaluation Metrics

- ❶ The metric chosen can lead to **very different models** (cobra effect).
- ❷ Accuracy is rarely the best fit for **unbalanced classes**.
- ❸ **Example:** If the positive class occurs 5% of the time, always predicting negative gives 95% accuracy but is **utterly useless**.
- ❹ There is **no one-size-fits-all** metric.
- ❺ Understand the metric's limits and trade-offs (**mathematics side**) and its impact on optimization (**business side**).

Choosing Evaluation Metrics

- ❶ The metric chosen can lead to **very different models** (cobra effect).
- ❷ **Accuracy** is rarely the best fit for **unbalanced classes**.
- ❸ **Example:** If the positive class occurs 5% of the time, always predicting negative gives 95% accuracy but is **utterly useless**.
- ❹ There is **no one-size-fits-all** metric.
- ❺ Understand the metric's limits and trade-offs (**mathematics side**) and its impact on optimization (**business side**).

Choosing Evaluation Metrics

- ❶ The metric chosen can lead to **very different models** (cobra effect).
- ❷ **Accuracy** is rarely the best fit for **unbalanced classes**.
- ❸ **Example:** If the positive class occurs 5% of the time, always predicting negative gives 95% accuracy but is **utterly useless**.
- ❹ There is **no one-size-fits-all** metric.
- ❺ Understand the metric's limits and trade-offs (**mathematics side**) and its impact on optimization (**business side**).

Choosing Evaluation Metrics

- ❶ The metric chosen can lead to **very different models** (cobra effect).
- ❷ **Accuracy** is rarely the best fit for **unbalanced classes**.
- ❸ **Example:** If the positive class occurs 5% of the time, always predicting negative gives 95% accuracy but is **utterly useless**.
- ❹ There is **no one-size-fits-all** metric.
- ❺ Understand the metric's limits and trade-offs (**mathematics side**) and its impact on optimization (**business side**).

Choosing Evaluation Metrics

- ❶ The metric chosen can lead to **very different models** (cobra effect).
- ❷ **Accuracy** is rarely the best fit for **unbalanced classes**.
- ❸ **Example:** If the positive class occurs 5% of the time, always predicting negative gives 95% accuracy but is **utterly useless**.
- ❹ There is **no one-size-fits-all** metric.
- ❺ Understand the metric's limits and trade-offs (**mathematics side**) and its impact on optimization (**business side**).

Cross-Testing and Cross-Validation

- ❶ **Cross-testing:** Evaluate the metric on a **holdout dataset** not used for training.
- ❷ Data can be held out at multiple steps (metric evaluation, hyperparameter optimization).
- ❸ **k-fold cross-validation:** Rotate held-out parts to train and evaluate multiple times; costs training time but shows **metric stability**.
- ❹ The holdout set can be the **most recent records**, giving more realistic estimates for future predictions.
- ❺ Recent-data holdout also helps assess whether data **drifted** between training and holdout.

Cross-Testing and Cross-Validation

- ❶ **Cross-testing:** Evaluate the metric on a **holdout dataset** not used for training.
- ❷ Data can be held out at multiple steps (metric evaluation, hyperparameter optimization).
- ❸ **k-fold cross-validation:** Rotate held-out parts to train and evaluate multiple times; costs training time but shows **metric stability**.
- ❹ The holdout set can be the **most recent records**, giving more realistic estimates for future predictions.
- ❺ Recent-data holdout also helps assess whether data **drifted** between training and holdout.

Cross-Testing and Cross-Validation

- ❶ **Cross-testing:** Evaluate the metric on a **holdout dataset** not used for training.
- ❷ Data can be held out at multiple steps (metric evaluation, hyperparameter optimization).
- ❸ **k-fold cross-validation:** Rotate held-out parts to train and evaluate multiple times; costs training time but shows **metric stability**.
- ❹ The holdout set can be the **most recent records**, giving more realistic estimates for future predictions.
- ❺ Recent-data holdout also helps assess whether data **drifted** between training and holdout.

Cross-Testing and Cross-Validation

- ❶ **Cross-testing:** Evaluate the metric on a **holdout dataset** not used for training.
- ❷ Data can be held out at multiple steps (metric evaluation, hyperparameter optimization).
- ❸ **k-fold cross-validation:** Rotate held-out parts to train and evaluate multiple times; costs training time but shows **metric stability**.
- ❹ The holdout set can be the **most recent records**, giving more realistic estimates for future predictions.
- ❺ Recent-data holdout also helps assess whether data **drifted** between training and holdout.

Cross-Testing and Cross-Validation

- ❶ **Cross-testing:** Evaluate the metric on a **holdout dataset** not used for training.
- ❷ Data can be held out at multiple steps (metric evaluation, hyperparameter optimization).
- ❸ **k-fold cross-validation:** Rotate held-out parts to train and evaluate multiple times; costs training time but shows **metric stability**.
- ❹ The holdout set can be the **most recent records**, giving more realistic estimates for future predictions.
- ❺ Recent-data holdout also helps assess whether data **drifted** between training and holdout.

Dataset Split for Model Evaluation

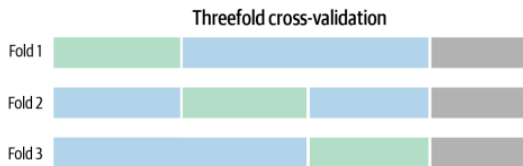


Figure: An example of dataset split for model evaluation

- ❶ **Gray block:** The **test (holdout) dataset**, set aside for final evaluation.
- ❷ The remaining data is split into **three parts** (threefold cross-validation).
- ❸ For each hyperparameter combination, the model is trained on the **blue** parts and validated on the **green** part — three times.
- ❹ Blue/green datasets are used with **all** combinations.
- ❺ The gray dataset is used **only once**, with the best hyperparameter combination.

Dataset Split for Model Evaluation

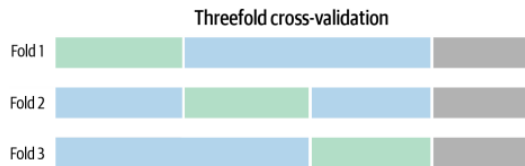


Figure: An example of dataset split for model evaluation

- ❶ **Gray block:** The **test (holdout) dataset**, set aside for final evaluation.
- ❷ The remaining data is split into **three parts** (threefold cross-validation).
- ❸ For each hyperparameter combination, the model is trained on the **blue** parts and validated on the **green** part — three times.
- ❹ Blue/green datasets are used with **all** combinations.
- ❺ The gray dataset is used **only once**, with the best hyperparameter combination.

Dataset Split for Model Evaluation

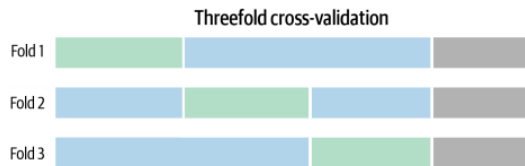


Figure: An example of dataset split for model evaluation

- ❶ **Gray block:** The **test (holdout) dataset**, set aside for final evaluation.
- ❷ The remaining data is split into **three parts** (threefold cross-validation).
- ❸ For each hyperparameter combination, the model is trained on the **blue** parts and validated on the **green** part — three times.
- ❹ Blue/green datasets are used with **all** combinations.
- ❺ The gray dataset is used **only once**, with the best hyperparameter combination.

Dataset Split for Model Evaluation

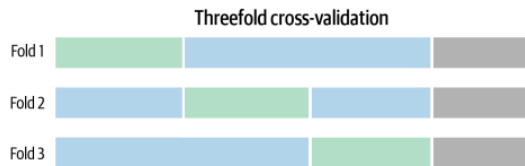


Figure: An example of dataset split for model evaluation

- ① **Gray block:** The **test (holdout) dataset**, set aside for final evaluation.
- ② The remaining data is split into **three parts** (threefold cross-validation).
- ③ For each hyperparameter combination, the model is trained on the **blue** parts and validated on the **green** part — three times.
- ④ Blue/green datasets are used with **all** combinations.
- ⑤ The gray dataset is used **only once**, with the best hyperparameter combination.

Dataset Split for Model Evaluation

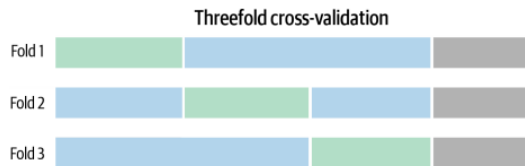


Figure: An example of dataset split for model evaluation

- ❶ **Gray block:** The **test (holdout) dataset**, set aside for final evaluation.
- ❷ The remaining data is split into **three parts** (threefold cross-validation).
- ❸ For each hyperparameter combination, the model is trained on the **blue** parts and validated on the **green** part — three times.
- ❹ Blue/green datasets are used with **all** combinations.
- ❺ The gray dataset is used **only once**, with the best hyperparameter combination.

Comparing Retrained Models

- ❶ Models are often **periodically retrained** with the same algorithms, hyperparameters, and features on more recent data.
- ❷ Next, **compare** the new version against the previous one.
- ❸ Also verify that **previous assumptions still hold**:
 - ▶ The problem has not fundamentally shifted.
 - ▶ Earlier modeling choices still fit the data.
- ❹ This is part of **performance and drift monitoring** (Chapter 7).

Comparing Retrained Models

- ❶ Models are often **periodically retrained** with the same algorithms, hyperparameters, and features on more recent data.
- ❷ Next, **compare** the new version against the previous one.
- ❸ Also verify that **previous assumptions still hold**:
 - ▶ The problem has not fundamentally shifted.
 - ▶ Earlier modeling choices still fit the data.
- ❹ This is part of **performance and drift monitoring** (Chapter 7).

Comparing Retrained Models

- ❶ Models are often **periodically retrained** with the same algorithms, hyperparameters, and features on more recent data.
 - ❷ Next, **compare** the new version against the previous one.
 - ❸ Also verify that **previous assumptions still hold**:
 - ▶ The problem has not fundamentally shifted.
 - ▶ Earlier modeling choices still fit the data.
- ❹ This is part of **performance and drift monitoring** (Chapter 7).

Comparing Retrained Models

- ❶ Models are often **periodically retrained** with the same algorithms, hyperparameters, and features on more recent data.
- ❷ Next, **compare** the new version against the previous one.
- ❸ Also verify that **previous assumptions still hold**:
 - ▶ The problem has not fundamentally shifted.
 - ▶ Earlier modeling choices still fit the data.
- ❹ This is part of **performance and drift monitoring** (Chapter 7).

Cross-Checking Model Behavior

- ➊ Beyond raw metrics, understand **how the model behaves**, in proportion to its impact.
- ➋ Ensure the model is not **actively harmful**: predicting that all patients need a doctor scores high on prevention but fails resource allocation.
- ➌ Reasonable steps:
 - ▶ Cross-check **different metrics**, not only the optimized one.
 - ▶ Check reactions to different inputs (plot average prediction vs. input values).
 - ▶ Compare behavior across **subpopulations** (e.g., error rate for males vs. females).
- ➍ These analyses show **correlation, not causality**; use with care for what-if analysis on unseen values.
- ➎ Comparisons need the **full environment** (tooling, data): same settings/different data for drift, same data/different settings for performance.

Cross-Checking Model Behavior

- ➊ Beyond raw metrics, understand **how the model behaves**, in proportion to its impact.
- ➋ Ensure the model is not **actively harmful**: predicting that all patients need a doctor scores high on prevention but fails resource allocation.
- ➌ Reasonable steps:
 - ▶ Cross-check **different metrics**, not only the optimized one.
 - ▶ Check reactions to different inputs (plot average prediction vs. input values).
 - ▶ Compare behavior across **subpopulations** (e.g., error rate for males vs. females).
- ➍ These analyses show **correlation, not causality**; use with care for what-if analysis on unseen values.
- ➎ Comparisons need the **full environment** (tooling, data): same settings/different data for drift, same data/different settings for performance.

Cross-Checking Model Behavior

- ❶ Beyond raw metrics, understand **how the model behaves**, in proportion to its impact.
- ❷ Ensure the model is not **actively harmful**: predicting that all patients need a doctor scores high on prevention but fails resource allocation.
- ❸ **Reasonable steps:**
 - ▶ Cross-check **different metrics**, not only the optimized one.
 - ▶ Check reactions to different inputs (plot average prediction vs. input values).
 - ▶ Compare behavior across **subpopulations** (e.g., error rate for males vs. females).
- ❹ These analyses show **correlation, not causality**; use with care for what-if analysis on unseen values.
- ❺ Comparisons need the **full environment** (tooling, data): same settings/different data for drift, same data/different settings for performance.

Cross-Checking Model Behavior

- ❶ Beyond raw metrics, understand **how the model behaves**, in proportion to its impact.
- ❷ Ensure the model is not **actively harmful**: predicting that all patients need a doctor scores high on prevention but fails resource allocation.
- ❸ **Reasonable steps:**
 - ▶ Cross-check **different metrics**, not only the optimized one.
 - ▶ Check reactions to different inputs (plot average prediction vs. input values).
 - ▶ Compare behavior across **subpopulations** (e.g., error rate for males vs. females).
- ❹ These analyses show **correlation, not causality**; use with care for what-if analysis on unseen values.
- ❺ Comparisons need the **full environment** (tooling, data): same settings/different data for drift, same data/different settings for performance.

Cross-Checking Model Behavior

- ❶ Beyond raw metrics, understand **how the model behaves**, in proportion to its impact.
- ❷ Ensure the model is not **actively harmful**: predicting that all patients need a doctor scores high on prevention but fails resource allocation.
- ❸ **Reasonable steps:**
 - ▶ Cross-check **different metrics**, not only the optimized one.
 - ▶ Check reactions to different inputs (plot average prediction vs. input values).
 - ▶ Compare behavior across **subpopulations** (e.g., error rate for males vs. females).
- ❹ These analyses show **correlation, not causality**; use with care for what-if analysis on unseen values.
- ❺ Comparisons need the **full environment** (tooling, data): same settings/different data for drift, same data/different settings for performance.

Impact of Responsible AI on Modeling

- ❶ Individual predictions may need to be **explainable**: which features pushed the prediction one way or another.
- ❷ **Shapley value**: Average marginal contribution of a feature value across all possible coalitions.
- ❸ **Individual Conditional Expectation (ICE)**: Shows the dependence between the target function and features of interest.
- ❹ **Example**: A hormone level may indicate a health issue in general, but for a **pregnant woman** it indicates no risk.
- ❺ Some legal frameworks mandate explainability for decisions affecting humans (e.g., **loan denial**).

Impact of Responsible AI on Modeling

- ❶ Individual predictions may need to be **explainable**: which features pushed the prediction one way or another.
- ❷ **Shapley value**: Average marginal contribution of a feature value across all possible coalitions.
- ❸ **Individual Conditional Expectation (ICE)**: Shows the dependence between the target function and features of interest.
- ❹ **Example**: A hormone level may indicate a health issue in general, but for a **pregnant woman** it indicates no risk.
- ❺ Some legal frameworks mandate explainability for decisions affecting humans (e.g., **loan denial**).

Impact of Responsible AI on Modeling

- ❶ Individual predictions may need to be **explainable**: which features pushed the prediction one way or another.
- ❷ **Shapley value**: Average marginal contribution of a feature value across all possible coalitions.
- ❸ **Individual Conditional Expectation (ICE)**: Shows the dependence between the target function and features of interest.
- ❹ **Example**: A hormone level may indicate a health issue in general, but for a **pregnant woman** it indicates no risk.
- ❺ Some legal frameworks mandate explainability for decisions affecting humans (e.g., **loan denial**).

Impact of Responsible AI on Modeling

- ❶ Individual predictions may need to be **explainable**: which features pushed the prediction one way or another.
- ❷ **Shapley value**: Average marginal contribution of a feature value across all possible coalitions.
- ❸ **Individual Conditional Expectation (ICE)**: Shows the dependence between the target function and features of interest.
- ❹ **Example**: A hormone level may indicate a health issue in general, but for a **pregnant woman** it indicates no risk.
- ❺ Some legal frameworks mandate explainability for decisions affecting humans (e.g., loan denial).

Impact of Responsible AI on Modeling

- ❶ Individual predictions may need to be **explainable**: which features pushed the prediction one way or another.
- ❷ **Shapley value**: Average marginal contribution of a feature value across all possible coalitions.
- ❸ **Individual Conditional Expectation (ICE)**: Shows the dependence between the target function and features of interest.
- ❹ **Example**: A hormone level may indicate a health issue in general, but for a **pregnant woman** it indicates no risk.
- ❺ Some legal frameworks mandate explainability for decisions affecting humans (e.g., **loan denial**).

Dimensions of Explainability

- ❶ Deep learning networks are called **black-box models** due to complexity.
- ❷ Their formula is conceptually simple but **very large**, impossible to apprehend intuitively.
- ❸ Global and local tools (partial dependence plots, Shapley values) give **insights**, but do not make the model intuitive.
- ❹ Communicating a rigorous, intuitive understanding requires **limiting model complexity**.
- ❺ **Fairness requirements** can also impose dimensioning constraints on model development.

Dimensions of Explainability

- ❶ Deep learning networks are called **black-box models** due to complexity.
- ❷ Their formula is conceptually simple but **very large**, impossible to apprehend intuitively.
- ❸ Global and local tools (partial dependence plots, Shapley values) give **insights**, but do not make the model intuitive.
- ❹ Communicating a rigorous, intuitive understanding requires **limiting model complexity**.
- ❺ **Fairness requirements** can also impose dimensioning constraints on model development.

Dimensions of Explainability

- ❶ Deep learning networks are called **black-box models** due to complexity.
- ❷ Their formula is conceptually simple but **very large**, impossible to apprehend intuitively.
- ❸ Global and local tools (partial dependence plots, Shapley values) give **insights**, but do not make the model intuitive.
- ❹ Communicating a rigorous, intuitive understanding requires **limiting model complexity**.
- ❺ **Fairness requirements** can also impose dimensioning constraints on model development.

Dimensions of Explainability

- ❶ Deep learning networks are called **black-box models** due to complexity.
- ❷ Their formula is conceptually simple but **very large**, impossible to apprehend intuitively.
- ❸ Global and local tools (partial dependence plots, Shapley values) give **insights**, but do not make the model intuitive.
- ❹ Communicating a rigorous, intuitive understanding requires **limiting model complexity**.
- ❺ *Fairness requirements can also impose dimensioning constraints on model development.*

Dimensions of Explainability

- ❶ Deep learning networks are called **black-box models** due to complexity.
- ❷ Their formula is conceptually simple but **very large**, impossible to apprehend intuitively.
- ❸ Global and local tools (partial dependence plots, Shapley values) give **insights**, but do not make the model intuitive.
- ❹ Communicating a rigorous, intuitive understanding requires **limiting model complexity**.
- ❺ **Fairness requirements** can also impose dimensioning constraints on model development.

Fairness Example: Weequay and Togruta

- ❶ A US-based organization trains a model to predict worker performance and hires based on it.
- ❷ Hypothetically, everyone belongs to one of two groups: Weequay or Togruta.
- ❸ More Weequay attend university, so Weequay applicants are more numerous and more qualified.
- ❹ **Equal opportunity:** Hiring should not depend on group affiliation, all things being equal — but this leads to more Weequay hires.
- ❺ **Disparate impact:** Avoid practices that adversely affect one protected group more; assessed on subpopulation proportions.

Fairness Example: Weequay and Togruta

- ❶ A US-based organization trains a model to predict worker performance and hires based on it.
- ❷ Hypothetically, everyone belongs to one of two groups: **Weequay** or **Togruta**.
- ❸ More Weequay attend university, so Weequay applicants are more numerous and more qualified.
- ❹ **Equal opportunity:** Hiring should not depend on group affiliation, all things being equal — but this leads to more Weequay hires.
- ❺ **Disparate impact:** Avoid practices that adversely affect one protected group more; assessed on subpopulation proportions.

Fairness Example: Weequay and Togruta

- ❶ A US-based organization trains a model to predict worker performance and hires based on it.
- ❷ Hypothetically, everyone belongs to one of two groups: **Weequay** or **Togruta**.
- ❸ More Weequay attend university, so Weequay applicants are **more numerous and more qualified**.
- ❹ **Equal opportunity:** Hiring should not depend on group affiliation, all things being equal — but this leads to more Weequay hires.
- ❺ **Disparate impact:** Avoid practices that adversely affect one protected group more; assessed on subpopulation proportions.

Fairness Example: Weequay and Togruta

- ❶ A US-based organization trains a model to predict worker performance and hires based on it.
- ❷ Hypothetically, everyone belongs to one of two groups: **Weequay** or **Togruta**.
- ❸ More Weequay attend university, so Weequay applicants are **more numerous and more qualified**.
- ❹ **Equal opportunity:** Hiring should not depend on group affiliation, all things being equal — but this leads to more Weequay hires.
- ❺ **Disparate impact:** Avoid practices that adversely affect one protected group more; assessed on subpopulation proportions.

Fairness Example: Weequay and Togruta

- ❶ A US-based organization trains a model to predict worker performance and hires based on it.
- ❷ Hypothetically, everyone belongs to one of two groups: **Weequay** or **Togruta**.
- ❸ More Weequay attend university, so Weequay applicants are **more numerous and more qualified**.
- ❹ **Equal opportunity:** Hiring should not depend on group affiliation, all things being equal — but this leads to more Weequay hires.
- ❺ **Disparate impact:** Avoid practices that adversely affect one protected group more; assessed on **subpopulation proportions**.

The Fairness Trade-Off

- ❶ The two objectives are **mutually exclusive**.
- ❷ Equal opportunity $\rightarrow$ 60% (or more) Weequay and 40% (or fewer) Togruta hired $\rightarrow$ **disparate impact**.
- ❸ Forcing 40% Togruta $\rightarrow$ some rejected Weequay were predicted more qualified $\rightarrow$ **contradicts equal opportunity**.
- ❹ **80% rule**: The Togruta hiring rate should be $\geq 80\%$ of the Weequay hiring rate (here, up to 65% Weequay is acceptable).
- ❺ Defining these objectives **cannot be a decision made by data scientists alone**.

The Fairness Trade-Off

- ❶ The two objectives are **mutually exclusive**.
- ❷ Equal opportunity $\rightarrow$ 60% (or more) Weequay and 40% (or fewer) Togruta hired $\rightarrow$ **disparate impact**.
- ❸ Forcing 40% Togruta $\rightarrow$ some rejected Weequay were predicted more qualified $\rightarrow$ contradicts equal opportunity.
- ❹ 80% rule: The Togruta hiring rate should be $\geq$ 80% of the Weequay hiring rate (here, up to 65% Weequay is acceptable).
- ❺ Defining these objectives cannot be a decision made by data scientists alone.

The Fairness Trade-Off

- ❶ The two objectives are **mutually exclusive**.
- ❷ Equal opportunity $\rightarrow$ 60% (or more) Weequay and 40% (or fewer) Togruta hired $\rightarrow$ **disparate impact**.
- ❸ Forcing 40% Togruta $\rightarrow$ some rejected Weequay were predicted more qualified $\rightarrow$ contradicts **equal opportunity**.
- ❹ 80% rule: The Togruta hiring rate should be $\geq$ 80% of the Weequay hiring rate (here, up to 65% Weequay is acceptable).
- ❺ Defining these objectives cannot be a decision made by data scientists alone.

The Fairness Trade-Off

- ❶ The two objectives are **mutually exclusive**.
- ❷ Equal opportunity $\rightarrow$ 60% (or more) Weequay and 40% (or fewer) Togruta hired $\rightarrow$ **disparate impact**.
- ❸ Forcing 40% Togruta $\rightarrow$ some rejected Weequay were predicted more qualified $\rightarrow$ contradicts **equal opportunity**.
- ❹ **80% rule:** The Togruta hiring rate should be $\geq 80\%$ of the Weequay hiring rate (here, up to 65% Weequay is acceptable).
- ❺ Defining these objectives cannot be a decision made by data scientists alone.

The Fairness Trade-Off

- ❶ The two objectives are **mutually exclusive**.
- ❷ Equal opportunity $\rightarrow$ 60% (or more) Weequay and 40% (or fewer) Togruta hired $\rightarrow$ **disparate impact**.
- ❸ Forcing 40% Togruta $\rightarrow$ some rejected Weequay were predicted more qualified $\rightarrow$ contradicts **equal opportunity**.
- ❹ **80% rule:** The Togruta hiring rate should be $\geq 80\%$ of the Weequay hiring rate (here, up to 65% Weequay is acceptable).
- ❺ Defining these objectives **cannot be a decision made by data scientists alone**.

Implementing Fairness Objectives

- ❶ Without guidance, data scientists build **equal opportunity models** (the world as it is); most tools do this.
- ❷ Two independent models (one per group) may address overrepresentation, but cause **disparate treatment** — possibly discriminatory.
- ❸ **Post-processing**: Different thresholds per group adjust the trade-off, but may still be **disparate treatment**.
- ❹ Bias is as much a **business question** as a technical one; there may be many protected attributes.
- ❺ The solution depends on context: **privileges** (hiring), **negative impact** (fraud rejection), or **neutral** (disease prediction).

Implementing Fairness Objectives

- ❶ Without guidance, data scientists build **equal opportunity models** (the world as it is); most tools do this.
- ❷ **Two independent models** (one per group) may address overrepresentation, but cause **disparate treatment** — possibly discriminatory.
- ❸ **Post-processing**: Different thresholds per group adjust the trade-off, but may still be **disparate treatment**.
- ❹ Bias is as much a **business question** as a technical one; there may be many protected attributes.
- ❺ The solution depends on context: **privileges** (hiring), **negative impact** (fraud rejection), or **neutral** (disease prediction).

Implementing Fairness Objectives

- ❶ Without guidance, data scientists build **equal opportunity models** (the world as it is); most tools do this.
- ❷ **Two independent models** (one per group) may address overrepresentation, but cause **disparate treatment** — possibly discriminatory.
- ❸ **Post-processing:** Different thresholds per group adjust the trade-off, but may still be **disparate treatment**.
- ❹ Bias is as much a **business question** as a technical one; there may be many protected attributes.
- ❺ The solution depends on context: **privileges** (hiring), **negative impact** (fraud rejection), or **neutral** (disease prediction).

Implementing Fairness Objectives

- ❶ Without guidance, data scientists build **equal opportunity models** (the world as it is); most tools do this.
- ❷ **Two independent models** (one per group) may address overrepresentation, but cause **disparate treatment** — possibly discriminatory.
- ❸ **Post-processing:** Different thresholds per group adjust the trade-off, but may still be **disparate treatment**.
- ❹ Bias is as much a **business question** as a technical one; there may be many protected attributes.
- ❺ The solution depends on context: **privileges** (hiring), **negative impact** (fraud rejection), or **neutral** (disease prediction).

Implementing Fairness Objectives

- ❶ Without guidance, data scientists build **equal opportunity models** (the world as it is); most tools do this.
- ❷ **Two independent models** (one per group) may address overrepresentation, but cause **disparate treatment** — possibly discriminatory.
- ❸ **Post-processing**: Different thresholds per group adjust the trade-off, but may still be **disparate treatment**.
- ❹ Bias is as much a **business question** as a technical one; there may be many protected attributes.
- ❺ The solution depends on context: **privileges** (hiring), **negative impact** (fraud rejection), or **neutral** (disease prediction).

Version Management and Reproducibility

- ❶ Data scientists build, test, and iterate on **many model versions** and must keep them straight.
- ❷ **Need 1 — Experimentation:** Go back and forth on decisions, revert, and restore previous branches of experiments at dead ends.
- ❸ **Need 2 — Audit:** Data scientists, auditors, or managers may need to **replay computations** years later.
- ❹ Versioning is largely solved for code via **source version control**; platforms offer similar features for pipelines and configurations.
- ❺ Merging data/model branches is harder than code, but the basic need is to **return to a specific experiment** and copy its settings.

Version Management and Reproducibility

- ❶ Data scientists build, test, and iterate on **many model versions** and must keep them straight.
- ❷ **Need 1 — Experimentation:** Go back and forth on decisions, revert, and restore previous **branches** of experiments at dead ends.
- ❸ **Need 2 — Audit:** Data scientists, auditors, or managers may need to **replay computations** years later.
- ❹ Versioning is largely solved for code via **source version control**; platforms offer similar features for pipelines and configurations.
- ❺ Merging data/model branches is harder than code, but the basic need is to **return to a specific experiment** and copy its settings.

Version Management and Reproducibility

- ❶ Data scientists build, test, and iterate on **many model versions** and must keep them straight.
- ❷ **Need 1 — Experimentation:** Go back and forth on decisions, revert, and restore previous **branches** of experiments at dead ends.
- ❸ **Need 2 — Audit:** Data scientists, auditors, or managers may need to **replay computations** years later.
- ❹ Versioning is largely solved for code via **source version control**; platforms offer similar features for pipelines and configurations.
- ❺ Merging data/model branches is harder than code, but the basic need is to **return to a specific experiment** and copy its settings.

Version Management and Reproducibility

- ❶ Data scientists build, test, and iterate on **many model versions** and must keep them straight.
- ❷ **Need 1 — Experimentation:** Go back and forth on decisions, revert, and restore previous **branches** of experiments at dead ends.
- ❸ **Need 2 — Audit:** Data scientists, auditors, or managers may need to **replay computations** years later.
- ❹ Versioning is largely solved for code via **source version control**; platforms offer similar features for pipelines and configurations.
- ❺ Merging data/model branches is harder than code, but the basic need is to **return to a specific experiment** and copy its settings.

Version Management and Reproducibility

- ❶ Data scientists build, test, and iterate on **many model versions** and must keep them straight.
- ❷ **Need 1 — Experimentation:** Go back and forth on decisions, revert, and restore previous **branches** of experiments at dead ends.
- ❸ **Need 2 — Audit:** Data scientists, auditors, or managers may need to **replay computations** years later.
- ❹ Versioning is largely solved for code via **source version control**; platforms offer similar features for pipelines and configurations.
- ❺ Merging data/model branches is harder than code, but the basic need is to **return to a specific experiment** and copy its settings.

Reproducibility: Facets to Document (1/2)

- ❶ Operationalization needs model reproduction in **another environment** and possibly from a different starting point; repeatability also makes **debugging** possible.
- ❷ **Assumptions:** Decisions about the problem, scope, and data must be explicit and logged to check against new information.
- ❸ **Randomness:** Control pseudo-randomness via the generator seed; same seed gives the same sequence.
- ❹ **Data:** The same data must be available; data versioning storage can be prohibitive, and data branching tools are less mature than code.
- ❺ **Settings:** All processing must be reproducible with the same settings.

Reproducibility: Facets to Document (1/2)

- ❶ Operationalization needs model reproduction in **another environment** and possibly from a different starting point; repeatability also makes **debugging** possible.
- ❷ **Assumptions:** Decisions about the problem, scope, and data must be explicit and logged to check against new information.
- ❸ **Randomness:** Control pseudo-randomness via the generator seed; same seed gives the same sequence.
- ❹ **Data:** The same data must be available; data versioning storage can be prohibitive, and data branching tools are less mature than code.
- ❺ **Settings:** All processing must be reproducible with the same settings.

Reproducibility: Facets to Document (1/2)

- ❶ Operationalization needs model reproduction in **another environment** and possibly from a different starting point; repeatability also makes **debugging** possible.
- ❷ **Assumptions:** Decisions about the problem, scope, and data must be explicit and logged to check against new information.
- ❸ **Randomness:** Control pseudo-randomness via the generator **seed**; same seed gives the same sequence.
- ❹ **Data:** The same data must be available; data versioning storage can be prohibitive, and data branching tools are less mature than code.
- ❺ **Settings:** All processing must be reproducible with the same settings.

Reproducibility: Facets to Document (1/2)

- ❶ Operationalization needs model reproduction in **another environment** and possibly from a different starting point; repeatability also makes **debugging** possible.
- ❷ **Assumptions:** Decisions about the problem, scope, and data must be explicit and logged to check against new information.
- ❸ **Randomness:** Control pseudo-randomness via the generator **seed**; same seed gives the same sequence.
- ❹ **Data:** The same data must be available; data versioning storage can be prohibitive, and data branching tools are less mature than code.
- ❺ **Settings:** All processing must be reproducible with the same settings.

Reproducibility: Facets to Document (1/2)

- ➊ Operationalization needs model reproduction in **another environment** and possibly from a different starting point; repeatability also makes **debugging** possible.
- ➋ **Assumptions:** Decisions about the problem, scope, and data must be explicit and logged to check against new information.
- ➌ **Randomness:** Control pseudo-randomness via the generator **seed**; same seed gives the same sequence.
- ➍ **Data:** The same data must be available; data versioning storage can be prohibitive, and data branching tools are less mature than code.
- ➎ **Settings:** All processing must be reproducible with the same settings.

Reproducibility: Facets to Document (2/2)

- ❶ **Results:** Compare in-depth analyses (confusion matrices to partial dependence plots), not just text diffs.
- ❷ **Implementation:** Slightly different implementations can yield different predictions on close calls; batch vs. live API scoring may need different implementations.
- ❸ **Environment:** A model is more than its algorithm and parameters; a different Python package version can change results — **freeze computation environments**.
- ❹ Part of documentation can be **automated**; integrated design-and-deployment platforms reduce reproducibility cost.
- ❺ Version management and reproducibility are **critical** where governance and audits matter.

Reproducibility: Facets to Document (2/2)

- ❶ **Results:** Compare in-depth analyses (confusion matrices to partial dependence plots), not just text diffs.
- ❷ **Implementation:** Slightly different implementations can yield different predictions on close calls; batch vs. live API scoring may need different implementations.
- ❸ **Environment:** A model is more than its algorithm and parameters; a different Python package version can change results — **freeze computation environments**.
- ❹ Part of documentation can be **automated**; integrated design-and-deployment platforms reduce reproducibility cost.
- ❺ Version management and reproducibility are **critical** where governance and audits matter.

Reproducibility: Facets to Document (2/2)

- ❶ **Results:** Compare in-depth analyses (confusion matrices to partial dependence plots), not just text diffs.
- ❷ **Implementation:** Slightly different implementations can yield different predictions on close calls; batch vs. live API scoring may need different implementations.
- ❸ **Environment:** A model is more than its algorithm and parameters; a different Python package version can change results — **freeze computation environments**.
- ❹ Part of documentation can be **automated**; integrated design-and-deployment platforms reduce reproducibility cost.
- ❺ Version management and reproducibility are **critical** where governance and audits matter.

Reproducibility: Facets to Document (2/2)

- ❶ **Results:** Compare in-depth analyses (confusion matrices to partial dependence plots), not just text diffs.
- ❷ **Implementation:** Slightly different implementations can yield different predictions on close calls; batch vs. live API scoring may need different implementations.
- ❸ **Environment:** A model is more than its algorithm and parameters; a different Python package version can change results — **freeze computation environments**.
- ❹ Part of documentation can be **automated**; integrated design-and-deployment platforms reduce reproducibility cost.
- ❺ Version management and reproducibility are **critical** where governance and audits matter.

Reproducibility: Facets to Document (2/2)

- ❶ **Results:** Compare in-depth analyses (confusion matrices to partial dependence plots), not just text diffs.
- ❷ **Implementation:** Slightly different implementations can yield different predictions on close calls; batch vs. live API scoring may need different implementations.
- ❸ **Environment:** A model is more than its algorithm and parameters; a different Python package version can change results — **freeze computation environments**.
- ❹ Part of documentation can be **automated**; integrated design-and-deployment platforms reduce reproducibility cost.
- ❺ Version management and reproducibility are **critical** where governance and audits matter.

Closing Thoughts

- ❶ Model development is one of the **most critical and consequential** steps of MLOps.
- ❷ Technical decisions made here have **big repercussions** throughout the life of the model.
- ❸ **Exposure, transparency, and collaboration** are crucial to long-term success.
- ❹ It is the stage most practiced by data scientists.
- ❺ In the pre-MLOps world, it often represented the **whole ML effort**, producing a model used as is, with all its consequences and limitations.

Closing Thoughts

- ❶ Model development is one of the **most critical and consequential** steps of MLOps.
- ❷ Technical decisions made here have **big repercussions** throughout the life of the model.
- ❸ **Exposure, transparency, and collaboration** are crucial to long-term success.
- ❹ It is the stage most practiced by data scientists.
- ❺ In the pre-MLOps world, it often represented the **whole ML effort**, producing a model used as is, with all its consequences and limitations.

Closing Thoughts

- ❶ Model development is one of the **most critical and consequential** steps of MLOps.
- ❷ Technical decisions made here have **big repercussions** throughout the life of the model.
- ❸ **Exposure, transparency, and collaboration** are crucial to long-term success.
- ❹ It is the stage most practiced by data scientists.
- ❺ In the pre-MLOps world, it often represented the **whole ML effort**, producing a model used as is, with all its consequences and limitations.

Closing Thoughts

- ❶ Model development is one of the **most critical and consequential** steps of MLOps.
- ❷ Technical decisions made here have **big repercussions** throughout the life of the model.
- ❸ **Exposure, transparency, and collaboration** are crucial to long-term success.
- ❹ It is the stage most practiced by data scientists.
- ❺ In the pre-MLOps world, it often represented the **whole ML effort**, producing a model used as is, with all its consequences and limitations.

Closing Thoughts

- ❶ Model development is one of the **most critical and consequential** steps of MLOps.
- ❷ Technical decisions made here have **big repercussions** throughout the life of the model.
- ❸ **Exposure, transparency, and collaboration** are crucial to long-term success.
- ❹ It is the stage most practiced by data scientists.
- ❺ In the pre-MLOps world, it often represented the **whole ML effort**, producing a model used as is, with all its consequences and limitations.